APPLICATIONS OF THE SCHWINGER VARIATIONAL PRINCIPLE TO ELECTRON–MOLECULE COLLISIONS AND MOLECULAR PHOTOIONIZATION

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Abstract:

In this article we present a detailed overview of our studies of molecular photoionization and electron–molecule collisions in which we have used Schwinger-like variational principles and several important extensions of these principles. The various variational functionals and formulations, the interrelationships between these formulations, and a detailed discussion of the numerical and computational procedures which have been used in applications are presented.

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1. Introduction

1.1. Background and scope of review

The last decade has witnessed a significant increase in experimental activity in the general area of molecular photoionization processes [1, 2] and electron–molecule collisions [3]. In the area of molecular photoionization, this progress has been primarily due to the increasing availability of synchrotron radiation sources. Such experimental studies have in turn stimulated the development of theoretical methods which are needed both for the quantitative prediction of the appropriate cross sections and for a fundamental understanding of the underlying dynamics of these collision processes [4]. These methods naturally involve several additional concepts and techniques related to the strictly molecular aspects of these problems, particularly the anisotropy of the multicenter molecular force fields. For example, several of these [4] attempt to utilize discrete basis functions in the solution of these low-energy collision problems. Such basis set methods have been successfully used in bound state molecular electronic structure applications. Moreover, the evolution of several of these methods has led to hybrid procedures which combine basis set approaches with numerical techniques developed for the related atomic collision problems [5–8].

In our studies of molecular photoionization processes and electron–molecule collisions over the last several years we have made extensive use of methods and techniques which are based on, or in some way related to, the Schwinger variational principle and extensions of this variational principle [5, 9–12]. In the course of this work, various formulations and numerical procedures for the solution of the related collisional equations for both single and multichannel scattering [5, 10–13] have been developed and published in several papers. For obvious reasons, such numerical procedures and often the interrelationships between formulations have not been discussed in great detail in any single publication. The purpose of this article is to present a review of our studies of molecular photoionization and low-energy electron–molecule collisions in which we have used Schwinger-like variational principles. We emphasize that in this article we focus primarily on our contributions in this area and do not attempt to review or assess the many contributions of others to this field [13].

1.2. Outline of contents

An outline of the article is as follows. In section 2, we discuss the Schwinger variational principle for elastic scattering and the development of related higher-order variational functionals from the Lippmann–Schwinger integral equation. This is followed by a discussion of variational functionals for general matrix elements which are again based on the integral equation. We then go on to clarify the relationship between the Kohn and Schwinger variational methods by means of some simple arguments. Next we discuss several approaches to solving these variational functionals and, implicitly, the related integral equations. These include a procedure we refer to as the iterative Schwinger method, another procedure based on Padé approximants, and a hybrid method which is particularly suitable for long-range potentials. This is followed by a discussion of our recent formulation of a multichannel extension of the Schwinger principle. Next we discuss some numerical techniques for the explicit solution of these equations. These techniques include the single-center expansion method and basis set approaches. In this section our discussion of the single-center expansion method gives several important details of current applications of this technique to molecular problems. In the next section we discuss many examples of our recent studies of molecular photoionization where these techniques have been applied, e.g., N_2, CO, CO_2 and C_2H_2. We then go on to a similar discussion of applications of these numerical techniques and multichannel formulation to elastic and electronically inelastic scattering of low-energy electrons by molecules. We end with a few concluding remarks and some discussion of future work.

2. General formulation

2.1. Schwinger variational principle for elastic scattering

Variational methods provide a powerful way of obtaining approximate solutions to a variety of differential and integral equations. Variational principles can be conveniently classified into differential or integral forms. A variational principle of differential form is based directly on the differential equation and requires trial functions satisfying the boundary conditions of the problem. On the other hand, a variational principle of integral form is based on the integral equation and therefore the boundary conditions, which are taken into account through the Green's function, need not be incorporated in the trial function. The Kohn variational principle [14] is probably the most widely used variational principle in quantum collision phenomena. It is of differential form and is based on Schrödinger's equation. In spite of several desirable features, the Schwinger variational principle [15, 16], which is based on the Lippmann–Schwinger equation has received less attention as a practical method for collision problems. This is undoubtedly due primarily to the occurrence of the Green's function in this variational principle.

For convenience, consider the simple example of potential scattering defined by the Schrödinger equation

$$(\tfrac{1}{2}\nabla^2 + k^2/2)\, \Psi_{\boldsymbol{k}}(\boldsymbol{r}) = V(\boldsymbol{r})\, \Psi_{\boldsymbol{k}}(\boldsymbol{r}) \tag{2.1}$$

where $\Psi_{\boldsymbol{k}}(\boldsymbol{r})$ must satisfy specific asymptotic boundary conditions, e.g.,

$$\Psi_{\boldsymbol{k}}^{(+)}(\boldsymbol{r}) \xrightarrow[r\to\infty]{} A\left(\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}) + f(\theta,\phi)\frac{\exp(\mathrm{i}kr)}{r}\right). \tag{2.2}$$

Here and elsewhere we use atomic units. Equation (2.1) and the boundary condition of eq. (2.2) can both be incorporated in a single integral or Lippmann–Schwinger equation of the form

$$\Psi_{\boldsymbol{k}}^{(+)}(\boldsymbol{r}) = S_{\boldsymbol{k}}(\boldsymbol{r}) + \int G_0^{(+)}(\boldsymbol{r},\boldsymbol{r}')\, V(\boldsymbol{r}')\, \Psi_{\boldsymbol{k}}^{(+)}(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}'\,. \tag{2.3}$$

In eq. (2.3) $G_0^{(+)}(\boldsymbol{r},\boldsymbol{r}')$ is the free-particle Green's function with outgoing-wave boundary conditions defined by

$$\tfrac{1}{2}(\nabla^2 + k^2)\, G_0^{(+)}(\boldsymbol{r},\boldsymbol{r}') = \delta(\boldsymbol{r}-\boldsymbol{r}') \tag{2.4}$$

and $S_{\boldsymbol{k}}(\boldsymbol{r})$ is the regular solution of the corresponding homogeneous equation. By examining the asymptotic form of eq. (2.3) we see that the Green's function enforces the boundary condition of eq. (2.2) on the solutions of eq. (2.3).

The Schwinger variational principle is a variational principle for the scattering amplitude f of eq. (2.2) which is based on eq. (2.3). To see this we consider the transition matrix element T_{fi} which is defined as

$$T_{\mathrm{fi}} = \langle S_{\boldsymbol{k}_\mathrm{f}}|T|S_{\boldsymbol{k}_\mathrm{i}}\rangle = \langle S_{\boldsymbol{k}_\mathrm{f}}|V|\Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}}\rangle \tag{2.5}$$

and is related to the scattering amplitude through

$$f = -4\pi^2 T_{\mathrm{fi}}. \tag{2.6}$$

This transition matrix element can also be written as

$$T_{\mathrm{fi}} = \langle \Psi^{(-)}_{\boldsymbol{k}_\mathrm{f}}|V|S_{\boldsymbol{k}_\mathrm{i}}\rangle \tag{2.7}$$

and

$$T_{\mathrm{fi}} = \langle \Psi^{(-)}_{\boldsymbol{k}_\mathrm{f}}|V - VG_0^{(+)}V|\Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}}\rangle\,. \tag{2.8}$$

Equation (2.8) follows from eq. (2.7) by using the Lippmann–Schwinger equation (eq. (2.3)), i.e.,

$$S_{\boldsymbol{k}_\mathrm{i}} = \Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}} - G_0^{(+)}V\Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}}\,. \tag{2.9}$$

From eqs. (2.5), (2.7) and (2.8) we obtain

$$T_{\mathrm{fi}} = \langle S_{\boldsymbol{k}_\mathrm{f}}|V|\Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}}\rangle\langle \Psi^{(-)}_{\boldsymbol{k}_\mathrm{f}}|V|S_{\boldsymbol{k}_\mathrm{i}}\rangle/\langle \Psi^{(-)}_{\boldsymbol{k}_\mathrm{f}}|V - VG_0^{(+)}V|\Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}}\rangle\,. \tag{2.10}$$

Variation of the right-hand side of eq. (2.10) due to small changes in the exact $\Psi^{(+)}_{\boldsymbol{k}_\mathrm{i}}$ and $\Psi^{(-)}_{\boldsymbol{k}_\mathrm{f}}$ shows that this expression for T_{fi} is stationary. Equation (2.10) is the normalization-independent form of the Schwinger variational principle [15, 16].

The Schwinger variational principle has several important and practical features. For example, this principle automatically incorporates the correct boundary conditions and does not require trial wave functions which satisfy these boundary conditions as in the case of the Kohn method. Since the trial functions are directly associated with the potential V in eq. (2.10), the method only requires trial wave functions in the region where the scattering potential is nonvanishing. This feature is particularly desirable in applications to the scattering of low-energy electrons by the nonspherical force fields of molecules and molecular ions where discrete basis functions may be used exclusively in the trial wave function. These basis functions such as Cartesian Gaussian and Slater orbitals can be very effective in representing the multicenter nature of molecular continuum wave functions and have been used extensively in molecular electronic structure calculations [17]. Finally, the Schwinger method is not generally expected to be troubled by the spurious singularities which arise in the Kohn variational method [18]. That this is true is suggested by the following simple argument. From eq. (2.10) it follows that the matrix which must be inverted for a finite basis in the Schwinger variational method will be $V(1 - G_0V)$. But this matrix is expected to be singular only when G_0V has an eigenvalue near 1 which is the condition for a true bound state pole or resonance pole in the T-matrix. Although highly unlikely, if a resonance other than these two cases should ever arise for a finite basis, it will disappear in the Schwinger method as the size of the basis is increased. In contrast, in the Kohn method such a spurious resonance can persist even as the basis approaches completeness [18]. On the other hand, the Kohn principle requires matrix elements of the Hamiltonian whereas in the Schwinger principle one must evaluate integrals containing the more complicated operator $VG_0^{(+)}V$. We believe, however, that with current computer capabilities the occurrence of the $VG_0^{(+)}V$ term in this variational principle is no longer a serious drawback.

2.2. Higher-order variational principles

Iterative schemes for the solution of the Lippmann–Schwinger equation lead very conveniently to higher-order variational functionals for the transition matrix which are related to the Schwinger variational principle. In general, only the lower-order functionals among these are of any practical use. To study these functionals we first consider two iterative forms of the Lippmann–Schwinger equation. We can either multiply the integral equation for Ψ directly by G_0V

$$(G_0V)^n\Psi = (G_0V)^n S + (G_0V)^{n+1}\Psi \tag{2.11}$$
*

or we can introduce the function C_n

$$C_n = \Psi - \sum_{i=0}^{n-1} (G_0V)^i S \tag{2.12}$$

where $C_0 = \Psi$, and observe that, since C_n and $(G_0V)^n\Psi$ are equivalent, this function satisfies the integral equation

$$C_n = (G_0V)^n S + G_0VC_n\,. \tag{2.13}$$

The integral equations for Ψ and C_n can be solved iteratively to obtain

$$(G_0V)^n \Psi^{[m]} = (G_0V)^n \sum_{i=0}^{m} (G_0V)^i S + (G_0V)^{(n+m+1)} \Psi^{[m]} \tag{2.14}$$

$$C_n^{[m]} = (G_0V)^n \sum_{i=0}^{m} (G_0V)^i S + (G_0V)^{m+1} C_n^{[m]} \tag{2.15}$$

where the superscript $[m]$ indicates the mth iteration. These equations are merely different expressions for the same quantity. Variational functionals for these integral equations, eqs. (2.14) and (2.15), can be readily constructed, i.e.,

$$Y_n^m = \left|\langle \Psi^{[m]}|V(G_0V)^n \sum_{i=0}^{m} (G_0V)^i|S\rangle\right|^2 \Big/ \langle \Psi^{[m]}|V\{(G_0V)^n - (G_0V)^{(n+m+1)}\}|\Psi^{[m]}\rangle \tag{2.16}$$

* Where no ambiguity arises we omit subscripts and superscripts for convenience.

and

$$X_n^m = \left|\langle C_n^{[m]}|V(G_0V)^n \sum_{i=0}^{m} (G_0V)^i|S\rangle\right|^2 \Big/ \langle C_n^{[m]}|V - V(G_0V)^{m+1}|C_n^{[m]}\rangle \tag{2.17}$$

respectively. Note that these functionals are not identical since Y_n^m and X_n^m have different values.

These variational principles are only solved approximately and hence the resulting K-matrices can, in principle, be different. For example, consider the case for $n = 1$ and $m = 0$. For these two functionals we have

$$X_1^0 = \langle C_1|VG_0V|S\rangle\langle S|VG_0V|C_1\rangle/\langle C_1|V - VG_0V|C_1\rangle \tag{2.18}$$

and

$$Y_1^0 = \langle\Psi|VG_0V|S\rangle\langle S|VG_0V|\Psi\rangle/\langle\Psi|VG_0V - VG_0VG_0V|\Psi\rangle\,. \tag{2.19}$$

Comparison of the denominators of eqs. (2.18) and (2.19) shows that the functional Y_1^0 should provide a higher-order result than X_1^0. Of course, Y_1^0 would be very difficult to handle in any practical studies.

In the case of $m = 0$, the relationships between these functionals can also be understood systematically within the framework of Padé approximants. We will discuss these relationships in section 2.4.2.

2.3. Comparison of specific variational functionals

2.3.1. The Schwinger and Kohn principles

The Kohn and Schwinger variational principles are typical examples of variational functionals of differential and integral forms respectively. In the previous section we have discussed a family of Schwinger-like variational functionals. Many variants of the Kohn variational principle can also be derived [18–21]. As stated earlier, Kohn-type methods have been more widely applied to collision problems than the Schwinger methods. Recent numerical experience has shown that the occurrence of the Green's function in the Schwinger-like expressions is not a serious drawback to their application and hence it is appropriate to clarify the formal relationship between these methods.

Altshuler's studies [22] were probably the first to compare the Kohn and Schwinger variational principles numerically. His conclusion was that for a given trial function the Schwinger seemed to provide a better estimate of the phase shift than the Kohn method. However, from an indirect comparison of these two variational methods, Schwartz [23] found that for comparable trial functions the Kohn method converged faster. More recently, Thirumalai and Truhlar [24] also concluded that the Kohn and Harris-Michels methods [25] showed much better convergence to the accurate result for a specific model problem than the Schwinger method. There have also been some theoretical studies of the relationship between these variational principles. For example, Kato [26] obtained the direct relationship between the Schwinger and Rubinow principles. Lieber [27] discussed the relationship between these two principles for three-body breakup processes. Without proof, Delves [28] also asserted that the Schwinger principle should provide higher-order results than the Kohn principle. To clarify the convergence characteristics of these variational principles, we have studied the formal hierarchical relationship between them [29]. In what follows we will outline this discussion and show that for a given trial function the Schwinger principle provides a more accurate estimate of the scattering matrix than does the Kohn method.

We begin by defining the asymmetric variational functional [29]

$$I(C_a, C_b) = \langle C_a|E - H|C_b\rangle - \langle C_a|V|S\rangle - \langle S|V|C_b\rangle - \langle S|V|S\rangle \tag{2.20}$$

in which the two trial functions for the C functions defined in eq. (2.12) satisfy the boundary conditions, e.g., in a one-dimensional case

$$C_a = \begin{cases} \lambda k^{-1/2} \cos kr, & r \to \infty \\ 0, & r \to 0 \end{cases} \tag{2.21}$$

where λ is the tangent of the phase shift. In the Kohn principle λ is given by the functional

$$\begin{aligned} \tfrac{1}{2}[\lambda]_k &= I(C_1^{(0)}, C_1^{(0)}) \\ &= \langle C_1^{(0)}|E - H|C_1^{(0)}\rangle - \langle C_1^{(0)}|V|S\rangle - \langle S|V|C_1^{(0)}\rangle - \langle S|V|S\rangle \end{aligned} \tag{2.22}$$

and $C_1^{(0)}$ is expanded in some basis

$$C_1^{(0)} = \Psi - S = \lambda\bar{C} + \sum_i a_i\eta_i \tag{2.23}$$

where $\bar{C}$ is an asymptotically irregular function such as the Neumann function which has been regularized at the origin. The η_i's are L^2-basis functions. We consider the integral equation satisfied by the exact C function

$$C_1 = G_0VS + G_0VC_1\,. \tag{2.24}$$

If $C_1^{(0)}$ is a reasonable trial function, insertion of the Kohn trial function $C_1^{(0)}$ into the right-hand side of eq. (2.24) should provide an improved trial function $C_1^{(1)}$, i.e.,

$$C_1^{(1)} = G_0VS + G_0VC_1^{(0)}\,. \tag{2.25}$$

These functions $C_1^{(0)}$ and $C_1^{(1)}$ are approximate and should not be confused with the exact functions of section 2.2. For example, $C_1^{(1)}$ is an approximation to $C_1^{[0]}$ of eq. (2.15). Replacement of one of the $C_1^{(0)}$ functions in eq. (2.22) with $C_1^{(1)}$ of eq. (2.25) gives

$$I(C_1^{(0)}, C_1^{(1)}) = \langle C_1^{(0)}|E - H|C_1^{(1)}\rangle - \langle C_1^{(0)}|V|S\rangle - \langle S|V|C_1^{(1)}\rangle - \langle S|V|S\rangle \tag{2.26}$$

which in turn can be written as

$$I(C_1^{(0)}, C_1^{(1)}) = \langle \Psi^{(0)}|(V - VG_0V)|\Psi^{(0)}\rangle - \langle \Psi^{(0)}|V|S\rangle - \langle S|V|\Psi^{(0)}\rangle \tag{2.27}$$

where $\Psi^{(0)} = S + C_1^{(0)}$. The right-hand side of eq. (2.27) is just the bilinear form of the Schwinger functional [30]. Hence we have

$$\tfrac{1}{2}[\lambda]_S = I(C_1^{(0)}, C_1^{(1)}) \tag{2.28}$$

where $[\lambda]_S$ is just $\tan\delta$ given by the Schwinger procedure. Comparison of eqs. (2.22) and (2.28) shows that for a specific trial function the Schwinger principle yields a higher-order estimate of the K-matrix than the Kohn principle.

Some additional insight into the convergence characteristics of these variational principles can be obtained by looking at the approximations implied in the solution of the Lippmann–Schwinger equation for the K-matrix in the two cases. It is well-known that the approximation to the K-operator obtained from the Schwinger variational principle is equivalent to exactly solving the integral equation for the K-operator

$$K = V + VG_0K \tag{2.29}$$

using a separable potential of the form

$$V^S = \sum_{\alpha,\beta} V|\alpha\rangle(V^{-1})_{\alpha,\beta}\langle\beta|V. \tag{2.30}$$

A separable potential of this form is known to be exact within the space spanned by the basis $\{\alpha\}$

$$V^S|\alpha\rangle = V|\alpha\rangle. \tag{2.31}$$

To discuss the relationship of this approximation to the Kohn principle, we write a variant of eq. (2.29)

$$f = VG_0V + VG_0f \tag{2.32}$$

where $f = K - V$. The Kohn method is equivalent to solving eq. (2.32) using the finite-rank Green's function

$$G_0^S = \sum_{\alpha,\beta} |\alpha\rangle(B^{-1})_{\alpha\beta}\langle\beta| \tag{2.33}$$

where

$$B_{\alpha\beta} = \langle\alpha|G_0^{-1}|\beta\rangle = \langle\alpha|E - H_0|\beta\rangle. \tag{2.34}$$

This finite-rank Green's function does not satisfy the relationship satisfied by V^S, i.e.,

$$G_0^S|\alpha\rangle \neq G_0|\alpha\rangle. \tag{2.35}$$

Finite-rank approximations with the property of eq. (2.31) are generally expected to be better approximations to operators than finite-rank approximations which do not satisfy this relation [31]. This observation again suggests that the Schwinger principle should yield a higher-order result than the Kohn method.

2.3.2. Numerical examples

We now consider two numerical examples which illustrate the relationships discussed above. Our first example [32] is the model two-channel problem originally devised by Huck [33]. The explicit form of the model Hamiltonian is

$$H = |\chi_1\rangle(-\tfrac{1}{2}\mathrm{d}^2/\mathrm{d}r^2)\langle\chi_1| + |\chi_2\rangle(-\tfrac{1}{2}\mathrm{d}^2/\mathrm{d}r^2 + \Delta E)\langle\chi_2| + \sum_{m \neq n}^{2} |\chi_m\rangle V_{mn} \langle\chi_n| \tag{2.36}$$

where

$$V_{12} = V_{21} = \begin{cases} \text{constant}, & r \leq a \\ 0, & r > a \end{cases} \tag{2.37}$$

and $\langle\chi_m|\chi_n\rangle = \delta_{mn}$. The scattering basis functions used in the variational calculation were [32]

$$\eta_i^m = |\chi_m\rangle r^i \mathrm{e}^{-\alpha r} \qquad (i = 1, 2, \ldots N). \tag{2.38}$$

The results of studies of this model using the Schwinger and other Kohn-type variational methods are summarized in table 2.1. These Kohn-type methods include the anomaly free (AF) [34], minimum norm (MN) [35], optimized anomaly free (OAF) [36] and the restricted interpolated anomaly free (RIAF) [37] procedures. As expected, these results clearly show that the Schwinger principle does provide K-matrices which are superior to those of the Kohn method. Furthermore, we note that the Schwinger trial function did not include the long-range basis functions required in the Kohn methods.

Our second example will be S-wave scattering by the exponential potential

$$V(r) = -\exp(-r). \tag{2.39}$$

Table 2.1
Comparison of the Schwinger and Kohn-like variational methods for the model two-channel problem of Huck [33]: Deviation of the variationally determined K-matrices from the exact values[a]

		AF[b]	MN[c]	OAF[d]	RIAF[e]	Schwinger ($\alpha^{f} = 2.5$)	Schwinger ($\alpha^{f} = 0.9$)
ΔK_{11}	$N =$ 1	–	−18.72853		–	−19.09277	−0.14259
	2	–	−58.41920	–	–	0.26131	1.53971
	4	−5.61743	−5.69784	−4.54847	–	−0.01000	−0.00004
	6	−2.99726	3.29061	−2.99989	−3.00448	0.0	0.0
	10	−1.39131	−1.40472	−1.37881	−1.39040	0.0	0.0
	25	−0.363330	–	−0.33532	−0.29985	0.0	0.0
ΔK_{12}	$N =$ 1	–	11.86603	–	–	12.10630	−0.09141
	2	–	37.84736	–		0.07861	0.96567
	4	3.57888	3.66193	2.89239	–	0.00600	0.00002
	6	1.91396	2.09620	1.91579	1.91839	0.0	0.0
	10	0.88909	0.89739	0.88115	0.88850	0.0	0.0
	25	0.23037	–	0.21363	0.19117	0.0	0.0
ΔK_{22}	$N =$ 1	–	−7.54469	–	–	−7.57920	0.19492
	2	–	24.52584	–	–	0.01638	−0.60478
	4	−2.28298	−2.35953	−1.84341	–	−0.00355	0.00002
	6	−1.22397	−1.33719	−1.22525	−1.22670	0.0	0.0
	10	−0.56896	−0.57410	−0.56393	−0.56858	0.0	0.0
	25	−0.14621	–	−0.13619	−0.12198	0.0	0.0

a) The exact values are: $K_{11} = 21.76525$, $K_{12} = K_{21} = -14.12742$ and $K_{22} = 8.73385$ (ref. [34]);
b) Ref. [34]; c) Ref. [35]; d) Ref. [36]; e) Ref. [37]; f) The exponent of the basis function in eq. (2.38).

For a meaningful and consistent comparison of these variational principles we use the same trial function in all calculations. This trial function is of the form

$$r\psi(r) = \alpha_0 \sin kr + \alpha_i(1 - e^{-\beta r}) \cos kr + \sum_{i=1}^{n} d_i r^i e^{-\alpha r} . \tag{2.40}$$

Results of Kohn calculations using the AF, OAF, MNK (minimum norm Kohn), and OMN (optimized minimum norm) methods are compared with the Schwinger results in table 2.2. This comparison again illustrates the higher-order convergence characteristic of the Schwinger variational principle.

An analysis of the role of long-range basis functions in these calculations is instructive. In the Kohn calculations of Thirumalai et al. the long-range basis functions were not included in the Schwinger trial function. To illustrate the role of such basis functions in these studies we look at S-wave scattering by a cut-off version of the potential of eq. (2.39), i.e.,

$$V_R(r) = \begin{cases} -e^{-r}, & r \leq R \\ 0, & r > R \end{cases} . \tag{2.41}$$

In these Schwinger calculations the trial function included only four discrete basis functions of the type shown in eq. (2.30) with an exponent of 2.5. The continuum functions were specifically excluded in these studies. The resulting K-matrices are shown in table 2.3 and illustrate that, as the cut-off radius increases, the long-range basis functions play an increasingly important role. For example, for $R \geq 3.0$ a.u. the short-range basis functions already fail to adequately represent the entire range of the potential.

2.3.3. The Schwinger and $\tilde{C}$-functionals

Among the variational functionals based on the Lippmann–Schwinger equation, the so-called $\tilde{C}$-functional [12, 38], i.e.,

$$X_{ij} = \langle C_i | VG_0V | S_j \rangle \langle S_i | VG_0V | C_j \rangle / \langle C_i | V - VG_0V | C_j \rangle \tag{2.42}$$

Table 2.2
Comparison of the Schwinger and Kohn methods for the model potential of eq. (2.39): Ratio of variational K-matrices to the accurate values for $k = 0.55$ a.u.[a]

n[b]	AF[c]	OAF	MNK	MNR	OMN	Schwinger[d]
0	0.9735	0.9733	0.9733	0.9735	0.9733	0.9972
2	0.9968	0.9940	0.9902	0.9969	0.9941	0.9999
4	0.9999	0.9970	0.9910	0.9999	0.9970	1.0000

[a] Accurate value is $K_0 = 2.2003827$.
[b] The number of discrete basis functions in the trial function. See eq. (2.40). For $n = 0$ no discrete basis functions are included in the trial function.
[c] The results in the AF, OAF, MNK, MNR and OMN columns are from ref. [24] except those for $n = 0$ which are from ref. [19].
[d] Results with the Schwinger variational principle.

Table 2.3
S-wave K-matrix elements for the potential of eq. (2.41) at $k = 0.55$ a.u.

R(a.u.)[a]	$\tilde{K}_0$[b]	$\tilde{K}_0/K_0$[c]
1.0	0.26265	1.0000
2.0	1.44933	0.9991
3.0	2.02861	0.9968
4.0	2.04780	0.9935
5.0	2.03975	0.9711
6.0	2.03853	0.9447
20.0	2.03843	0.9264

[a] Value of the cut-off radius in the potential of eq. (2.41).
[b] Schwinger variational results with 4 discrete basis functions with $\alpha = 2.5$ in eq. (2.40).
[c] Ratio of the K-matrices in the preceding column with the accurate values.

is particularly important. This functional was originally proposed by Kolsrud [39] and subsequently discussed more fully by us [12]. This functional is related to the K-matrix by

$$-\tfrac{1}{2}K_{ij} = \langle S_i|V|S_j\rangle + \langle S_i|VG_0V|S_j\rangle + X_{ij} \tag{2.43}$$

and can be very useful in dealing with long-range potentials. One obvious difference between the $\tilde{C}$-functional and the Schwinger principle is that the former contains the VG_0V term and not simply the potential in the numerator. However, since this term already occurs in the denominator of the Schwinger functional, its presence in the numerator of the $\tilde{C}$-functional does not significantly affect the overall computational effort. In fact, experience shows that the evaluation of the $\tilde{C}$-functional only requires about 15% more computer time than does the evaluation of the Schwinger principle. Furthermore, all the numerical procedures which have been developed for the Schwinger method are directly applicable to the $\tilde{C}$-functional [12].

An important feature of the $\tilde{C}$-functional is that it yields results comparable to those obtained after one iteration of the iterative Schwinger variational method [12]. This can be seen as follows. The K-matrix provided by the $\tilde{C}$-functional can be written as

$$-\tfrac{1}{2}K^{C}_{ij} = \langle S_i|V|S_j\rangle + \langle S_i|VG_0V|S_j\rangle + \langle S_i|VG_0V(V - VG_0V)^{-1}VG_0V|S_j\rangle \tag{2.44}$$

where $(V - VG_0V)^{-1}$ is taken in the sense of matrix inversion. This result is implicitly based on the integral equation

$$(V - VG_0V)C^{(1)}_j = VG_0VS_j\,. \tag{2.45}$$

We now use an iterative step in eq. (2.25) to write

$$C^{(2)}_j = G_0VS_j + G_0VC^{(1)}_j\,. \tag{2.46}$$

With this $C_j^{(2)}$ the total scattering function is given by

$$\phi_j^{(2)} = S_j + C_j^{(2)} = S_j + G_0 V S_j + G_0 V C_j^{(1)} \tag{2.47}$$

and the corresponding K-matrix by

$$-\tfrac{1}{2} K_{ij}^{\mathrm{C}} = \langle S_i | V | \phi_j^{(2)} \rangle . \tag{2.48}$$

This result immediately shows that the K-matrix of the $\tilde{\mathrm{C}}$-functional is one rank higher than that of the Schwinger method which, in turn, can be written as

$$-\tfrac{1}{2} K_{ij}^{\mathrm{S}} = \langle S_i | V | \phi_j^{(1)} \rangle . \tag{2.49}$$

The important practical implication of this result stems from our experience that a single iteration in the Schwinger procedure can lead to dramatic improvements in K-matrices obtained with quite small basis set expansions. This is certainly true for polar molecules [12]. The uniterated $\tilde{\mathrm{C}}$-functional method can hence be a useful alternative to the iterated Schwinger principle where large basis sets would make the iterative procedure impractical [12]. Finally, iterative improvements can also be included as well in the $\tilde{\mathrm{C}}$-functional approach [12]. Some results of applications of the $\tilde{\mathrm{C}}$-functional will be discussed in section 5.2.

2.4. Hierarchy of related variational functionals for general matrix elements

Variational principles for quantities other than the T-matrix are often needed. One such example arises in studies of molecular photoionization cross sections where electronic transition matrix elements are required. In general, variational principles can be written down for many quantities of physical interest [21]. In this section we will only be concerned with matrix elements of wave functions, Ψ, which satisfy the Lippmann–Schwinger equation

$$(1 - G_0 V) | \Psi \rangle = | S \rangle . \tag{2.50}$$

We then consider matrix elements involving Ψ of the form

$$M(R, S) = \langle R | \Psi \rangle . \tag{2.51}$$

Our task here is to construct a series of variational expressions for the matrix element $M(R, S)$ [40].

First M can be partially expanded in a Born series to give

$$M(R, S) = \sum_{k=0}^{n-1} \langle R | (G_0 V)^k | S \rangle + I^n(R, S) , \tag{2.52}$$

where $I^n(R, S)$ is the residual component of $M(R, S)$ defined by

$$I^n(R, S) = \langle R | (G_0 V)^n | \Psi \rangle . \tag{2.53}$$

Now if the terms in the sum in eq. (2.52) can be computed exactly, then a variational expression for M can be obtained from a variational expression for $I^n(R, S)$.

To construct a variational expression for $I^n(R, S)$ we first must consider the function χ_n which satisfies the equation [41, 42]

$$\langle \chi_n|(1 - G_0 V) = \langle R|(G_0 V)^n . \tag{2.54}$$

It follows that $I^n(R, S)$ is given by

$$I^n(R, S) = \langle R|(G_0 V)^n|\Psi\rangle = \langle \chi_n|(1 - G_0 V)|\Psi\rangle = \langle \chi_n|S\rangle . \tag{2.55}$$

Thus if we have trial functions χ_n^{t} and Ψ^{t}, a variational approximation $\tilde{I}^n(R, S)$ is given by

$$\tilde{I}^n(R, S) = \langle R|(G_0 V)^n|\Psi^{\mathrm{t}}\rangle + \langle \chi_n^{\mathrm{t}}|S\rangle - \langle \chi_n^{\mathrm{t}}|1 - G_0 V|\Psi^{\mathrm{t}}\rangle . \tag{2.56}$$

The approach we will take here is to construct the trial function as a linear combination of basis functions, and then obtain the expansion coefficients from the variational condition. The basis set used to expand the trial function Ψ^{t} is of the form

$$\mathrm{A}_{\mathrm{K}}^{j,n} = \{(G_0, V)_{j-2n-1}|\alpha_k\rangle : k = 1, 2, \ldots, \eta\} \tag{2.57}$$

and the basis set used for χ_n^{t} is

$$\mathrm{A}_{\mathrm{B}}^{i} = \{\langle \alpha_k|[G_0, V]_i : k = 1, 2, \ldots, \eta\} . \tag{2.58}$$

The basis set $\{|\alpha_k\rangle : k = 1, 2, \ldots, \eta\}$ is an arbitrary set of basis functions containing η elements used to construct both $\mathrm{A}_{\mathrm{K}}^{j,n}$ and $\mathrm{A}_{\mathrm{B}}^{i}$. The notation $(A, B)_i$ used in eq. (2.57) and $[A, B]_i$ used in eq. (2.58) is defined by

$$(A, B)_{2n} = (AB)^n \tag{2.59}$$

$$(A, B)_{2n+1} = (AB)^n A \tag{2.60}$$

and

$$[A, B]_n (A, B)_{-n} = 1 . \tag{2.61}$$

In eqs. (2.57) and (2.58) we have introduced two new indices i, j in addition to n which specifies the I^n of interest. To give explicit examples of the definitions in eqs. (2.59)–(2.61) consider $i = 3$, $j = 3$ and $n = 1$, then eqs. (2.57) and (2.58) become

$$\mathrm{A}_{\mathrm{K}}^{3,1} = \{|\alpha_k\rangle : k = 1, 2, \ldots, \eta\} \tag{2.62}$$

and

$$\mathrm{A}_{\mathrm{B}}^{3} = \{\langle \alpha_k| V G_0 V : k = 1, 2, \ldots, \eta\} . \tag{2.63}$$

Expanding the trial functions Ψ^{t} in basis set $A_K^{i,n}$ and χ_n^{t} in basis set A_B^{i} and requiring that $\tilde{I}^n(R, S)$ be stable with respect to first-order changes in the expansion coefficients yields for $\tilde{I}_{ij}^n$,

$$\tilde{I}_{ij}^n = \sum_{k,\,l=1}^{n} \langle R|(G_0V)_{j-1}|\alpha_k\rangle F_{kl}^{-1}\langle\alpha_l|[G_0, V]_i|S\rangle \tag{2.64}$$

where F_{kl}^{-1} are elements of $\mathbf{F}^{-1}$, the inverse of the matrix $\mathbf{F}$ which has elements

$$F_{kl} = \langle a_k|[G_0, V]_i(1 - G_0V)(G_0, V)_{j-2n-1}|\alpha_l\rangle\,. \tag{2.65}$$

Now combining the variational expression for I^n given in eq. (2.64) and the equation for $M(R, S)$ given in eq. (2.52) we obtain a variational expression for $M(R, S)$ given by [40]

$$\tilde{M}_{ij}^n(R, S) = \sum_{m=0}^{n-1} \langle R|(G_0V)^m|S\rangle + \sum_{k,\,l=1}^{\eta} \langle R|(G_0, V)_{j-1}|\alpha_k\rangle F_{kl}^{-1}\langle\alpha_l|[G_0, V]_i|S\rangle\,. \tag{2.66}$$

The variational expression for $\tilde{M}_{ij}^n(R, S)$ given in eq. (2.66) is very general and is equivalent to many other variational functionals which have been given in the literature [40]. In particular, $\tilde{M}_{11}^0(VS', S)$ is the Schwinger variational expression for the K-matrix. Also the four functions F_1, F_2, F_3 and F_c for the K-matrix discussed by Takatsuka and McKoy [38] are given by $\tilde{M}_{11}^1(VS', S)$, $\tilde{M}_{33}^1(VS', S)$, $\tilde{M}_{22}^1(VS', S)$ and $M_{33}^2(VS', S)$ respectively. The functionals $f_w^{(m)}$ given by Miraglia [43] are equivalent to the functionals $\tilde{M}_{2m-3,\,2m-3}^{m-2}(VS', S)$ when scattering without rearrangement is considered. The functionals $[f]_n$ of Gross and Runge [44] are given by $\tilde{M}_{2n+1,\,2n+1}^n(VS', S)$. Finally the hierarchy of functionals $I(\tilde{C}_m, \tilde{C}_n)$ discussed in section 2.3.1 is equivalent to the functionals $M_{2(m+n)-5,\,2(m+n)-5}^{m+n-3}(VS', S)$.

The functionals given in eq. (2.66) can also be used to compute matrix elements other than the scattering matrix. If the function $|R\rangle$ is replaced by the coordinate eigenfunction $|\boldsymbol{r}\rangle$ then the functionals $\tilde{M}_{ij}^n(\boldsymbol{r}, S)$ become functionals for the wave functions [45]. Additionally if $|R\rangle$ is taken to be $\boldsymbol{r}|\phi\rangle$, where ϕ is a bound state wave function then $\tilde{M}_{ij}^n(\boldsymbol{r}\phi, S)$ are variational expressions for dipole matrix elements. The functional $\tilde{M}_{13}^1(\boldsymbol{r}\phi, S)$ has been used to compute photoionization cross sections using the iterative Schwinger method [9], and the functional $M_{35}^3(\boldsymbol{r}\phi, S)$ has been used to compute photoionization cross sections in conjunction with the $\tilde{C}$-functional by Lee et al. [12]. Thus the functional $M_{ij}^n(R, S)$ given in eq. (2.66) provides a framework for discussing a great variety of variational functionals based upon the Lippmann–Schwinger equation for potential scattering.

2.5. Hartree–Fock continuum orbitals

2.5.1. Hartree–Fock equations

The Schwinger methods have been applied to a wide variety of systems at the Hartree–Fock level. These studies include both photoionization studies and electron–molecule scattering. A simple case for both processes is when the molecular system is initially in a closed shell state. We will restrict the present discussion to just these cases since the resulting Hartree–Fock equations are then of a particularly simple form and also since these are very commonly encountered situations. Scattering from more general, open shell, systems has been treated using the Schwinger method [46] but we will not consider it in detail here.

The Hartree–Fock (HF) wave function for a closed-shell molecule can be written as a single Slater determinant

$$\Psi = |\phi_1\alpha\phi_1\beta \cdots \phi_n\alpha\phi_n\beta| . \tag{2.67}$$

For an electron being scattered by this closed-shell target the HF wave function is

$$\Psi_k = |\phi_1\alpha\phi_1\beta \cdots \phi_n\alpha\phi_n\beta\phi_k\alpha| . \tag{2.68}$$

The HF equation for the continuum orbital ϕ_k is then determined by

$$\langle \delta\Psi_k|H - E|\Psi_k\rangle = 0 \tag{2.69}$$

where

$$\delta\Psi_k = |\phi_1\alpha\phi_1\beta \cdots \phi_n\alpha\phi_n\beta\, \delta\phi_k\alpha| \tag{2.70}$$

and where eq. (2.69) holds for all possible $\delta\phi_k$.

The electronic molecular Hamiltonian can be written as

$$H = \sum_{i=1}^{N} f(i) + \sum_{i<j}^{N} \frac{1}{r_{ij}} \tag{2.71}$$

with

$$f(i) = -\tfrac{1}{2}\nabla_i^2 - \sum_{\alpha} \frac{Z_\alpha}{r_{i\alpha}} \tag{2.72}$$

where Z_α are the nuclear charges, and N is the total number of electrons, in this case $N = 2n + 1$. The one-electron HF Hamiltonian can be written as

$$H^{\mathrm{HF}} = f + \sum_{i=1}^{N} (2J_i - K_i) \tag{2.73}$$

where f is the one-electron operator defined in eq. (2.72) and J_i and K_i are the usual Coulomb and exchange operators [47]. The HF orbitals then satisfy

$$H^{\mathrm{HF}}\phi_n = \varepsilon_n\phi_n \tag{2.74}$$

where ε_n is the orbital energy.

If we do not assume that the orbitals ϕ_k and $\delta\phi_k$ are orthogonal to each other or to the orthonormal set of occupied target orbitals, then eq. (2.69) can be expanded to yield

$$0 = \langle (P\,\delta\phi_k)|H^{\mathrm{HF}} - \varepsilon|P\phi_k\rangle \tag{2.75}$$

where

$$P = 1 - \sum_{i=1}^{n} |\phi_i\rangle\langle\phi_i| \tag{2.76}$$

and

$$\varepsilon = E - E^{\mathrm{HF}} \tag{2.77}$$

and where E^{HF} is the electronic energy of the HF wave function given in eq. (2.67). However, in scattering by closed-shell systems considered here, it is not necessary to include the projection operator P in eq. (2.75) since without the projection operator, eq. (2.75) reduces to the HF orbital equation

$$H^{\mathrm{HF}}\phi'_k = \varepsilon\phi'_k \tag{2.78}$$

which is identical to the HF equation given in eq. (2.74). Now the solutions of eq. (2.78) with $\varepsilon > 0$ (i.e., scattering solutions) will automatically be orthogonal to the solutions with $\varepsilon < 0$ (i.e., the HF orbitals) so that a solution to eq. (2.78), ϕ'_k, will also be a solution to the HF equation given in eq. (2.75). Thus for scattering from closed-shell systems in the Hartree–Fock approximation it is not necessary to include the orthogonality constraints, as is given in eq. (2.75). However, even for closed-shell systems, when approximate HF potentials are used it may be advantageous to use the orthogonality conditions. When single-center expansions have been used to approximate the HF potentials, it has been found that enforcing orthogonality between the continuum orbital and the occupied orbitals increases the accuracy of the scattering solutions [48].

The scattering equations for photoionization from a closed-shell molecule in the frozen-core Hartree–Fock (FCHF) approximation are similar. In the frozen-core approximation the core orbitals of the ion are not allowed to relax from their form in the neutral molecule [49]. Thus for photoionization the final-state wave function, where the ionized electron is removed from the orbital ϕ_n, is written as

$$\Psi_k = (\tfrac{1}{2})^{1/2}\{|\phi_1\alpha\phi_1\beta\cdots\phi_n\alpha\phi_k\beta| + |\phi_1\alpha\phi_1\beta\cdots\phi_k\alpha\phi_n\beta|\}, \tag{2.79}$$

where the orbitals ϕ_i are the HF orbitals of the neutral molecule which satisfy eq. (2.74). The equation for the continuum orbital ϕ_k is then obtained by solving eq. (2.69) with

$$\delta\Psi_k = (\tfrac{1}{2})^{1/2}\{|\phi_1\alpha\phi_1\beta\cdots\phi_n\alpha\,\delta\phi_k\beta| + |\phi_1\alpha\phi_1\beta\cdots\delta\phi_k\alpha\phi_n\beta|\} \tag{2.80}$$

for all possible $\delta\phi_k$. Expanding out eq. (2.69), gives for photoionization,

$$\begin{aligned}0 = {}&\langle(P\,\delta\phi_k)|H^{\mathrm{FC}} - \varepsilon|P\phi_k\rangle + 2\langle\delta\phi_k|\phi_n\rangle\langle\phi_n|H^{\mathrm{FC}} - K_n - \varepsilon|\phi_n\rangle\langle\phi_n|\phi_k\rangle \\ &+2\langle\delta\phi_k|\phi_n\rangle\langle\phi_n|H^{\mathrm{FC}} - K_n|P\phi_k\rangle + 2\langle(P\,\delta\phi_k)|H^{\mathrm{FC}} - K_n|\phi_n\rangle\langle\phi_n|\phi_k\rangle,\end{aligned} \tag{2.81}$$

where

$$H^{\mathrm{FC}} = f + \sum_{i=1}^{n-1}(2J_i - K_i) + J_n + K_n, \tag{2.82}$$

and

$$\varepsilon = E - (E^{\mathrm{HF}} - \varepsilon_n), \tag{2.83}$$

and with P as defined in eq. (2.76). Now since ϕ_n is a solution of eq. (2.74), eq. (2.81) can be simplified giving [46]

$$0 = \langle (P\,\delta\phi_k)|H^{\mathrm{FC}} - \varepsilon|P\phi_k\rangle + 2(\varepsilon_n - \varepsilon)\langle\delta\phi_k|\phi_n\rangle\langle\phi_n|\phi_k\rangle \tag{2.84}$$

which must hold for all $\delta\phi_k$. Thus if we consider $\delta\phi_k = \phi_n$ then eq. (2.84) requires that for $\varepsilon_n \neq \varepsilon$, $\langle\phi_n|\phi_k\rangle = 0$. So that if ϕ'_k satisfies

$$0 = \langle (P\,\delta\phi'_k)|H^{\mathrm{FC}} - \varepsilon|P\phi'_k\rangle \tag{2.85}$$

for all $\delta\phi'_k$, and $\varepsilon \neq \varepsilon_n$, then ϕ'_k will be a solution to eq. (2.84). Thus the solutions to eq. (2.85) with $\varepsilon \neq \varepsilon_n$ will also give us the solutions to the FCHF equation, eq. (2.84) [10].

Unlike the case of electron–neutral molecule scattering, it is necessary to include the orthogonality terms in eq. (2.85) in order to obtain the correct FCHF solution, unless the continuum orbitals are orthogonal to the HF orbitals by symmetry. Also the simple form of the scattering equations for photoionization given here are a result of using the FCHF approximation, and if we used any other set of target orbitals, such as those from a relaxed core, we would have to use the full expression given in eq. (2.81).

The most straightforward approach to putting HF equations of the form given in eqs. (2.75) and (2.85) into the form of a Lippmann–Schwinger equation as given in eq. (2.3) is to use a generalized Phillips–Kleinman pseudo-potential [50]. Thus for a one-electron Schrödinger equation of the form given in eqs. (2.75) and (2.85)

$$(1-Q)L(1-Q)\phi_k = 0 \tag{2.86}$$

where $Q = 1 - P$ and

$$L = -\tfrac{1}{2}\nabla^2 + V - \varepsilon, \tag{2.87}$$

the corresponding Lippmann–Schwinger equation can be written as

$$\phi_k = \phi_k^0 + G_0 V_Q \phi_k \tag{2.88}$$

with ϕ_k^0 being the free particle solutions, and G_0 the free particle Green's function. The pseudo-potential V_Q is defined as

$$V_Q = V - LQ - QL + QLQ, \tag{2.89}$$

and includes both the effects of the HF scattering potential and of requiring the continuum orbital to be orthogonal to the occupied target orbitals.

An alternative approach to solving an equation of the form given in eq. (2.86) is to use Lagrange undetermined multipliers [8, 51, 52], which then transforms eq. (2.86) into

$$L\phi_k = \sum_{i=1}^{n} \lambda_{i,k}\phi_i \tag{2.90}$$

where λ_i are undetermined multipliers, and ϕ_k is also subject to the conditions

$$\langle\phi_i|\phi_k\rangle = 0, \qquad i = 1, 2, \ldots, n. \tag{2.91}$$

Equation (2.90) can be transformed into an integral equation of the form

$$\phi_k = \phi_k^0 + \sum_{i=1}^{n} \lambda_{i,k} G_0\phi_i + G_0 V\phi_k. \tag{2.92}$$

The total wave function ϕ_k can then be constructed from the solution to

$$\psi_k = \phi_k^0 + G_0 V\psi_k \tag{2.93}$$

and

$$\psi_i = G_0\phi_i + G_0 V\psi_i \tag{2.94}$$

so that ϕ_k is given by

$$\phi_k = \psi_k + \sum_{i=1}^{n} \lambda_{i,k}\psi_i. \tag{2.95}$$

Then the undetermined multipliers can be determined using eq. (2.91) to give

$$\lambda_{i,k} = -\sum_j (A^{-1})_{ij}\langle\phi_j|\psi_k\rangle \tag{2.96}$$

where $(A^{-1})_{ij}$ are the matrix elements of the inverse of the matrix $\mathbf{A}$ with elements

$$A_{ij} = \langle\phi_i|\psi_j\rangle \tag{2.97}$$

so that ϕ_k is then given by

$$\phi_k = \psi_k + \sum_{i,j=1}^{n} \psi_i (A^{-1})_{ij}\langle\phi_j|\psi_k\rangle. \tag{2.98}$$

Thus in order to compute ϕ_k using the Lagrange undetermined multiplier form we must first solve the Lippmann–Schwinger equations given in eqs. (2.93) and (2.94), and then compute the appropriate matrix elements given in eqs. (2.97) and (2.98). It then follows that if a general matrix element of the scattering solution of the form

$$\langle R|\phi_k\rangle = \langle R|\psi_k\rangle + \sum_{i,j=1}^{n} \langle R|\psi_i\rangle (A^{-1})_{ij} \langle \phi_j|\psi_k\rangle \tag{2.99}$$

is required, the variational functionals for general matrix elements discussed in section 2.4 can be applied by using variational functionals for each of the matrix elements appearing in eq. (2.99) [18]. Explicitly, we can write down variational estimates for the matrix elements, $\langle R|\psi_k\rangle$, $\langle R|\psi_i\rangle$, $\langle\phi_i|\psi_j\rangle$ and $\langle\phi_i|\psi_k\rangle$.

2.5.2. General variational methods

As we have shown in the previous section any matrix element involving a continuum Hartree–Fock wave function can be reduced to matrix elements of the form $M(R, S) = \langle R|\psi\rangle$ where ψ satisfies the inhomogeneous equation

$$(1-K)|\psi\rangle = |S\rangle\,, \tag{2.100}$$

with $K = G_0V$. Any of the generalized variational expressions given in section 2.4 can be applied to such matrix elements. We can then study iterative approaches to improving variational expressions for $M(R, S)$ such as those given in eq. (2.66). Thus we need to develop a method for computing the error in the variational element of the form of $M(R', S) = \langle R'|\psi\rangle$ with $|R'\rangle = (VG_0)^n|R\rangle$ and with ψ satisfying an equation of the form of eq. (2.100). So we need only consider corrections to variational approximations of the general form $M(R, S)$ where ψ satisfies eq. (2.100).

A variational estimate for $M(R, S)$ can be obtained by considering the conjugate equation

$$\langle\chi|(1-K) = \langle R|\,, \tag{2.101}$$

then M is given by

$$M = \langle R|\psi\rangle = \langle\chi|S\rangle = \langle\chi|(1-K)|\psi\rangle\,. \tag{2.102}$$

Then a variational expression for M of the Schwinger type, given trial functions χ^{t} and ψ^{t}, is [41, 42]

$$M \approx M_0(R, S) = \langle R|\psi^{t}\rangle + \langle\chi^{t}|S\rangle - \langle\chi^{t}|(1-K)|\psi^{t}\rangle\,. \tag{2.103}$$

Note that eqs. (2.101)–(2.103) are analogous to eqs. (2.54)–(2.56) for the matrix element $I^n(R, S)$.

Now again as in the calculation of I^n, we expand the trial functions in basis sets

$$|\psi^{t}\rangle = \sum_{u_i\in A_K} a_i|u_i\rangle \tag{2.104}$$

$$\langle\chi^{t}| = \sum_{v_i\in A_B} b_i\langle v_i| \tag{2.105}$$

where A_K is a ket basis set

$$A_K = \{|\alpha_1\rangle, |\alpha_2\rangle, \ldots, |\alpha_\eta\rangle\} \tag{2.106}$$

and A_B is a bra basis set

$$A_B = \{\langle\beta_1|, \langle\beta_2|, \ldots, \langle\beta_\eta|\}. \tag{2.107}$$

Then requiring M_0 in eq. (2.103) to be variationally stable with respect to the expansion coefficients leads to

$$|\psi^t\rangle = \sum_{\substack{u_i \in A_K \\ v_j \in A_B}} |u_i\rangle D_{ij}^{-1} \langle v_j|S\rangle \tag{2.108}$$

and

$$\langle\chi^t| = \sum_{\substack{u_i \in A_K \\ v_j \in A_B}} \langle R|u_i\rangle D_{ij}^{-1} \langle v_i| \tag{2.109}$$

where D_{ij}^{-1} are elements of $\mathbf{D}^{-1}$, the inverse of the matrix $\mathbf{D}$ with elements

$$D_{ij} = \langle v_i|(1-K)|u_j\rangle. \tag{2.110}$$

The basis sets A_K and A_B are forced to have the same number of basis functions, η, since the matrix $\mathbf{D}$ must have both right and left inverses. The variational expression for M_0 given by eq. (2.103), analogous to eq. (2.64), is then

$$M_0(R, S) = \sum_{\substack{u_i \in A_K \\ v_j \in A_B}} \langle R|u_i\rangle D_{ij}^{-1} \langle v_j|S\rangle, \tag{2.111}$$

or defining the operator T as

$$T = \sum_{\substack{u_i \in A_K \\ v_j \in A_B}} |u_i\rangle D_{ij}^{-1} \langle v_j| \tag{2.112}$$

we again have for M_0

$$M_0 = \langle R|T|S\rangle. \tag{2.113}$$

Next we want to derive an expression for M_c, the error in M_0 defined by

$$M_c = M - M_0. \tag{2.114}$$

This is done by constructing zeroth-order solutions $|\psi_0\rangle$ and $\langle\chi_0|$ as [40]

$$|\psi_0\rangle = |S\rangle + K|\psi^{t}\rangle \tag{2.115}$$

and

$$\langle\chi_0| = \langle R| + \langle\chi^{t}|K\,. \tag{2.116}$$

Then it follows that the exact wave functions can be defined in terms of the zeroth-order solutions by [40]

$$[1-(K+KTK)(1-X)]|\psi\rangle = |\psi_0\rangle \tag{2.117}$$

and

$$\langle\chi|(1-X)[1-(K+KTK)(1-X)] = \langle\chi_0|(1-X)\,, \tag{2.118}$$

where X is the operator defined as

$$X = \sum_{\substack{u_i\in \mathrm{A_K}\\ v_j\in \mathrm{A_B}}} |u_i\rangle O_{ij}^{-1}\langle v_j| \tag{2.119}$$

where O_{ij}^{-1} are matrix elements of the matrix $\mathbf{O}^{-1}$ which is the inverse of the matrix $\mathbf{O}$ which has matrix elements

$$O_{ij} = \langle v_i|u_j\rangle\,. \tag{2.120}$$

It follows that the correction to the original variational estimate, M_0, is given by

$$M_c = \langle\chi_0|(1-X)|\psi\rangle = \langle\chi|(1-X)|\psi_0\rangle = \langle\chi|(1-X)[1-(K+KTK)(1-X)]|\psi\rangle\,. \tag{2.121}$$

Also note that using eqs. (2.117), (2.118) and (2.121) we can construct a Born series for the correction matrix element, M_c, as

$$M_c = \sum_{n=0}^{\infty} \langle\chi_0|(1-X)[(K+KTK)(1-X)]^n|\psi_0\rangle\,. \tag{2.122}$$

We will discuss two approaches which have been used to compute M_c [40]. The first approach is to use a sequence of variationally stable Padé approximants to M_c based on the Born series given in eq. (2.122) [40]. The second approach is the iterative Schwinger method in which the basis sets $\mathrm{A_B}$ and $\mathrm{A_K}$ are augmented with approximate solutions thus transforming eqs. (2.115) and (2.116) into implicit equations for $|\psi\rangle$ and $\langle\chi|$. These implicit equations can then be solved iteratively [40].

The Padé approximants to M_c can be formed by first considering a variational approximation to M_c

of the form

$$M_c \approx \langle \chi_0|(1-X)|\psi''\rangle + \langle \chi''|(1-X)|\psi_0\rangle - \langle \chi''|(1-X)[1-(K+KTK)(1-X)]|\psi''\rangle\,. \tag{2.123}$$

Then a sequence of $[N/N]$ Padé approximants is obtained when the trial function χ'' is constructed as a linear combination of the basis functions in the basis set $\mathrm{B}_{\mathrm{K}}^{N}$ defined as

$$\mathrm{B}_{\mathrm{K}}^{N} = \{[(K+KTK)(1-X)]^{i}|\psi_0\rangle\colon i=0,1,2,\ldots,N-1\} \tag{2.124}$$

and when χ'' is constructed from the basis functions in the set $\mathrm{B}_{\mathrm{B}}^{N}$ defined as

$$\mathrm{B}_{\mathrm{B}}^{N} = \{\langle \chi_0|[(1-X)(K+KTK)]^{i}\colon i=0,1,2,\ldots,N-1\}\,. \tag{2.125}$$

The $[N/N]$ variational Padé approximation to M_c, and thus to M, is then given by

$$M_N^{\mathrm{P}} = M_0 + \sum_{\substack{u_i\in \mathrm{B}_{\mathrm{K}}^{N}\\ v_j\in \mathrm{B}_{\mathrm{B}}^{N}}} \langle \chi_0|(1-X)|u_i\rangle E_{ij}^{-1}\langle v_j|(1-X)|\psi_0\rangle \tag{2.126}$$

where E_{ij}^{-1} are matrix elements of the matrix inverse of $\mathbf{E}$ with elements

$$E_{ij} = \langle v_i|(1-X)[1-(K+KTK)(1-X)]|u_i\rangle\,. \tag{2.127}$$

Although the general conditions are not known [42] under which a diagonal Padé sequence such as that given in eq. (2.126), will converge, experience has shown that paradiagonal Padé sequences are rapidly convergent for problems of physical interest [40, 53, 54]. It is useful to note that eq. (2.126) can be put into the form

$$M_N^{\mathrm{P}} = \sum_{\substack{u_i\in \mathrm{A_K}\cup \mathrm{B}_{\mathrm{K}}^{N}\\ v_i\in \mathrm{A_B}\cup \mathrm{B}_{\mathrm{B}}^{N}}} \langle R|u_i\rangle D_{ij}^{-1}\langle v_i|S\rangle\,. \tag{2.128}$$

Thus the basis sets $\mathrm{B}_{\mathrm{K}}^{N}$ and $\mathrm{B}_{\mathrm{B}}^{N}$ can be viewed as providing a systematic procedure for augmenting the initial basis sets $\mathrm{A_K}$ and $\mathrm{A_B}$, which should lead to a convergent sequence of variational approximations to M [40].

Another approach which has been used to obtain approximations to M_c is the iterative Schwinger method [5, 40]. At the Nth iteration of the iterative Schwinger method, the matrix element M is approximated by

$$M_N^{\mathrm{S}} = \sum_{\substack{u_i\in \mathrm{A_K}\cup \mathrm{C}_{\mathrm{K}}^{N-1}\\ v_i\in \mathrm{A_B}\cup \mathrm{C}_{\mathrm{B}}^{N-1}}} \langle R|u_i\rangle D_{ij}^{-1}\langle v_j|S\rangle \tag{2.129}$$

where the augmentation basis sets $\mathrm{C}_{\mathrm{K}}^{N}$ and $\mathrm{C}_{\mathrm{B}}^{N}$ are defined as

$$\mathrm{C}_{\mathrm{K}}^{N} = \{|\psi_N^{\mathrm{S}}\rangle\} \tag{2.130}$$

and

$$\mathrm{C}_{\mathrm{B}}^{N} = \{\langle \chi_N^{\mathrm{S}} |\} \tag{2.131}$$

where

$$|\psi_N^{\mathrm{S}}\rangle = |S\rangle + \sum_{\substack{u_i \in \mathrm{A_K} \cup \mathrm{C}_{\mathrm{K}}^{N-1} \\ v_j \in \mathrm{A_B} \cup \mathrm{C}_{\mathrm{B}}^{N-1}}} K|u_i\rangle D_{ij}^{-1} \langle v_j|S\rangle \tag{2.132}$$

and

$$\langle \chi_N^{\mathrm{S}}| = \langle R| + \sum_{\substack{u_i \in \mathrm{A_K} \cup \mathrm{C}_{\mathrm{K}}^{N-1} \\ v_j \in \mathrm{A_B} \cup \mathrm{C}_{\mathrm{B}}^{N-1}}} \langle R|u_i\rangle D_{ij}^{-1} \langle v_j|K, \tag{2.133}$$

and where $|\psi_0^{\mathrm{S}}\rangle$ and $\langle \chi_0^{\mathrm{S}}|$ are given by $|\psi_0\rangle$ and $\langle \chi_0|$ of eqs. (2.115) and (2.116).

Although the convergence of the iterative procedure given in eqs. (2.130)–(2.133) is not guaranteed, this procedure does converge in many cases. When the iterative procedure does converge and the relationships

$$\sum_{v_j \in \mathrm{A_B} \cup \mathrm{C}_{\mathrm{B}}^{N-1}} D_{ij}^{-1} \langle v_j|S\rangle = \begin{cases} 1, & \text{if } u_i \in \mathrm{C}_{\mathrm{K}}^{N-1} \\ 0, & \text{otherwise} \end{cases} \tag{2.134}$$

and

$$\sum_{u_i \in \mathrm{A_K} \cup \mathrm{C}_{\mathrm{K}}^{N-1}} \langle R|u_i\rangle D_{ij}^{-1} = \begin{cases} 1, & \text{if } v_i \in \mathrm{C}_{\mathrm{B}}^{N-1} \\ 0, & \text{otherwise} \end{cases} \tag{2.135}$$

are satisfied, then the converged functions ψ_N^{S} and χ_N^{S} are solutions of the original equations, eqs. (2.100) and (2.101), and M_N^{S} is exactly equal to M [5, 40]. The iterative procedure outlined here gives the same result as the Padé approximant for $N = 1$ (i.e., $M_1^{\mathrm{S}} = M_1^{\mathrm{P}}$). The results from these two methods will, however, in general be different for $N > 1$.

When a matrix of elements $M(R, S)$ are required (e.g., the partial-wave K-matrix) and the matrix is square so that

$$M_{ij} = M(R_i, S_j) \qquad \text{for } i, j = 1, 2, \ldots, p \tag{2.136}$$

where p is the dimension of the square matrix $\mathbf{M}$, then matrix analogs of the correction procedures discussed above can be developed. Thus for the Padé approximant correction method given in eq. (2.126) one would use the basis sets

$$\mathrm{B}_{\mathrm{K}}^{N} = \{[(K + KTK)(1 - X)]^j |\psi_{0,i}\rangle : j = 0, 1, 2, \ldots, N-1 \text{ and } i = 1, 2, \ldots, p\} \tag{2.137}$$

and

$$\mathrm{B}_{\mathrm{B}}^{N} = \{\langle \chi_{0,i}|[(1 - X)(K + KTK)]^j : j = 0, 1, 2, \ldots, N-1 \text{ and } i = 1, 2, \ldots, p\} \tag{2.138}$$

where

$$|\psi_{0,i}\rangle = (1 + KT)|S_i\rangle \tag{2.139}$$

and

$$\langle\chi_{0,i}| = \langle R_i|(1 + TK)\,. \tag{2.140}$$

With the basis sets $\mathrm{B}_{\mathrm{K}}^{N}$ and $\mathrm{B}_{\mathrm{B}}^{N}$ as defined in eqs. (2.137) and (2.138), eq. (2.126) then becomes the matrix-Padé approximant to the correction matrix M_c [40, 42].

Similarly, for the iterative Schwinger method, one can write the augmentation basis set as

$$\mathrm{C}_{\mathrm{K}}^{N} = \{|\psi_{N,i}^{\mathrm{S}}\rangle \colon i = 1, 2, \ldots, p\} \tag{2.141}$$

and

$$\mathrm{C}_{\mathrm{B}}^{N} = \{\langle\chi_{Ni}^{\mathrm{S}}| \colon i = 1, 2, \ldots, p\} \tag{2.142}$$

for use in eqs. (2.132) and (2.133). Then the convergence conditions become

$$\sum_{v_j \in \mathrm{A_B} \cup \mathrm{C_B}^{N-1}} D_{ij}^{-1}\langle v_j|S_k\rangle = \begin{cases} 1\,, & \text{if } |u_i\rangle = |\psi_{N,k}^{\mathrm{S}}\rangle \\ 0\,, & \text{otherwise} \end{cases} \tag{2.143}$$

and

$$\sum_{u_i \in \mathrm{A_K} \cup \mathrm{C_K}^{N-1}} \langle R_k|u_i\rangle D_{ij}^{-1} = \begin{cases} 1\,, & \text{if } \langle v_j| = \langle\chi_{N,k}^{\mathrm{S}}| \\ 0\,, & \text{otherwise.} \end{cases} \tag{2.144}$$

These matrix correction methods employ larger variational basis sets than do the individual matrix element methods, and hence should converge more rapidly. However, if the matrix $\mathbf{M}$ is not square, then the corrections to the matrix elements must be computed separately and the matrix approaches cannot be used.

2.5.3. *Modified methods*

The general solution methods described in section 2.5.2 can often be applied directly to the general variational principles of section 2.5.1 to obtain the quantity of physical interest [5, 40, 55–56]. However, there are special cases which we have encountered which require modified techniques. The first exceptional case is when different types of matrix elements of the same scattering wave function are required. The second exceptional case is when the general solution method is applied to the Lagrange multiplier formulation of the Hartree–Fock equations. In both these cases the straightforward application of the general solution methods is possible. However, for these cases, modified methods which are computationally simpler have actually been employed. Additionally, the form of the scattering potential sometimes dictates a different choice for ϕ_k^0 and G_0 than the free-particle functions used in the Lippmann–Schwinger equations given in eqs. (2.50), (2.54), (2.88) and (2.92).

In computing the photoionization cross section using standing partial-wave solutions, one must obtain both the scattering K-matrix and various dipole matrix elements involving the scattering solutions. In this case we need to know both the values of a square matrix (e.g., the K-matrix) as given in eq. (2.136) and also other matrix elements given by

$$M'_{ij} = M(R'_i, S_j) \qquad \text{for } i = 1, 2, \ldots, p' \text{ and } j = 1, 2, \ldots, p \tag{2.145}$$

with $p' \neq p$. The approach we have used is to iteratively solve the square matrix problem using eqs. (2.132), (2.133), (2.141) and (2.142) and then compute M'_{ij} using [9, 40]

$$(M'_N)_{ij} = \langle R'_i|\psi^S_{Nj}\rangle . \tag{2.146}$$

Although the matrix elements $(M'_N)_{ij}$ are different from those obtained if the iterative method is directly applied to $M(R'_i, S_j)$ the matrix elements given by eq. (2.146) are still variationally stable, and have been found to converge adequately in most cases considered to date. The variational stability of $(M'_N)_{ij}$ can be seen by considering the variational expression $M_0(KR'_i, S_j)$ given by eq. (2.111) using the basis sets $\mathrm{A_K} \cup \mathrm{C}^N_\mathrm{K}$ and $\mathrm{A_B} \cup \mathrm{C}^N_\mathrm{B}$. Then $(M'_N)_{ij}$ is just

$$(M'_N)_{ij} = \langle R'_i|S_j\rangle + M_0(KR'_i, S_j) \tag{2.147}$$

thus, $(M'_N)_{ij}$ is also variationally stable, i.e., does not contain any first-order errors due to errors in the basis sets.

The second modified solution technique has been used with the Lagrange multiplier form of the HF potential [8]. As was mentioned in section 2.5.1, with this form of the potential we need to compute four matrix elements:

$$I^{(1)}_{R,\boldsymbol{k}} = \langle R|\psi_{\boldsymbol{k}}\rangle , \qquad I^{(2)}_{R,i} = \langle R|\psi_i\rangle , \qquad I^{(3)}_{i,\boldsymbol{k}} = \langle \phi_i|\psi_{\boldsymbol{k}}\rangle \qquad \text{and} \qquad I^{(4)}_{ij} = \langle \phi_i|\psi_j\rangle .$$

We can approximate these matrix elements using $\tilde{I}^{(1)}_{R,\boldsymbol{k}} = M^1_{13}(R, \phi^0_{\boldsymbol{k}})$, $\tilde{I}^{(2)}_{R,i} = M^1_{13}(R, G_0\phi_i)$, $\tilde{I}^{(3)}_{i,\boldsymbol{k}} = M^1_{13}(\phi_i, \phi^0_{\boldsymbol{k}})$ and $\tilde{I}^{(4)}_{i,j} = M^1_{13}(\phi_i, G_0\phi_j)$, where M^N_{ij} is defined by eq. (2.64) and the same expansion basis sets are used in all four matrix elements. The result one obtains by substituting these variational expressions into eq. (2.99) is identical to that one would obtain by approximating V in eq. (2.92) by

$$V \approx \sum_{i,j=1}^{\eta} V|\alpha_i\rangle V^{-1}_{ij}\langle\alpha_j|V \tag{2.148}$$

where V^{-1}_{ij} is the inverse of the matrix with elements $\langle\alpha_i|V|\alpha_j\rangle$. Then the wave function $\phi'_{\boldsymbol{k}}$ which is the solution of eq. (2.92) with V approximated by eq. (2.148) is defined by

$$\phi'_{\boldsymbol{k}} = \phi^0_{\boldsymbol{k}} + \sum_{i=1}^{n} \lambda_{i,\boldsymbol{k}} G_0\phi_i + \sum_{i,j=1}^{\eta} G_0 V|\alpha_i\rangle V^{-1}_{ij}\langle\alpha_j|V|\phi'_{\boldsymbol{k}}\rangle . \tag{2.149}$$

The function $\phi'_{\boldsymbol{k}}$ can be obtained exactly since it is the solution of an integral equation which has a separable kernel. Then we have that [8]

$$\langle R|\phi_{\boldsymbol{k}}\rangle \approx \langle R|\phi'_{\boldsymbol{k}}\rangle = \tilde{I}^{(1)}_{R,\boldsymbol{k}} + \sum_{i,j=1}^{n} \tilde{I}^{(2)}_{R,i}[\tilde{I}^{(4)}]^{-1}_{ij}\tilde{I}^{(3)}_{j,\boldsymbol{k}}\,. \tag{2.150}$$

Thus for any basis set used in eq. (2.149), the matrix element $\langle R|\phi'_{\boldsymbol{k}}\rangle$ will be variationally stable. By a slight extension of the arguments of Ernst et al. [31], we see that if the basis set used in eq. (2.149) contained the exact solution $\phi_{\boldsymbol{k}}$, then we would find that $\phi'_{\boldsymbol{k}} = \phi_{\boldsymbol{k}}$. Thus we can utilize an iterative procedure of the same type as discussed in section 2.5.2, to solve for $\phi_{\boldsymbol{k}}$ and obtain accurate estimates of $\langle R|\phi_{\boldsymbol{k}}\rangle$. This approach proceeds by first constructing $\phi'^{(0)}_{\boldsymbol{k}}$ which is the solution of eq. (2.149) with the original basis set. Then we compute $\phi'^{(1)}_{\boldsymbol{k}}$, which is the solution of eq. (2.149) with the original basis set augmented by $\phi'^{(0)}_{\boldsymbol{k}}$. Further iterations can then be performed with $\phi'^{(n)}_{\boldsymbol{k}}$ being used to compute $\phi'^{(n+1)}_{\boldsymbol{k}}$. If this procedure converges with

$$\sum_{j=1}^{n} V^{-1}_{ij}\langle\alpha_j|V|\phi'^{(n+1)}_{\boldsymbol{k}}\rangle = \begin{cases} 1\,, & \text{if } |\alpha_i\rangle = |\phi'^{(n)}_{\boldsymbol{k}}\rangle \\ 0\,, & \text{otherwise} \end{cases} \tag{2.151}$$

then we have obtained the exact solution, i.e., $\phi'^{(n)}_{\boldsymbol{k}} = \phi_{\boldsymbol{k}}$. This iterative procedure has been found to yield accurate results for several types of scattering potentials [8].

In all the equations above we have used the free-particle Green's function and the total potential when writing down the Lippmann–Schwinger equation. This can easily be generalized by splitting the potential into two parts

$$V = V_1 + V_2 \tag{2.152}$$

then the Schrödinger equation for the scattering wave function (ignoring orthogonality constraints) is given by

$$(-\tfrac{1}{2}\nabla^2 + V_1 + V_2 - \varepsilon)\varepsilon_{\boldsymbol{k}} = 0\,. \tag{2.153}$$

Then the corresponding Lippmann–Schwinger equation is given by

$$\phi_{\boldsymbol{k}} = \phi^{(1)}_{\boldsymbol{k}} + G_1 V_2 \phi_{\boldsymbol{k}} \tag{2.154}$$

where

$$(-\tfrac{1}{2}\nabla^2 + V_1 - \varepsilon)\phi^{(1)}_{\boldsymbol{k}} = 0 \tag{2.155}$$

and

$$(-\tfrac{1}{2}\nabla^2 + V_1 - \varepsilon)\, G_1(\boldsymbol{r}, \boldsymbol{r}') = -\delta^3(\boldsymbol{r} - \boldsymbol{r}')\,. \tag{2.156}$$

The discussions we have given for the various solution techniques in section 2.5.2 remain the same with these split potentials, except that V is replaced by V_2, $\phi^0_{\boldsymbol{k}}$ is replaced by $\phi^{(1)}_{\boldsymbol{k}}$ and G_0 is replaced by G_1.

It is essential to employ this two-potential approach when the potential is Coulombic at large r [30], which is the case for electron–molecular ion scattering or molecular photoionization. In this case V_1 is taken to be $-\alpha/r$ and V_2 is then the residual potential, $V+\alpha/r$. The solutions $\phi_k^{(1)}$ are then Coulomb waves and G_1 is the Coulomb Green's function. We will defer a detailed discussion of how these functions are computed to section 3.1.3.

The two-potential approach has also been used with V_1 taken to be the local (or static) term in the HF potential [8]. Thus for the electron–neutral scattering, V_1 is given by

$$V_{\mathrm{d}} = f + \sum_{i=1}^{n} 2J_i \tag{2.157}$$

and the residual exchange potential V_2 is given by

$$V_{\mathrm{ex}} = -\sum_{i=1}^{n} K_i . \tag{2.158}$$

This separation of the potential is motivated by the long range nature of the scattering potentials found when the molecule has a dipole moment. The static portion of these potentials is very difficult to represent with only L^2-basis sets. The iterative procedures we have discussed above go beyond using only L^2-basis sets by including continuum functions in the expansions basis sets. However these methods still tend to converge slowly for strongly polar systems [54]. We found that putting only the short-range exchange potentials on a discrete basis, the convergence behavior of these iterative procedures is significantly enhanced [8]. The scattering solutions ϕ_k^{d} and the Green's function G_{d} can be obtained using the integral equations approach of Sams and Kouri [7, 8, 57, 58]. A detailed discussion of this technique will be given in section 3.1.2.

2.6. Multichannel extension of the Schwinger variational principle

2.6.1. Introduction

At the static-exchange level the Schwinger principle has proved to be a very effective approach to electron–molecule collisions. For relatively high impact energies or in molecular photoionization of outermost electronic levels the static-exchange approximation is expected to work quite well and, in fact, the results we will discuss in sections 4 and 5.1 will show this to be the case. On the other hand, at low collision energies the energetically closed channels are needed to describe the polarization of the target molecule as the incident electron approaches. Moreover, one must include the energetically open channels in any formulation of this collision problem. For example, in these low-energy collisions optically forbidden excitations become an important channel. To date progress in the theoretical studies of electron–molecule collisions beyond the static-exchange approximation has been quite limited. With a few exceptions [59–65] this is particularly true in the area of electronically inelastic collisions. This is undoubtedly due to the significant complexities arising from the inclusion of closed channels and additional open channels in electron–molecule collisions.

In this section we discuss a multichannel extension of the Schwinger variational principle which takes into account the role of both open and closed electronic channels [11]. We note that the inclusion of polarization effects and the extension to electronically inelastic scattering is relatively straightforward for the Kohn method.

2.6.2. Coupled equations and the Schwinger principle

A multichannel extension of the Schwinger principle can be obtained by simply applying this variational principle to the usual coupled equations. Such an extension has been considered by Maleki and Macek [66]. In addition to the complexity of these coupled equations, one faces the problem of how to deal with the Green's functions for the closed channels. For electron collisions, an expansion of the closed-channel Green's function would, moreover, have to include the electronic continuum states of the target.

One possible way of bypassing this difficulty is to use the Feshbach partitioning technique [67]. In this technique one introduces the projection operators P_0 and Q_0 for the open and closed channels respectively. These operators are defined in the $(N+1)$ particle space and hence act on all coordinates. As is well-known [67], the complete Schrödinger equation can then be converted to an equation for the open channel space only, i.e.,

$$[P_0\hat{H}P_0 + P_0\hat{H}Q_0(Q_0\hat{H}Q_0)^{-1}Q_0\hat{H}P_0]\Psi = 0 \tag{2.159}$$

where $\hat{H} = E - H$. This approach has generally been used in multichannel formulations of the Kohn-type variational principles [18]. Nesbet [18] has developed a multichannel Schwinger functional based on the coupled equations derived from eq. (2.159). This formulation is rather complicated. In general, however, the price one pays for using the Feshbach partitioning technique is (i) for inelastic collisions singularities can arise in the optical potential, i.e., $P_0\hat{H}Q_0(Q_0\hat{H}Q_0)^{-1}Q_0\hat{H}P_0$, at real energies. Such a singularity can be related to a true resonance of the system but it may also be spurious. Although such singularities can be removed, it would certainly be desirable to develop a multichannel extension of the Schwinger variational principle which is free of singularities on the real energy axis as is the case for potential scattering, and (ii) the definition of the projection operators P_0 and Q_0 are not unique [68–70]. In our formulation of the multichannel extension of the Schwinger functional we try to retain some advantages of the original Schwinger variational principle [16] and to avoid some of these difficulties.

2.6.3. A new functional [11, 71]

As usual, the Hamiltonian we consider can be written as

$$H = (H_N + T_{N+1}) + V \tag{2.160}$$

where H_N is the target Hamiltonian and T_{N+1} is the kinetic energy operator for the incident electron and V is the interaction potential between the incident electron and the target, i.e.,

$$V = \sum_{i=1}^{N} \frac{1}{r_{i,N+1}} - \sum_{\alpha} \frac{Z_\alpha}{r_{\alpha,N+1}}\,. \tag{2.161}$$

In eq. (2.161) the first and second terms are the electron repulsion and electron–nuclei attraction respectively.

We define a projection operator P which specifies the open target eigenstates of the collision. P is an N-body operator and, unlike P_0 of Feshbach formalism, is unique. With Φ_m as a target eigenstate P is simply

$$P = \sum_{m}^{\text{open}} |\Phi_m\rangle\langle\Phi_m| \,. \tag{2.162}$$

Note that this projection operator is different from the operator defined in eq. (2.76). With this operator we obtain a projected Lippmann–Schwinger equation for the mth scattering state

$$P\Psi_m^{(+)} = S_m + G_P^{(+)} V \Psi_m^{(+)} \tag{2.163}$$

where $\Psi_m^{(+)}$ has the asymptotic form of an incoming plane wave and a scattered component. S_m is hence a product of the target wave function Φ_m and an incident plane wave, i.e.,

$$S_m = \Phi_m(1, 2, \ldots, N) \exp(\mathrm{i}\boldsymbol{k}_m \cdot \boldsymbol{r}_{N+1}) \,. \tag{2.164}$$

The Green's function of eq. (2.163) is uniquely defined on the open channel components

$$G_P^{(+)} = \sum_{m}^{\text{open}} |\Phi_m\rangle g_m^{(+)}(\boldsymbol{r}_{N+1}, \boldsymbol{r}'_{N+1}) \langle\Phi_m| \,. \tag{2.165}$$

In this equation $g_m^{(+)}(\boldsymbol{r}_{N+1}, \boldsymbol{r}'_{N+1})$ denotes the outgoing-wave Green's function for T_{N+1} at energy $E - E_m$, where E and E_m are the total energy and eigenvalue of Φ_m respectively.

We next apply the potential V to both sides of eq. (2.163)

$$(VP - VG_P^{(+)} V)\, \Psi_m^{(+)} = VS_m \,. \tag{2.166}$$

Unlike the original Schwinger principle the operator on the left-hand side of this equation is not symmetric. This is simply because we have already projected the closed-channel components out of eq. (2.163). The closed-channel contributions can, however, be taken into account through a projected Schrödinger equation

$$[\hat{H} - a(P\hat{H} + \hat{H}P)]\, \Psi_m^{(+)} = a(VP - PV)\, \Psi_m^{(+)} \tag{2.167}$$

where, at this stage, the parameter a is arbitrary. We now note that combining eqs. (2.166) and (2.167) can lead to an integro-differential equation with a symmetric operator, i.e.,

$$[\tfrac{1}{2}(PV + VP) - VG_P^{(+)} V + \frac{1}{2a}\{\hat{H} - a(P\hat{H} + \hat{H}P)\}]\, \Psi_m^{(+)} = VS_m \,. \tag{2.168}$$

Closer examination of the operators in this equation shows that for Fermion collisions the parameter a must be chosen to be $(N+1)/2$ to make the operator $\hat{H} - a(P\hat{H} + \hat{H}P)$ on the left-hand side Hermitian. We note that in a general inhomogeneous equation

$$A^{(+)} x^{(+)} = b \,, \tag{2.169}$$

where $A^{(+)}$ is a scattering operator having outgoing-wave boundary conditions and $x^{(+)}$ is an unknown function, if the operator $A^{(+)}$ does not satisfy the condition $\langle x^{(-)}|A^{(+)}x^{(+)}\rangle = \langle A^{(-)}x^{(-)}|x^{(+)}\rangle$, the corresponding variational functional

$$[X] = \langle x^{(-)}|b\rangle \langle b|x^{(+)}\rangle / \langle x^{(-)}|A^{(+)}|x^{(+)}\rangle \tag{2.170}$$

is neither stable nor does it have unique solutions in general [71]. The choice of $a = (N+1)/2$ ensures that the above condition is satisfied.

Based on eq. (2.168) we can write a multichannel variational functional for the scattering amplitude

$$[f_{mn}] = -\frac{1}{2\pi} \cdot \frac{\langle \Psi_m^{(-)}|V|S_n\rangle \langle S_m|V|\Psi_n^{(+)}\rangle}{\langle \Psi_m^{(-)}|[\frac{1}{2}(PV+VP) - VG_P^{(+)}V + (N+1)^{-1}\{\hat{H} - \frac{1}{2}(N+1)(P\hat{H}+\hat{H}P)\}]|\Psi_n^{(+)}\rangle} \tag{2.171}$$

which is an extension of the usual Schwinger principle. The variational expression for the scattering amplitude can then be written as

$$f_{mn} = -\tfrac{1}{2}\pi \sum_{i,j} \langle S_m|V|\psi_i\rangle \Delta_{ij}^{-1} \langle \psi_j|V|S_n\rangle \tag{2.172}$$

where the ψ_i's are basis functions defined by $(N+1)$-body Slater determinants. The Δ is the matrix representation of the operator of the denominator of eq. (2.171) in this basis. This formalism is obviously based on the variation of the total wave function rather than that of a specific scattering orbital.

2.6.4. Static-exchange approximation to this functional

As an example of the use of this functional, we now use it to obtain the Schwinger variational principle for the static-exchange potential. The total wave function is taken as

$$\Psi_0^{(+)} = A[\Phi_0 \mu^{(+)}] \tag{2.173}$$

where Φ_0 is the Hartree–Fock wave function for the target state and $\mu^{(+)}$ denotes the scattering orbital. A is the antisymmetrizing operator. Integration over the coordinates of the first N particles gives

$$[f_{ij}] = -\frac{1}{2\pi} \cdot \frac{\langle \mu_i^{(-)}|V_{se}|s_i\rangle\langle s_i|V_{se}|\mu_j^{(+)}\rangle}{\langle \mu_i|V_{se} - V_{se}g_0^{(+)}V_{se}|\mu_j^{(+)}\rangle} \tag{2.174}$$

where V_{se} is the static-exchange potential of the target. The matrix elements in eq. (2.171) arising from the Hamiltonian in the denominator vanish identically. Equation (2.174) is just the Schwinger variational principle for elastic scattering by the static-exchange potential.

2.6.5. A two-potential formalism

In electron–molecule collisions and in the related problem of molecular photoionization, the full scattering potential generally has components of significantly different range. Based on this physical distinction, one can divide configuration space into different regions which can then be treated by the most suitable technique. This is obviously the underlying physical idea of R-matrix theory [72]. The long-range component of these potentials can arise from the direct or local part of the molecular

Hartree–Fock potential, e.g., the dipole and Coulomb potentials. For electron impact excitation of dipole-allowed the transition matrix elements have the range of a dipole potential. The exchange and polarization potentials are of much shorter range and it would hence seem very appropriate to treat these potentials and the transition dipole potential separately using a two-potential formalism. Such a two-potential approach has been used in Kohn-type variational procedures more recently, in studies of elastic electron–molecule collisions [73].

In this subsection we discuss a two-potential adaptation of the multichannel extension of the Schwinger variational principle described above. This formulation can be useful in studies of electronic excitation of dipole-allowed states.

We consider an unperturbed Hamiltonian containing some direct-potential, e.g.,

$$H_0 = H_N + T_{N+1} + PVP \tag{2.175}$$

for which the corresponding scattering problem can be readily solved by a numerical procedure. The interaction potential V is defined in eq. (2.161) and hence

$$PVP = \sum_{m,n} |\Phi_m\rangle v_{mn}(\boldsymbol{r})\langle\Phi_n| \tag{2.176}$$

with

$$v_{mn}(\boldsymbol{r}) = \langle\Phi_m|V|\Phi_n\rangle \tag{2.177}$$

which is a direct potential. Other choices can be made for V. For example, a local exchange and model polarization potential could be included in V. The scattering solutions of this H_0 which does not contain a nonlocal potential can be obtained by one of many techniques [6–8].

As in the multichannel extension of the Schwinger principle the projected wave function satisfies

$$P\Psi_m^{(+)} = \chi_m^{(+)} + G_{\mathrm{R}}^{(+)} V_{\mathrm{R}} \Psi_m^{(+)} \tag{2.178}$$

where $\chi_m^{(+)}$ is the scattering solution for H_0 with the outgoing-wave boundary condition. The residual interaction V_{R} is $V - PVP$. $G_{\mathrm{R}}^{(+)}$ is the Green's function associated with H_0, e.g.,

$$G_{\mathrm{R}}^{(+)} = \sum_{m,n} |\Phi_m\rangle\, g_{mn}^{\mathrm{R}(+)}(\boldsymbol{r}_{N+1}, \boldsymbol{r}'_{N+1})\,\langle\Phi_n| \tag{2.179}$$

where the one-particle Green's function $g_{mn}^{\mathrm{R}(+)}$ satisfies the equation

$$\sum_l [(E - E_m - T)\delta_{ml} - v_{ml}]\, g_{ln}^{\mathrm{R}(+)} = \delta_{mn}\,\delta(\boldsymbol{r} - \boldsymbol{r}') \tag{2.180}$$

where T is the kinetic energy operator and v_{mi} is defined in eq. (2.177). This Green's function $g_{mn}^{\mathrm{R}(+)}$ is related to the free-particle Green's function $g_{mn}^{0(+)}$ through the Dyson equation [30]

$$g_{mn}^{\mathrm{R}(+)}(\boldsymbol{r}, \boldsymbol{r}) = g_{mn}^{0(+)}(\boldsymbol{r}, \boldsymbol{r}') + \sum_{l,l'} \int \mathrm{d}\boldsymbol{r}''\, g_{ml}^{0(+)}(\boldsymbol{r}, \boldsymbol{r}'')\, v_{ll'}(\boldsymbol{r}'')\, g_{l'n}^{\mathrm{R}(+)}(\boldsymbol{r}'', \boldsymbol{r}') \tag{2.181}$$

with

$$g^0_{mn} = 0 \qquad \text{if } m \neq n \tag{2.182}$$

or

$$\mathbf{g}^{\mathrm{R}} = \mathbf{g}^0 + \mathbf{g}^0 \boldsymbol{\upsilon} \mathbf{g}^{\mathrm{R}} \tag{2.183}$$

written in an obvious matrix notation. We will return to this Green's function later.

In analogy to eq. (2.168) we can readily obtain an equation for the total wave function

$$[\tfrac{1}{2}(PV_{\mathrm{R}} + V_{\mathrm{R}}P) - V_{\mathrm{R}} G_{\mathrm{R}}^{(+)} V_{\mathrm{R}} + (N+1)^{-1}\{\hat{H} - \tfrac{1}{2}(N+1)(P\hat{H} + \hat{H}P)\}]\, \Psi_m^{(+)} = V_{\mathrm{R}} \chi_m^{(+)} . \tag{2.184}$$

We can now construct a variational functional for the scattering amplitude due to the short-range potential V_{R}, i.e.,

$$[f_{mn}]_{\mathrm{s}} = -\frac{1}{2\pi} \frac{\langle \Psi_m^{(-)} | V_{\mathrm{R}} | \chi_n^{(+)} \rangle \langle \chi_m^{(-)} | V_{\mathrm{R}} | \Psi_n^{(+)} \rangle}{\langle \Psi_m^{(-)} | [\tfrac{1}{2}(PV_{\mathrm{R}} + V_{\mathrm{R}}P) - V_{\mathrm{R}} G_{\mathrm{R}}^{(+)} V_{\mathrm{R}}] + (N+1)^{-1}\{\hat{H} - \tfrac{1}{2}(N+1)(P\hat{H} + \hat{H}P)\} | \Psi_n^{(+)} \rangle} . \tag{2.185}$$

Since the residual potential can always be assumed to be short-ranged, the trial wave function can be expanded in short-range basis functions.

The one-particle Green's function g^{R}_{mn} of eq. (2.179) is associated with a short-range interaction and hence, in applications, we only need a matrix representation of $g^{\mathrm{R}(+)}_{mn}$ in some suitable discrete basis such as Cartesian Gaussian functions. We can conveniently obtain such a representation of g^{R}_{mn} from Dyson's equation, i.e., eq. (2.183),

$$\mathbf{g}^{\mathrm{R}(+)} = (1 - \mathbf{g}^{0(+)} \boldsymbol{\upsilon})^{-1} \mathbf{g}^{0(+)} . \tag{2.186}$$

A procedure for variationally correcting the $g^{\mathrm{R}(+)}$ of eq. (2.186) has been discussed by Newton [30]. In an application to potential scattering Watson and McKoy [74] used this relation to obtain a matrix representation of the full Green's function. The resulting scattering amplitudes compared well with those given by the Schwinger principle. This approximation to the Dyson equation obviously neglects the coupling between the long- and short-range components of the potential υ.

Finally, the matrix element in the numerator of the functional, eq. (2.185) contains the scattering solutions $\chi_m^{(\pm)}$ of the zeroth-order Hamiltonian H_0 but these solutions are associated with the short-range potential V_{R} and hence need not be generated outside the range of this potential. These solutions $\chi_m^{(+)}$ may be generated numerically or in a basis. Moreover, in view of the short-range nature of this potential we may also use a separable representation of V_{R} in evaluating this matrix element, e.g., $\langle \Psi_m^{(-)} | V_{\mathrm{R}} | \chi_n^{(+)} \rangle$.

2.6.6. Feshbach resonances

We now look at the particular form of this multichannel extension of the Schwinger variational principle when a Feshbach resonance arises. For such a case the dominant configuration describing the resonant wave function should almost be orthogonal to the projection operator P of eq. (2.62). As an example consider the resonant state of the e–He system with the dominant configuration (1s2s2s). The main configuration in P is (1s1s) and hence they are orthogonal to each other. With such a resonant configuration ψ_r and assuming complete orthogonality of ψ_r to P we obtain for the resonant part of the T-matrix

$$[T_{mn}]_{\mathrm{Res}} \simeq \frac{(N+1)\langle\chi_m^{(-)}|V_{\mathrm{R}}|\psi_r\rangle\langle\psi_r|V_{\mathrm{R}}|\chi_m^{(+)}\rangle}{\langle\psi_r|(E-E_r)-(N+1)V_{\mathrm{R}}G_{\mathrm{R}}^{(+)}V_{\mathrm{R}}|\psi_r\rangle} \tag{2.187}$$

where E_r is the total energy of ψ_r. By inspection we obtain for the width, Γ, and energy shift, ΔE, of this resonance

$$\tfrac{1}{2}\Gamma = -(N+1)\,\mathrm{Im}\langle\psi_r|V_{\mathrm{R}}G_{\mathrm{R}}^{(+)}V_{\mathrm{R}}|\psi_r\rangle \tag{2.188a}$$

and

$$\Delta E = -(N+1)\,\mathrm{Re}\langle\psi_r|V_{\mathrm{R}}G_{\mathrm{R}}^{(+)}V_{\mathrm{R}}|\psi_r\rangle. \tag{2.188b}$$

Note that the factor $(N+1)$ comes from the normalization factors of the antisymmetrizers A_N and A_{N+1}. This T_{Res} can interfere with the background T-matrix of $\chi_m^{(+)}$. These results are approximations to the Feshbach formalism.

2.6.7. A numerical example

To assess the convergence characteristics of our multichannel Schwinger formulation we have studied the elastic scattering of electrons by the hydrogen atom below the inelastic threshold. In these studies we look at the convergence of the procedure with respect to both the number of closed channels, i.e., the number of exact- or pseudo-states representing polarization, and the size of the basis used in the expansion of the scattering orbital.

The scattering wave function is expanded in the following basis

$$A(\phi_{1s}\bar{s}\Theta_{1,3}) \qquad \text{and} \qquad A(\phi_{1s}\bar{c}\Theta_{1,3}) \tag{2.189a}$$

$$A(\phi_{1s}\phi_{1s}\Theta_1) \tag{2.189b}$$

$$A(\phi_{1s}f_i\Theta_{1,3})\ (i=1,2,\ldots,M) \tag{2.189c}$$

$$A(\phi_t f_j\Theta_{1,3})\ (j=1,2,\ldots,M_t) \tag{2.189d}$$

where ϕ_{1s} is the 1s eigenfunction of the H-atom and ϕ_t are the exact- or pseudo-states representing the closed channels. The $\bar{s}$ and $\bar{c}$ represent the long-range functions $(1-e^{-r})j_0$ and $(1-e^{-r})n_0$, respectively, and j_0 and n_0 are the Bessel and Neumann functions. Θ_1 and Θ_3 denote the singlet and triplet spin functions. The scattering orbital basis f_i are taken to be the Slater functions $r^i e^{-\alpha r} Y_{lm}$. Rescigno [75] has stressed the importance of the configuration of eq. (2.189b) in this open-shell system.

First we examine the convergence of the scattering orbital expansion for a given closed channel. In table 2.4 this convergence is shown for a 1s–2p calculation. As in the static-exchange case, the convergence for the scattering orbital associated with a closed channel is fast. Only three to five basis

Table 2.4
Convergence of the singlet (δ_s) and triplet (δ_t) phase shifts in a 1s–2p calculation[a] at $k^2 = 0.55$ for e–H atom scattering

N_{2p}[b]	δ_s	δ_t
0[c]	0.701	1.691
1	0.704	1.691
2	0.716	1.697
3	0.733	1.706
4	0.739	1.709
5	0.739	1.709
6	0.740	1.710
7	0.742	1.713

[a] The close-coupling result of a 1s–2p calculation of ref. [76] is $\delta_s = 0.734$. Schwartz's "exact" value of ref. [77] is $\delta_s = 0.908$.
[b] The number of basis functions for the closed channel. For the open channel orbital $M = 2$ (see eq. (2.189c)).
[c] The static-exchange value by the close-coupling method (ref. [78]) is $\delta_s = 0.700$.

functions are needed to obtain the converged results. Another important observation from these results is that our phase shift is better converged than that of the close-coupling method for the same excited state function (see the footnote of table 2.4). This can be so because we do not apply the Schwinger principle to the close-coupling equations.

Next we look at the phase shifts obtained using a 1s–2s–$\overline{2p}$ state expansion. The $\overline{2p}$ pseudo-state is the one proposed by Damburg and Karule [79] to describe the dipole polarizability of the H atom and its radial part is $r^2(1 + r/2)\exp(-r)$. Burke et al. [80] have carried out close-coupling calculations with this expansion and, in addition, with the 1s–2s–2p–$\overline{\overline{2p}}$ expansion. Table 2.5 shows our calculated phase shifts along with the close-coupling results of Burke [80]. Except for $k^2 = 0.64$, where our results are apparently not sufficiently converged, our calculated phase shifts are more accurate than those of the corresponding 1s–2s–$\overline{2p}$ close-coupling calculation. Moreover, our results, except again at $k^2 = 0.64$, are

Table 2.5
Singlet phase shifts for e–H atom collisions

Methods	$k^2 =$ 0.01	0.09	0.25	0.49	0.64
Static-exchange (CC[a,b])	2.396	1.508	1.031	0.744	0.651
1s–2s–$\overline{2p}$(CC)	2.529	1.657	1.155	0.875	0.823
1s–2s–2p–$\overline{\overline{2p}}$	2.532	1.663	1.162	0.881	0.832
1s–2s–$\overline{2p}$ (this work)	2.550	1.684	1.179	0.895	0.818[c]
"exact"[d]	2.556	1.696	1.201	0.930	0.887

[a] CC denotes the close-coupling calculation.
[b] Burke, Gallaher and Geltman [80].
[c] $\delta_s = 0.821$ if another p-type basis function is added.
[d] Shimamura [81]; Schwartz [77]; Abdel-Raouf and Belschner [82].

even better than those of the 1s–2s–2p–$\overline{\overline{2p}}$ close-coupling calculation which was devised to improve the 1s–2s–$\overline{2p}$ study. These results suggest that our variational functional has better convergence characteristics than the close-coupling procedure with respect to the number of exact- or pseudo-states.

3. Numerical techniques

3.1. Single-center expansion method

Single-center expansion techniques have been used on a wide variety of small molecular systems [83]. Such systems have ranged from the simplest molecular system, H_2^+ [84], to molecules as complex as CO_2 [56] and CH_4 [85]. Accurate results can be obtained using single-center expansions, provided the dependence of the physical quantity of interest on the expansion parameters has been carefully considered. In this respect, special attention must be given to the case of resonances since resonances are generally very sensitive to the parameters used in the expansion of the scattering potential [10, 56].

A single-center expansion of a function of Cartesian coordinates $f(\boldsymbol{r})$ or equivalently of a function of the spherical coordinates $f(r, \theta, \phi)$ is obtained by using the spherical harmonics $Y_{lm}(\theta, \phi)$, a complete set of angular functions [86]. The function $f(r, \theta, \phi)$ is then written as

$$f(r, \theta, \phi) = \sum_{lm} \frac{1}{r} f_{lm}(r)\, Y_{lm}(\Omega_{\hat{r}}) \tag{3.1}$$

and

$$f_{lm}(r) = r \int \mathrm{d}\Omega_{\hat{r}}\, Y^*_{lm}(\Omega_{\hat{r}}) f(r, \theta, \phi) \tag{3.2}$$

where $\mathrm{d}\Omega_{\hat{r}} = \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi$, and $\Omega_{\hat{r}}$ stands for (θ, ϕ). An overlap integral between two functions $f(\boldsymbol{r})$ and $g(\boldsymbol{r})$ is then given as a sum of radial integrals

$$\int \mathrm{d}^3\boldsymbol{r}\, f^*(\boldsymbol{r})\, g(\boldsymbol{r}) = \sum_{lm} \int_0^\infty \mathrm{d}r\, f^*_{lm}(r)\, g_{lm}(r)\,. \tag{3.3}$$

This expansion technique can also be applied to kernels of integral operators such as G_0 and V as well as to simple functions such as $\phi_{\boldsymbol{k}}(\boldsymbol{r})$ and $\alpha(\boldsymbol{r})$.

3.1.1. Single-center expansions for Hartree–Fock potentials

The wave function $\phi_{\boldsymbol{k}}(\boldsymbol{r})$ can be expanded in terms of $\Omega_{\hat{k}}$, the direction of $\boldsymbol{k}$, as

$$\phi_{\boldsymbol{k}}(\boldsymbol{r}) = \left(\frac{2}{\pi}\right)^{1/2} \sum_{lm} \frac{\mathrm{i}^l}{k} \phi_{klm}(\boldsymbol{r})\, Y^*_{lm}(\Omega_{\hat{k}})\,, \tag{3.4}$$

where $\phi_{klm}(\boldsymbol{r})$ are the partial-wave scattering functions. Then the HF Lippmann–Schwinger equation for the wave function,

$$\phi_{\boldsymbol{k}} = \phi^0_{\boldsymbol{k}} + G_0 V \phi_{\boldsymbol{k}}, \tag{3.5}$$

can be rewritten as a partial-wave Lippmann–Schwinger equation

$$\phi_{klm} = \phi^0_{klm} + G_0 V \phi_{klm}. \tag{3.6}$$

The single-center expansion can be used to evaluate the various matrix elements needed to compute variational expressions such as that given in eq. (2.66). The functions we then need to expand are the basis functions $\alpha_i(\boldsymbol{r})$, the static-exchange potential V, the inhomogeneities $S(r)$ and $R(r)$, and the Green's function G_0. Once all of these expansions have been computed, any matrix element can be obtained as a sum of various radial integrals.

We have employed three different types of functions to describe target wave functions and as initial scattering basis functions. These three types are the Slater function defined as

$$\chi_{\mathrm{S}}^{\alpha,\,l,\,m,\,n,\,\boldsymbol{A}}(\boldsymbol{r}) = N^{\mathrm{S}} |\boldsymbol{r} - \boldsymbol{A}|^{n-1} \exp(-\alpha|\boldsymbol{r} - \boldsymbol{A}|)\, Y_{lm}(\Omega_{\boldsymbol{r}-\boldsymbol{A}}), \tag{3.7}$$

the Cartesian Gaussian function defined as

$$\chi_{\mathrm{CG}}^{\alpha,\,l,\,m,\,n,\,\boldsymbol{A}}(\boldsymbol{r}) = N^{\mathrm{CG}} (x - A_x)^l (y - A_y)^m (z - A_z)^n \exp(-\alpha|\boldsymbol{r} - \boldsymbol{A}|^2) \tag{3.8}$$

and the spherical Gaussian function

$$\chi_{\mathrm{SG}}^{\alpha,\,l,\,m,\,\boldsymbol{A}}(\boldsymbol{r}) = N^{\mathrm{SG}} |\boldsymbol{r} - \boldsymbol{A}|^l \exp(-\alpha|\boldsymbol{r} - \boldsymbol{A}|^2)\, Y_{lm}(\Omega_{\boldsymbol{r}-\boldsymbol{A}}). \tag{3.9}$$

A detailed discussion of single-center expansions for Slater functions has been given by Harris and Michels [87] and single-center expansions for Cartesian Gaussian functions have been given by Fliflet and McKoy [88]. The main utility of the spherical Gaussian function is the ease with which expressions for functions with high l can be computed. The single-center expansion of eq. (3.9) can be obtained by noting that a spherical Gaussian is the product of a simple s-type Cartesian Gaussian function and a solution to Laplace's equation, both of which have simple expressions for their expansion about another center [88, 89]. The single-center expansion of a spherical Gaussian function as defined in eq. (3.9) is then given by

$$\chi_{\mathrm{SG}}^{\alpha,\,l,\,m,\,\boldsymbol{A}}(\boldsymbol{r}) = N^{\mathrm{SG}} \sum_{l'=0}^{l} \sum_{l'''=0}^{\infty} \sum_{J'=|l-l'-l'''|}^{l+l'+l'''} \sum_{M'=-J'}^{J'} \sum_{J''=|l'-l'''|}^{l'+l'''} \sum_{M''=-J''}^{J''} C(l, m, l', l''', J', M', J'', M'')$$

$$\times A^l (r/A)^{l'} \exp[-\alpha(r^2 + A^2)]\, i_{l'''}(2\alpha A r)\, Y^*_{J'M'}(\Omega_{\boldsymbol{A}})\, Y_{J''M''}(\Omega_r), \tag{3.10}$$

where C is defined as

$$C(l, m, l', l''', J', M', J'', M'') = \sum_{m', m''=-l'}^{l'} \sum_{J=|l-l'|}^{l+l'} \sum_{M=-J}^{J} \sum_{m'''=-l'''}^{l'''} \frac{(-1)^{l+m-(l'+m')}(2l'''+1)}{(l-l')!}$$

$$\times \left[\frac{4\pi(2l+1)(l-m')!(l+m')!}{(2J'+1)(2J''+1)(l'-m')!(l'+m')!}\right]^{1/2} \langle l'lm''-m|JM\rangle \langle l'lm'-m'|J0\rangle$$

$$\times \langle Jl'''00|J'0\rangle \langle Jl''Mm'''|J'M'\rangle \langle l'l'''00|J''0\rangle \langle l'l'''m''m'''|J''M''\rangle , \tag{3.11}$$

and where $i_{l'''}(2\alpha Ar)$ is a modified spherical Bessel function and $\langle ll'mm'|l''m''\rangle$ is a Clebsch–Gordon coefficient.

The single-center expansions for the static and exchange potentials can be easily obtained using the single-center expansion of $1/r_{12}$ which is

$$\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} = \sum_{lm} \frac{4\pi}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}^*(\Omega_{r_1})\, Y_{lm}(\Omega_{r_2}) \tag{3.12}$$

where $r_<(r_>)$ is the lesser (greater) of $|\boldsymbol{r}_1|$ and $|\boldsymbol{r}_2|$. The nuclear attraction term $Z_\alpha/r_{i\alpha}$ is then given by

$$\frac{Z_\alpha}{|\boldsymbol{r}_i - \boldsymbol{A}_\alpha|} = \sum_{lm} \frac{4\pi}{2l+1} \left(\frac{r_<^l}{r_>^{l+1}}\right)_A Y_{lm}^*(\Omega_{\boldsymbol{A}_\alpha})\, Y_{lm}(\Omega_{r_i}) \tag{3.13}$$

where

$$\left(\frac{r_<^l}{r_>^{l+1}}\right)_A = \begin{cases} A^l/r^{l+1}, & A<r \\ r^l/A^{l+1}, & A>r. \end{cases} \tag{3.14}$$

Then the Coulomb potential defined as

$$J_i(\boldsymbol{r}) = \int \mathrm{d}^3\boldsymbol{r}' \, \frac{\phi_i(\boldsymbol{r}')\,\phi_i(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} \tag{3.15}$$

can be obtained by taking the product of ϕ_i^*, ϕ_i and $1/r_{12}$ and integrating the appropriate radial integrals. The products can be formed by observing that the product of any two functions $f(\boldsymbol{r})$ and $g(\boldsymbol{r})$ can be expressed in terms of their single-center expansions defined in eq. (3.1) above as

$$f(\boldsymbol{r})\,g(\boldsymbol{r}) = \sum_{lm} \frac{1}{r} \left\{ \sum_{l'm'} \sum_{l''m''} \left[\frac{(2l''+1)(2l'+1)}{4\pi(2l+1)}\right]^{1/2} \langle l''l'00|l0\rangle \langle l''l'm''m'|lm\rangle \frac{1}{r} f_{l'm'}(r)\, g_{l''m''}(r) \right\} Y_{lm}(\Omega_r) . \tag{3.16}$$

The single-center expansions of the molecular orbital ϕ_i can be obtained from the single-center expansions of the basis functions discussed above since the molecular orbitals are usually represented as a linear combination of such orbitals.

The exchange operator K_i, defined as

$$(K_i\phi_k)(\boldsymbol{r}) = \phi_i(\boldsymbol{r}) \int \mathrm{d}^3r' \frac{\phi_i^*(\boldsymbol{r}')\, \phi_k(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} \tag{3.17}$$

can also be obtained using the product formula given in eq. (3.16) and by performing the appropriate radial integrals. More explicit expressions for the J and K operators for linear molecular systems have been given elsewhere [46, 88].

Finally, both the free-particle and Coulomb Green's functions can be readily expanded about a single center. The expansion of the kernel of the principal-value free-particle Green's function, G_0, is given by

$$G_0(r, r) = \frac{-\cos(k|\boldsymbol{r} - \boldsymbol{r}'|)}{2\pi|\boldsymbol{r} - \boldsymbol{r}'|}$$

$$= -\frac{2}{k} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \hat{j}_l(kr_<)\, \hat{n}_l(kr_>)\, Y_{lm}(\Omega_r)\, r^{-1}\, (r')^{-1}\, Y^*_{lm}(\Omega_{r'}) \tag{3.18}$$

and the associated free-particle partial wave function is

$$\phi^0_{klm}(\boldsymbol{r}) = \frac{1}{r} \hat{j}_l(kr)\, Y_{lm}(\Omega_r) \tag{3.19}$$

where $\hat{j}_l(kr)$ is a Riccati–Bessel function and $\hat{n}_l(kr)$ is a Riccati–Neumann function. Likewise the principal-value Coulomb Green's function can be expanded about a single center using [90]

$$G_c(\boldsymbol{r}, \boldsymbol{r}') = -\frac{2}{k} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} F_l(\gamma; kr_<)\, G_l(\gamma; kr_>)\, r^{-1}\, (r')^{-1}\, Y_{lm}(\Omega_r)\, Y^*_{lm}(\Omega_{r'}) \tag{3.20}$$

and the corresponding pure Coulomb scattering partial wave functions are

$$\phi^c_{klm}(\boldsymbol{r}) = \frac{1}{r} F_l(\gamma; kr)\, Y_{lm}(\Omega_r)\,, \tag{3.21}$$

where $\gamma = -\alpha/k$ with α being the charge of the ion, and $F_l(\gamma; kr)$ and $G_l(\gamma; kr)$ are regular and irregular Coulomb functions [91].

The radial integrals are performed numerically by evaluating the functions on a grid of points, and using an appropriate quadrature. In our studies we have used the extended Simpsons rule for simple overlap integrals. Thus for overlap integrals of the form given in eq. (3.13) we approximate the integral over the interval (r_1, r_2) by

$$I(r_1, r_2) = \int_{r_1}^{r_2} \mathrm{d}r\, f^*_{lm}(r)\, g_{lm}(r) \approx \frac{h}{3} [(fg)_0 + 4\{(fg)_1 + (fg)_3 + \cdots + (fg)_{2n-1}\}$$

$$+2\{(fg)_2 + (fg)_4 + \cdots + (fg)_{2n-2}\} + (fg)_{2n}] \tag{3.22}$$

where the step size is $h = (r_2 - r_1)/2n$ and

$$(fg)_p = f^*_{lm}(r_1 + hp)\, g_{lm}(r_1 + hp)\,. \tag{3.23}$$

In general, the value of the integral is then obtained on the interval $(0, \infty)$ as approximately equal to the integral on the interval $(0, r_{\max})$. The integrals over the interval $(0, \infty)$ can be approximated in this fashion when the integrands asymptotically approach zero at least as fast as $1/r^2$. The interval $(0, r_{\max})$ is then further subdivided into intervals in which different step sizes can be used. In general, for molecular problems we use a step size of 0.01 a.u. in the region near the nuclei, and at larger r we usually have at least 20 steps per half wave, or

$$h < \pi/20k$$

thus for k up to 1 a.u. we have $h = 0.16$ a.u. for the step size at large r. In between these two regions, there are intervals with intermediate step sizes. In sections 4 and 5 we will give the grids and cut-offs needed in particular applications.

The simple integration formula given in eq. (3.22) cannot be applied directly to radial integrals involving Green's functions, such as the free-particle Green's function, the Coulomb Green's function, or $1/r_{12}$, the Green's function for the ∇^2 operator. Such integrals are two-dimensional integrals of the form

$$I_{\alpha\beta} = \int_0^\infty \mathrm{d}r \int_0^\infty \mathrm{d}r'\, \alpha(r)\, f(r_<)\, g(r_>)\, \beta(r')\,. \tag{3.24}$$

In general, such an integral is computed using a quadrature formula of the form

$$I_{\alpha\beta} \approx \sum_{i=1}^{n} \sum_{j=1}^{n} W^{(2)}_{ij}\, \alpha(r_i)\, f(r_{\min(i,j)})\, g(r_{\max(i,j)})\, \beta(r_j)\,. \tag{3.25}$$

The most straightforward application of a one-dimensional integral formula to this two-dimensional integral, would be to take the weights $W^{(2)}_{ij}$ to be the products of two one-dimensional weights, $W^{(1)}_i$ and $W^{(1)}_j$, and the quadrature points would then be the corresponding points of the one-dimensional integration formula. However, when the integrand has a discontinuity in the slope, as in eq. (3.24), the resulting error in eq. (3.25) will be proportional to a lower power of h, the step size, than if the integrand had had a continuous slope.

To treat the discontinuity in the slope at $r = r'$ of the integrand of eq. (3.24) we use a special integration formula for regions which contain the discontinuity. Since we want the integration formula to be compatible with the use of Simpson's formula, we have used equally spaced integration points. Thus we want to obtain the integral of an integrand $\gamma(r, r')$ over a region of space bounded by the matrix of points

$$\begin{matrix} (-1, 1) & (0, 1) & (1, 1) \\ (-1, 0) & (0, 0) & (1, 0) \\ (-1, -1) & (0, -1) & (1, -1) \end{matrix} \tag{3.26}$$

where the spacing between the points is h, the step size, and where γ has a discontinuity in its slope along $r=r'$. To obtain the appropriate formula, we divide the region into two triangles in which the integrand does have continuous derivatives, then one of these regions will contain the points

$$\begin{array}{ccc} & & (1,1) \\ & (0,0) & (1,0) \\ (-1,-1) & (0,-1) & (1,-1)\,. \end{array} \tag{3.27}$$

We obtain a quadrature formula by approximating γ in this region by

$$\gamma(r,r') \approx a + br + cr' + dr^2 + e(r')^2 + frr'\,. \tag{3.28}$$

Using the six values of the function γ at the points given in eq. (3.27), the six constants, a–f in eq. (3.28) can be obtained. Using eq. (3.28) the integral of $\gamma(r,r')$ over the region defined by eq. (3.27) can be computed, and when added to the integral for the other triangle, we find for the total integral

$$I^{r=r'} = h^2\{\tfrac{4}{3}\gamma(r_0,r_0) + \tfrac{2}{3}(\gamma(r_0,r_1) + \gamma(r_1,r_0) + \gamma(r_{-1},r_0) + \gamma(r_0,r_{-1})\} + \mathrm{O}(h^5)\,. \tag{3.29}$$

For regions which do not contain $r=r'$ we use the product of two Simpson's rule quadratures to give

$$\begin{aligned} I^{r\neq r'} = h^2\{&\tfrac{16}{9}\gamma(r_0,r_0) + \tfrac{4}{9}[\gamma(r_0,r_1) + \gamma(r_1,r_0) + \gamma(r_{-1},r_0) + \gamma(r_0,r_{-1})] + \tfrac{1}{9}[\gamma(r_1,r_1) \\ &+ \gamma(r_1,r_{-1}) + (r_{-1},r_{-1}) + \gamma(r_{-1},r_1)]\} + \mathrm{O}(h^6)\,. \end{aligned} \tag{3.30}$$

These formulas are then repeatedly used to compute the integral given in eq. (3.24) so that there are on the order of n applications of eq. (3.29) and n^2 applications of eq. (3.30). The total error in the integral of eq. (3.24) is then given by the errors in both types of integration formula, and will thus be on the order of $\mathrm{O}(h^4)$.

All of the single-center expansions which we have discussed here are naturally truncated at some suitable maximum value for the problem under consideration. These maximum expansion l's and the grid step size h and maximum $r_{\max}$ then constitute all of the parameters used in the single-center expansion technique. These parameters can be enumerated as follows:

(1) l_m, maximum l included in the expansion of scattering functions, variational basis functions, scattering Green's functions, and orbitals used in constructing orthogonality constraints,

(2) l_s^x, maximum l included in the scattering functions in exchange terms,

(3) l_i^x, maximum l included in the expansion of the occupied orbitals in the exchange terms,

(4) l_i^d, maximum l included in the expansion of the occupied orbitals in the direct potential,

(5) λ_m^x, maximum l included in the expansion of $1/r_{12}$ in the exchange terms,

(6) λ_m^d, maximum l included in the expansion of $1/r_{12}$ in the direct electronic potential,

(7) λ_m^n, maximum l included in the expansion of $1/r_{12}$ in the electron–nuclei electrostatic interactions,

(8) l_p, maximum l include in the expansion of $\phi_k(\boldsymbol{r})$ as given in eq. (3.4),

(9) r_m, maximum grid point,

(10) h_i, step sizes used in the grid.

To insure the accuracy of a single-center expansion calculation, the convergence of the calculation with respect to each of these parameters should be checked. The rate of convergence will depend on the system under consideration. One empirical result which has been observed is that the energy of the peak in the photoionization cross section $E^{(l_m)}$ of a linear molecule due to a scattering shape resonance of σ symmetry approaches its asymptotic value as

$$E^{(l_m)} - E^{(\infty)} \propto 1/l_m^3 . \tag{3.31}$$

This relationship has been verified for resonances in the photoionization of N_2 [10] and CO_2 [92] and is believed to be governed by the rate of convergence of the electron–nuclear electrostatic interaction potential [10].

3.1.2. Two-potential technique applied to the static potential

To apply the two-potential approach described in section 2.5.3, to the case where V_1 is given by the direct potential V_d as defined in eq. (2.157), we must be able to compute the solutions, $\phi_{\boldsymbol{k}}^{d}$, to the direct potential scattering which satisfy

$$(-\tfrac{1}{2}\nabla^2 + V_d - \varepsilon)\phi_{\boldsymbol{k}}^{d} = 0 \tag{3.32}$$

and matrix elements of the direct Green's function

$$\phi_\alpha^d(\boldsymbol{r}) = \langle \boldsymbol{r}|G_d|\phi_\alpha\rangle \tag{3.33}$$

where G_d satisfies

$$(-\tfrac{1}{2}\nabla^2 + V_d - \varepsilon)\, G_d(\boldsymbol{r}, \boldsymbol{r}') = -\delta^3(\boldsymbol{r} - \boldsymbol{r}') . \tag{3.34}$$

The function ϕ_α^d then satisfies an inhomogeneous differential equation

$$(-\tfrac{1}{2}\nabla^2 + V_d - \varepsilon)\, \phi_\alpha^d(\boldsymbol{r}) = -\phi_\alpha(\boldsymbol{r}) . \tag{3.35}$$

Now eq. (3.32) can be rewritten as a partial-wave Lippmann–Schwinger equation to give

$$\phi_{klm}^{d} = \phi_{klm}^{0} + G_0 V_d \phi_{klm}^{d} \tag{3.36}$$

and eq. (3.35) can be rewritten in a similar fashion to obtain the expression for a particular solution

$$\phi_\alpha^d = G_0\phi_\alpha + G_0 V_d \phi_\alpha^d . \tag{3.37}$$

Expanding eq. (3.36) using single-center expansions and the fact that V_d is a local potential we obtain

$$\phi_{ll'}^{d}(r) = \hat{j}_l(kr)\, \delta_{ll'} - \frac{2}{k} \int_0^\infty \mathrm{d}r' \sum_{l''} \hat{n}_l(kr_>)\, \hat{j}_l(kr_<)\, V_{ll''}^{d}(r')\, \phi_{l''l'}^{d}(r') \tag{3.38}$$

where we have suppressed the subscripts k and m and the sums over m for clarity. Expansion of eq.

(3.37) yields

$$\phi^{\mathrm{d}}_{l\alpha}(r) = -\frac{2}{k}\int_0^\infty \mathrm{d}r'\, \hat{n}_l(kr_>)\, \hat{j}_l(kr_<)\, \phi_{l\alpha}(r') - \frac{2}{k}\int_0^\infty \mathrm{d}r' \sum_{l'} \hat{n}_l(kr_>)\, \hat{j}_l(kr_<)\, V^{\mathrm{d}}_{ll'}(r')\, \phi^{\mathrm{d}}_{l'\alpha}(r')\,. \tag{3.39}$$

As is well-known [7, 8, 57, 58, 93], these integral equations can be solved by using the solutions to the related Volterra integral equations defined by

$$\psi^{\mathrm{d}}_{ll'}(r) = \hat{j}_l(kr)\delta_{ll'} + \frac{2}{k}\sum_{l''} \hat{j}_l(r)\int_0^r \mathrm{d}r'\, \hat{n}_l(r')\, V^{\mathrm{d}}_{ll''}(r')\, \psi^{\mathrm{d}}_{l''l'}(r') - \frac{2}{k}\sum_{l''} \hat{n}_l(r)\int_0^r \mathrm{d}r'\, \hat{j}_l(r')\, V^{\mathrm{d}}_{ll''}(r')\, \psi^{\mathrm{d}}_{l''l}(r')\,, \tag{3.40}$$

and

$$\psi^{\mathrm{d}}_{l\alpha}(r) = -\frac{2}{k}\int_0^\infty \mathrm{d}r'\, \hat{n}_l(kr_>)\, \hat{j}_l(kr_<)\, \phi_{l\alpha}(r') + \frac{2}{k}\sum_{l'} \hat{j}_l(r)\int_0^r \mathrm{d}r'\, \hat{n}_l(r')\, V^{\mathrm{d}}_{ll'}(r')\, \psi^{\mathrm{d}}_{l'\alpha}(r')$$

$$-\frac{2}{k}\sum_{l'} \hat{n}_l(r)\int_0^r \mathrm{d}r'\, \hat{j}_l(r')\, V^{\mathrm{d}}_{ll'}(r')\, \psi^{\mathrm{d}}_{l'\alpha}(r')\,. \tag{3.41}$$

Then $\phi^{\mathrm{d}}_{ll'}$ and $\phi^{\mathrm{d}}_{l\alpha}$ are related to $\psi^{\mathrm{d}}_{ll'}$ and $\psi^{\mathrm{d}}_{l\alpha}$ through

$$\phi^{\mathrm{d}}_{ll'} = \sum_{l''} \psi^{\mathrm{d}}_{ll''}(M^{-1})_{l''l'} \tag{3.42}$$

and

$$\phi^{\mathrm{d}}_{l\alpha} = \psi^{\mathrm{d}}_{l\alpha} - \sum_{l'} \phi_{ll'} M'_{l'\alpha} \tag{3.43}$$

where $(M^{-1})_{l''l'}$ are the matrix elements of the matrix inverse of the matrix **M** defined by

$$M_{ll'} = \delta_{ll'} + \frac{2}{k}\sum_{l''}\int_0^\infty \mathrm{d}r\, \hat{n}_l(r)\, V^{\mathrm{d}}_{ll''}(r)\, \psi^{\mathrm{d}}_{l''l'}(r)\,, \tag{3.44}$$

and where M' is defined by

$$M'_{l\alpha} = \frac{2}{k}\sum_{l'}\int_0^\infty \mathrm{d}r\, \hat{n}_l(r)\, V^{\mathrm{d}}_{ll'}(r)\, \psi^{\mathrm{d}}_{l'\alpha}(r)\,. \tag{3.45}$$

The solutions of the Volterra integral equations given in eqs. (3.40) and (3.41) are obtained on a grid of points, so that at a point r_i the value of the solution is given by an integration over the solution for the points r_j with $j<i$. Then using a series expansion about $r=0$ one can obtain the value of the solutions at r_1, which then allows the value of the wave functions to be obtained at all subsequent values of r_i. The numerical integrations from 0 to r_i are performed using the extended Simpson's rule. For the integrals where r_i is a midpoint of an ordinary Simpson's rule quadrature, we have used the Simpson's "$\frac{3}{8}$ rule" to evaluate the integral over the last four quadrature points. Knirk [94] has shown this quadrature scheme should be generally more accurate than the overlapped Simpson's rule or the trapezoidal rule.

Finally, when a large number of l's are included in the expansions, it becomes essential to include a procedure to stabilize the propagated solutions [93, 95]. Instabilities occur due to the nature of the mixing of the various solutions as given by the matrix M^{-1}. The stabilization procedure we have used [9, 95] then overcomes this difficulty by transforming the solutions $\psi^{\mathrm{d}}_{ll'}$ and $\psi^{\mathrm{d}}_{l\alpha}$ according to the equations

$$[\psi^{\mathrm{d}}_{ll'}]_{\mathrm{ST}} = \sum_{l''} \psi^{\mathrm{d}}_{ll''} T_{l''l'} \tag{3.46}$$

where $[\psi]_{\mathrm{ST}}$ is a stabilized function, and

$$[\psi^{\mathrm{d}}_{l\alpha}]_{\mathrm{ST}} = \psi^{\mathrm{d}}_{l\alpha} + \sum_{l'} \psi^{\mathrm{d}}_{ll'} T_{l'\alpha}\,, \tag{3.47}$$

so that they have the forms

$$[\psi^{\mathrm{d}}_{ll'}(r_{\mathrm{s}})]_{\mathrm{ST}} = \hat{j}_l(r_{\mathrm{s}})\delta_{ll'} + \hat{n}_l(r_{\mathrm{s}})P_{ll'} \tag{3.48}$$

and

$$[\psi^{\mathrm{d}}_{l\alpha}(r_{\mathrm{s}})]_{\mathrm{ST}} = -\frac{2}{k}\int_0^\infty \mathrm{d}r'\, \hat{n}_l(kr_>)\,\hat{j}_l(kr_<)\,\phi_{l\alpha}(r') + \hat{n}_l(r_{\mathrm{s}})\,P_{l\alpha} \tag{3.49}$$

where r_{s} is the stabilization point and $T_{ll'}$, $P_{ll'}$, $T_{l\alpha}$ and $P_{l\alpha}$ are obtained by satisfying eqs. (3.48) and (3.49).

3.2. Basis set expansion method

One of the attractive features of the multichannel Schwinger principle of eq. (2.171) is that, for the full body-frame scattering amplitude, all the required matrix elements except those of $VG^{(+)}_P V$ can be evaluated analytically for any molecule provided the trial scattering function is expanded in Cartesian Gaussian functions and plane waves. For the full scattering amplitude the free-particle solution S_m of H_0, i.e., of $H_N + T_{N+1}$ of eq. (2.160), is

$$S_m = \Phi_m \exp(\mathrm{i}\boldsymbol{k}_m \cdot \boldsymbol{r}_{N+1}) \tag{3.50}$$

and we further assume that the molecular orbitals of the Slater determinant Φ_m are also represented by a Gaussian basis. The analytical expressions for such matrix elements arising from terms such as $\langle\Psi_m|V|S_n\rangle$, $\langle\Psi_m|PV|\Psi_n\rangle$, $\langle\Psi_m|\hat{H}|\Psi_n\rangle$, and others, except $\langle\Psi_m|VG_P^{(+)}V|\Psi_n\rangle$, and containing combinations of Cartesian Gaussian functions and plane waves have been discussed by several authors [74, 96]. The matrix elements of $VG_P^{(+)}V$ can also be obtained in analytical form if an approximate closure relation is inserted around $G_P^{(+)}$

$$\langle\Psi_m|VG_P^{(+)}V|\Psi_n\rangle \simeq \sum_{\gamma,\delta}\langle\Psi_m|V|\gamma\rangle\langle\gamma|G_P^{(+)}|\delta\rangle\langle\delta|V|\Psi_n\rangle\,. \tag{3.51}$$

For Cartesian Gaussian functions the integrals arising from $\langle\gamma|G_P^{(+)}|\delta\rangle$ can again be evaluated analytically [96, 97].

In this form our multichannel Schwinger principle can be used to study the elastic and inelastic scattering of electrons by both linear and nonlinear molecules. This is because all the matrix elements can be evaluated analytically for these targets. Although the single-center expansion technique could also be applied to this multichannel theory the numerical effort required can be substantial for targets other than very simple molecules. Of course, if the potentials are long-ranged, e.g., for excitation of dipole-allowed states, either very large Gaussian basis sets must be used or plane waves will have to be included in the expansion basis. In section 5.3, we will discuss the results of application of this multichannel theory to e–H_2 collisions.

4. Application to molecular photoionization

4.1. Iterative Schwinger method

4.1.1. Photoionization of H_2

As a first example we discuss results of applications of these methods based on the Schwinger variational expression to the photoionization of H_2 [9]. For this system, small basis sets gave very rapid convergence of the iterative Schwinger procedure outlined in sections 2.5.2 and 2.5.3. As illustrated in table 4.1, with a basis of s and z type Cartesian Gaussian functions of exponents 0.3 and 1.0 on each

Table 4.1
Photoionization of H_2: Convergence of eigenphase sums and cross sections using the iterative Schwinger variational method[a]

Iteration number N	Eigenphase sum δ (radians)	Cross section σ (Mb)[b] Length	Velocity
0	0.181	4.59	2.62
1	0.213	4.62	2.64
2	0.213	4.62	2.64

[a] The results are for $^1\Sigma_u^+$ scattering at $k = 0.4287$ a.u. or $E = 18.9$ eV.
[b] In megabarns (10^{-18} cm^2).

nuclei and a z type Cartesian Gaussian function of exponent 1.0 at the midpoint, the photoionization cross section in the $1\sigma_g \to k\sigma_u$ channel, was well converged within one iteration.

In the studies of molecular photoionization considered here, the cross sections are obtained using the frozen core Hartree–Fock approximation for the final ionized state wave functions. This implies that many important electron correlation effects have been neglected. One approach for obtaining an estimate of the errors due to correlation is to compare the length form of the photoionization cross section which is proportional to the square of the dipole matrix elements

$$I^{\mathrm{L}}_{\boldsymbol{k},\hat{n}} = (k)^{1/2}\langle \Psi_{\mathrm{i}}|\boldsymbol{r}\cdot\hat{n}|\Psi^{(-)}_{\mathrm{f},\boldsymbol{k}}\rangle \tag{4.1}$$

and the velocity form of the photoionization cross section which is proportional to the square of the dipole matrix element

$$I^{\mathrm{V}}_{\boldsymbol{k},\hat{n}} = \frac{(k)^{1/2}}{E}\langle \Psi_{\mathrm{i}}|\nabla\cdot\hat{n}|\Psi^{(-)}_{\mathrm{f},\boldsymbol{k}}\rangle\,. \tag{4.2}$$

In eqs. (4.1) and (4.2), E is the photon energy, $\hat{n}$ is the direction of polarization of the light, $\boldsymbol{k}$ is the momentum of the photoelectron, Ψ_{i} is the initial state wave function, and $\Psi^{(-)}_{\mathrm{f},\boldsymbol{k}}$ is the final state wave function. The doubly differential photoionization cross section in the length (or velocity) form in the body-fixed frame is then

$$\frac{\mathrm{d}^2\sigma^{\mathrm{L(V)}}}{\mathrm{d}\Omega_{\hat{k}}\,\mathrm{d}\Omega_{\hat{n}}} = \frac{4\pi^2 E}{c}|I^{\mathrm{L(V)}}_{\boldsymbol{k},\hat{n}}|^2\,, \tag{4.3}$$

where c is the speed of light. Then the total cross sections in the length (or velocity) form is obtained by averaging over all possible orientations of the molecule relative to the polarization of the light source, $\hat{n}$, and by integrating over all final directions of the photoelectrons, $\hat{k}$, to yield

$$\sigma^{\mathrm{L(V)}} = \frac{\pi E}{c}\int \mathrm{d}\Omega_{\hat{k}}\int \mathrm{d}\Omega_{\hat{n}}\,|I^{\mathrm{L(V)}}_{\boldsymbol{k},\hat{n}}|^2\,. \tag{4.4}$$

If the wave functions (both initial and final states) used to calculate the photoionization cross section were exact eigenfunctions of the electronic Hamiltonian then the dipole length and dipole velocity forms of the cross section would be equivalent. The equality of these two forms is hence a necessary but not sufficient condition that the computed cross sections are accurate. In this sense, the difference between the length and velocity forms can be viewed as an estimate of the minimum error in the calculation [98, 99].

Figure 1 shows length and velocity forms of the H_2 photoionization cross section obtained with the iterative Schwinger method [9]. The basis set used is that given above, and the expansion parameters were $l_{\mathrm{m}} = l^{\mathrm{x}}_{\mathrm{s}} = l^{\mathrm{x}}_{\mathrm{i}} = l^{\mathrm{d}}_{\mathrm{i}} = 13$, $\lambda^{\mathrm{x}}_{\mathrm{m}} = \lambda^{\mathrm{d}}_{\mathrm{m}} = \lambda^{\mathrm{n}}_{\mathrm{m}} = 26$, $l_{\mathrm{p}} = 7$, $r_{\mathrm{m}} = 66.2$ a.u., and $h = 0.01$–0.16 a.u., with a total of 780 grid points. For photon energies above 18 eV, the fixed-nuclei, frozen-core Hartree–Fock results agree well with experimental data. Also, from fig. 1, it can be seen that the difference between the length and velocity forms of the FCHF cross section does give a good estimate of the error in the calculation. In this simple nonresonant case, the length and velocity forms of the cross section bracket the experimental results. This is not always true.

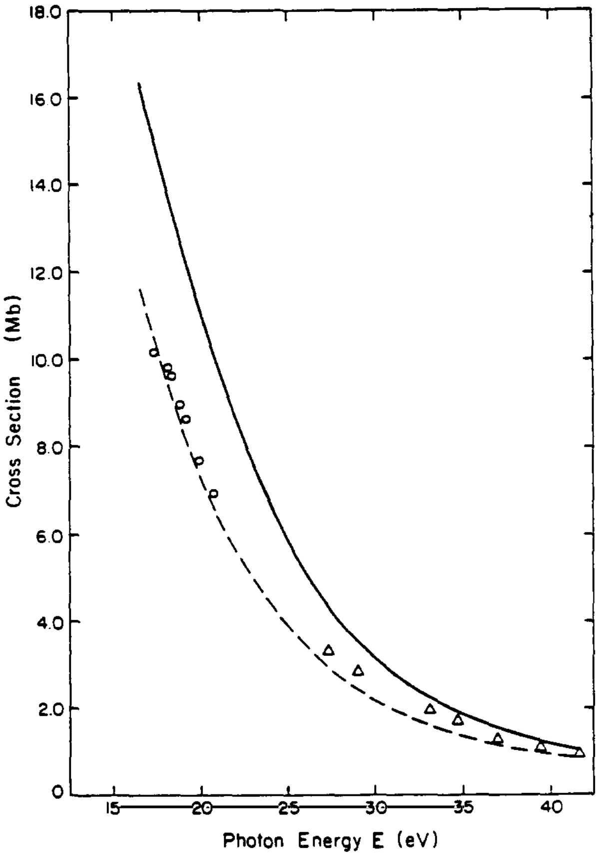

Fig. 1. Total photoionization cross section of H_2 in megabarns ($10^{-18}\,cm^2$): —— frozen-core Hartree–Fock dipole length; ---- frozen-core Hartree–Fock dipole velocity; ○ experimental data [G.R. Cook and P.H. Metzger, J. Opt. Soc. Am. 54 (1964) 968]; △ experimental data [J.A.R. Samson and R.B. Cairns, J. Opt. Soc. Am. 55 (1965) 1035]. The ionization potential for H_2 was taken to be 16.4 eV.

4.1.2. Photoionization of N_2

Extensive studies of the outer valence shell photoionization of N_2 have been performed using the iterative Schwinger method [10, 100]. The most interesting feature in this system is the shape resonance which occurs in the $3\sigma_g \to k\sigma_u$ channel leading to the $X\,^2\Sigma_g^+$ state of N_2^+. The nature of the shape resonant enhancement in this channel can be seen in fig. 2. The theoretical results given in this figure use two different initial-state wave functions (i.e., Ψ_i of eqs. (4.1) and (4.2)). These two initial-state wave functions are the usual Hartree–Fock (HF) wave function, and a configuration interaction (CI)-type wave function constructed from 386 spatial configurations [10] of the "singles-plus-doubles" excitations type [101]. The single-center expansion parameters for these calculations are $l_m = l_i^d = \lambda_m^d = 30$, $l_s^x = \lambda_m^x = 20$, $l_i^x = 16(1\sigma_g)$, $10(2\sigma_g)$, $10(3\sigma_g)$, $15(1\sigma_u)$, $9(2\sigma_u)$, $9(1\pi_u)$, $\lambda_m^n = 60$, $l_p = 7(k\sigma_u)$, $6(k\sigma_g, k\pi_u, k\pi_u)$, $r_m = 64.0$ a.u., $h_i = 0.01$–0.16 a.u., with 800 total grid points. The initial basis set contained 18 Gaussian functions in the σ_u symmetry and 8 Gaussian functions in the π_u symmetry [10]. The calculations are well converged by the first iteration as was the case with H_2 discussed above.

Figure 2 shows that in the region of a shape resonance, inclusion of initial state correlation does bring the length and velocity forms of the cross section closer together and in good agreement with

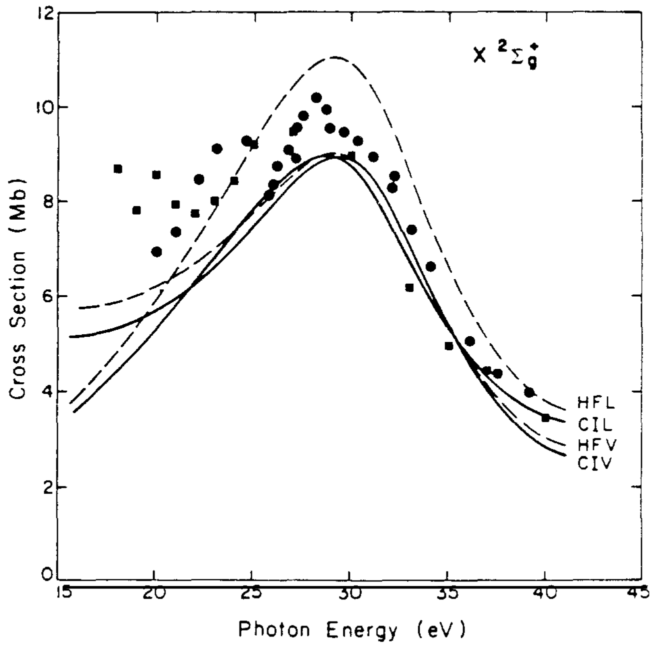

Fig. 2. Photoionization cross section of N_2 leading to the $X^2\Sigma_g^+$ state of N_2^+: HFL, dipole length form using HF initial state and FCHF final state; HFV, dipole velocity form using HF initial state and FCHF final state; CIL, dipole length form using CI initial state and FCHF final state; CIV, dipole velocity form using CI initial state and FCHF final state; ●, experimental data [E.W. Plummer, T. Gustafsson, W. Gudat and D.E. Eastman, Phys. Rev. A 15 (1977) 2339]; ■, experimental data [A. Hamnett, W. Stoll and C.E. Brion, J. Electron. Spectrosc. Relat. Phenom. 8 (1976) 367].

experimental data. This behavior of the cross section is similar to that observed by Swanson and Armstrong in the study of atomic photoionization [99]. The differences between the FCHF results and the experimental data in fig. 2 in the photon energy region around 23 eV can be attributed to autoionization from Rydberg states leading to the $C\,^2\Sigma_u^+$ state of N_2^+ [102]. To study such features, the effects of final-state correlations not included in the FCHF model must be considered.

We have also studied the sensitivity of the resonant photoionization cross section to the parameters in the single-center expansions. In fig. 3 the position of the peak in the resonant cross section is plotted as a function of $l = l_m = l_i^d = \lambda_m^d$ with $l_s^x = \lambda_m^x = 30$ and with all other parameters fixed as given above. The results show that the peak energy varies as l^{-3}, and allows us to extrapolate to infinite l and obtain the estimate of $E^{\infty} = 28.7$ eV. The error position of the peak of the cross section in fig. 2 is hence less than 0.5 eV.

The total photoionization cross section is obtained by averaging over all orientations of the molecule and integrating over all photoelectron directions. In the gas phase, the photoelectron angular distribution, known as the Integrated Target Angular Distribution (ITAD), is obtained by averaging over all molecular orientations with respect to the polarization of the light source. To obtain expressions for these angular distributions, we first expand the dipole matrix elements in spherical harmonics,

$$I_{k,\hat{n}}^{L(V)} = \left(\frac{4\pi}{3}\right)^{1/2} \sum_{lm\mu} I_{lm\mu}^{L(V)}\, Y_{lm}^{*}(\Omega_{\hat{k}})\, Y_{1\mu}^{*}(\Omega_{\hat{n}})\,. \tag{4.5}$$

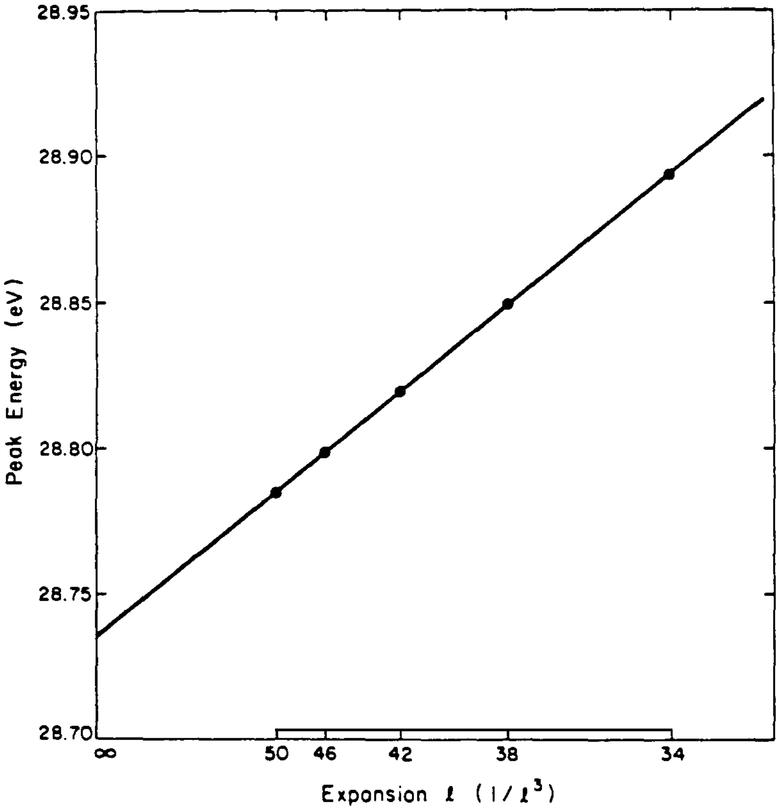

Fig. 3. Dependence of the peak photoionization cross section on $l = l_m = l_i^d = \lambda_m^d$ (see section 3.1) for the $3\sigma_g \to k\sigma_u$ channel of N_2 photoionization.

The partial-wave matrix elements are then given by

$$I^{L}_{lm\mu} = (k)^{1/2} \langle \Psi_i | r_\mu | \Psi^{(-)}_{f,\,klm} \rangle \tag{4.6}$$

and

$$I^{V}_{lm\mu} = \frac{(k)^{1/2}}{E} \langle \Psi_i | \nabla_\mu | \Psi^{(-)}_{f,\,klm} \rangle \tag{4.7}$$

where

$$r_\mu = \begin{cases} \mp(x \pm \mathrm{i}y)/2^{1/2} & \text{for } \mu = \pm 1 \\ z & \text{for } \mu = 0 \end{cases} \tag{4.8}$$

and

$$\nabla_\mu = \begin{cases} \mp[(\partial/\partial x) \pm \mathrm{i}(\partial/\partial y)]/2^{1/2} & \text{for } \mu = \pm 1 \\ \partial/\partial z & \text{for } \mu = 0\,. \end{cases} \tag{4.9}$$

Then the total cross section given in eq. (4.4) can be rewritten in terms of the partial-wave matrix elements as

$$\sigma^{\mathrm{L(V)}} = \frac{4\pi^2}{3c} E \sum_{\mu} D_{\mu}^{\mathrm{L(V)}} \tag{4.10}$$

where

$$D_{\mu}^{\mathrm{L(V)}} = \sum_{lm} |I_{l,m,\mu}^{\mathrm{L(V)}}|^2 . \tag{4.11}$$

Now the ITAD is obtained by averaging eq. (4.3) over all target orientations to give [103]

$$\frac{\mathrm{d}\sigma^{\mathrm{L(V)}}}{\mathrm{d}\Omega_{\hat{k}}} = \frac{\sigma^{\mathrm{L(V)}}}{4\pi} [1 + \beta_{\hat{k}}^{\mathrm{L(V)}} P_2(\cos\theta)] \tag{4.12}$$

where θ is the angle between the direction of polarization of the light and the momentum of the electron, $P_2(\cos\theta)$ is the Legendre polynomial of degree 2, and $\beta_{\hat{k}}$ is the asymmetry parameter given by

$$\beta_{\hat{k}}^{\mathrm{L(V)}} = \frac{4\pi^2}{5c\sigma^{\mathrm{L(V)}}} \sum_{\substack{lm\mu \\ l'm'\mu' \\ \mu''}} (-1)^{m-\mu'} I_{lm\mu}^{\mathrm{L(V)}} [I_{l'm'\mu'}^{\mathrm{L(V)}}]^*$$

$$\times [(2l+1)(2l'+1)]^{1/2} \langle 1100|20\rangle \langle ll'00|20\rangle \langle 11-\mu\mu'|2\mu''\rangle \langle ll'-m-m'|2-\mu''\rangle . \tag{4.13}$$

A second possible angular distribution is the Integrated Detector Angular Distribution (IDAD) suggested by Wallace and Dill [104], which is the cross section integrated over all possible photoelectron emission directions for a fixed molecular orientation. The IDAD might be useful in determining the orientation of a fixed photoionized target in the laboratory frame [104]. The IDAD cross section is given by integrating eq. (4.3) over all $\Omega_{\hat{k}}$ to give

$$\frac{\mathrm{d}\sigma^{\mathrm{L(V)}}}{\mathrm{d}\Omega_{\hat{n}}} = \frac{\sigma^{L(V)}}{4\pi} [1 + \beta_{\hat{n}}^{\mathrm{L(V)}} P_2(\cos\theta)] \tag{4.14}$$

where θ is the angle between the polarization of the light and the molecular z axis, and the asymmetry parameter $\beta_{\hat{n}}$ is given by

$$\beta_{\hat{n}}^{\mathrm{L(V)}} = [2D_0^{\mathrm{L(V)}} - (D_{-1}^{\mathrm{L(V)}} + D_{+1}^{\mathrm{L(V)}})] \Big/ \Big(\sum_{\mu} D_{\mu}^{\mathrm{L(V)}} \Big) . \tag{4.15}$$

Figure 4 shows the ITAD and IDAD asymmetry parameters for the photoionization of N_2 leading to the $X\,^2\Sigma_g^+$ state of the ion. The calculated ITAD asymmetry parameters agree well with experimental data, except for energies less than 23 eV where autoionization effects are known to be important. The results for

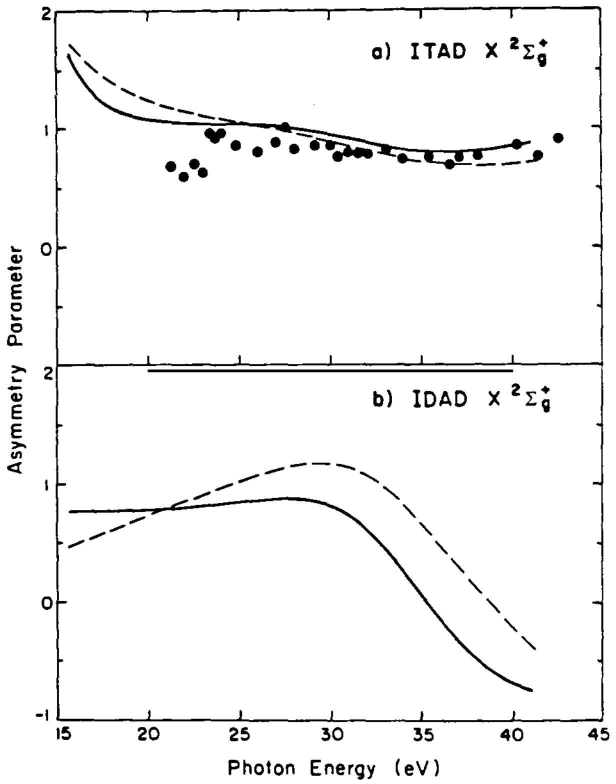

Fig. 4. Photoionization asymmetry parameters for the production of the $X^2\Sigma_g^+$ state of N_2^+: (a) ITAD asymmetry parameter $\beta_{\hat{k}}$; (b) IDAD asymmetry parameter $\beta_{\hat{n}}$; —— dipole length form using CI initial state and FCHF final state; ---- dipole velocity form using CI initial state and FCHF final state; ●, experimental data [G.V. Marr, J.M. Morton, R.M. Holmes and D.G. McCoy, J. Phys. B 12 (1979) 43].

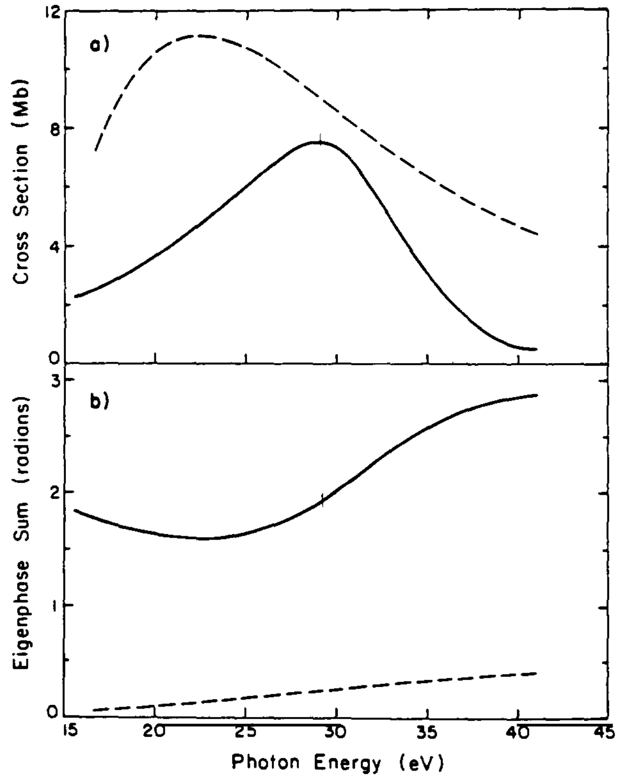

Fig. 5. Comparison of the photoionization in the $3\sigma \to k\sigma_u$ channel with the photoionization in the $1\pi_u \to k\delta_g$ channel of N_2; (a) comparison of photoionization cross sections; (b) comparison of eigenphase sums; ——, $3\sigma_g \to k\sigma_u$ channel; ----, $1\pi_u \to k\delta_g$ channel.

the IDAD asymmetry parameter show that above the resonance energy, the contribution from the $k\sigma_u$ continuum channel drops off rapidly leaving only the contribution from the $k\pi_u$ channel.

The behavior of the eigenphase sum of the scattering matrix can also be very important in characterizing shape resonances in molecular photoionization. In fig. 5 the FCHF results for the $3\sigma_g \to k\sigma_u$ and $1\pi_u \to k\delta_g$ photoionization channels of N_2 are shown. As seen in fig. 5a, both of these channels have peaked structures in total cross section, but fig. 5b shows that only the $3\sigma_g \to k\sigma_u$ channel exhibits resonant behavior in the eigenphase sums. The shape of the nonresonant $1\pi_u \to k\delta_g$ channel is determined by the usual energy dependence of the dipole matrix element in the same manner as in the $2p \to kd$ photoionization cross section of Ne [105].

Figure 6 gives the total photoionization cross section leading to the $X^2\Sigma_g^+$, $A^2\Pi_u$ and $B^2\Sigma_u^+$ states of N_2^+. From fig. 6 we see that the effect of initial state correlation is to lower the length form of the cross section and not to alter the velocity form appreciably. The CI length form of the cross section agrees well with the experimental data except for the features due to autoionization below 23 eV. The shape of the shoulder in the cross section due to the $3\sigma_g \to k\sigma_u$ resonance, and the high-energy fall-off of the cross section are both well reproduced using the FCHF model with initial-state correlation included.

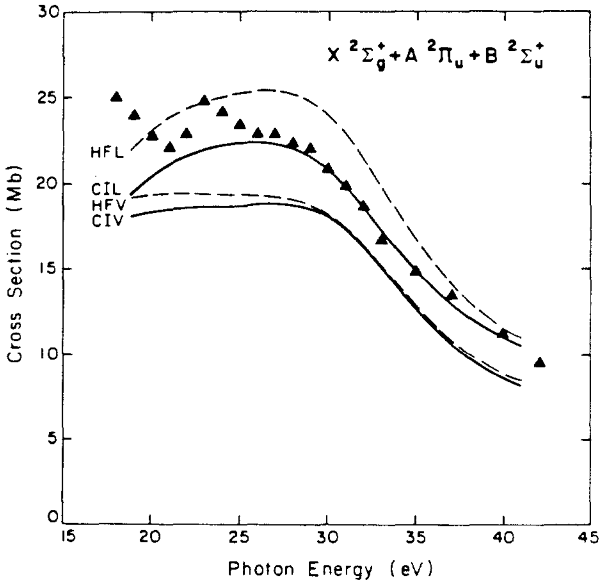

Fig. 6. Total photoionization cross section of N_2 for the production of the $X^2\Sigma_g^+$, $A^2\Pi_u$ and $B^2\Sigma_u^+$ states of N_2^+ (same designations as in fig. 2): ▲, total experimental cross section [G.R. Wight, M.J. Van der Wiel and C.E. Brion, J. Phys. B 9 (1976) 675] corrected to only include the contribution from these three channels using experimental branching ratios [A. Hamnett, W. Stoll and C.E. Brion, J. Electron. Spectrosc. Relat. Phenom. 8 (1976) 367].

In fig. 7 the results for the photoionization of N_2 obtained using the single-center-expansion iterative Schwinger method are compared with the results of both the Stieltjes–Tchebycheff moment theory (STMT) approach [106, 107] and of the continuum multiple scattering method (CMSM) [108]. The results in fig. 7 were obtained using the dipole length approximation and a HF initial state. For N_2 all three methods can be seen to be in qualitative agreement, with the STMT results in better quantitative agreement with the results of the Schwinger method than are the CMSM results. The oscillatory behavior of the STMT results when compared to the exact FCHF results obtained from the iterative Schwinger method is seen often in the moment theory approach and will occur again in other systems considered here. This behavior is probably an indication that lower-order moments are obtained more accurately than higher-order moments in this method.

We have also considered the effect of the shape resonance in the $3\sigma_g \to k\sigma_u$ photoionization channel of N_2 on the vibrational distribution of the product molecular ion [100]. If the dependence of the dipole matrix elements on the internuclear separation is considered, then in the adiabatic nuclei approximation the vibrationally resolved photoionization cross sections are given by

$$\sigma_{\nu,\nu'}^{L(V)} = \frac{4\pi^2}{3c} E \sum_{lm\mu} |\langle \chi_i^{\nu} | I_{lm\mu}^{L(V)}(R) | \chi_f^{\nu'} \rangle|^2 \tag{4.16}$$

where χ_i is an initial state vibrational wave function and χ_f is a final state vibrational wave function. As a percentage, the vibrational branching ratio, $R_{\nu\nu'}^0$, is given by

$$R_{\nu\nu'}^0 = \frac{\sigma_{0,\nu'}}{\sigma_{0,\nu}} \times 100 . \tag{4.17}$$

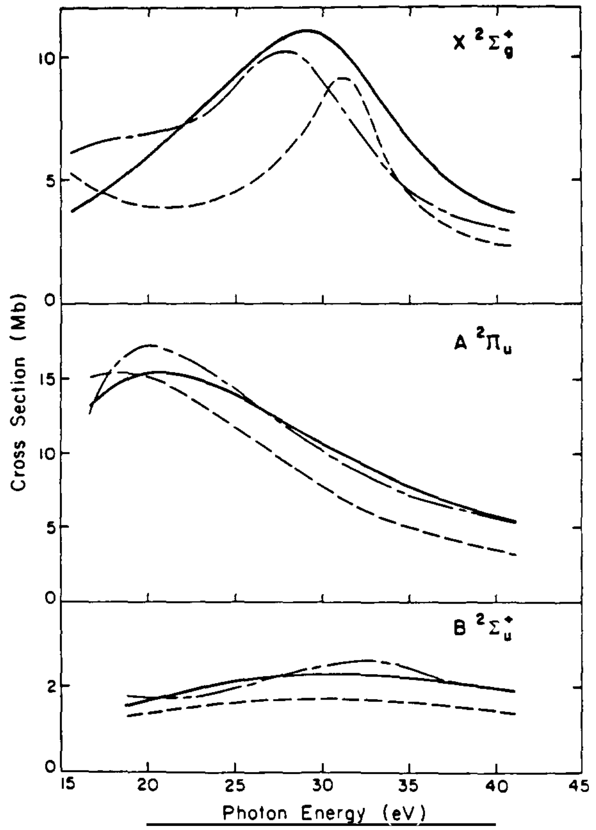

Fig. 7. Comparison of different theoretical photoionization cross sections of N_2 for the production of the $X^2\Sigma_g^+$, $A^2\Pi_u$ and $B^2\Sigma_u^+$ states of N_2^+: ——, single-center FCHF using iterative Schwinger method (ref. [10]); — — — —, FCHF results obtained using the STMT approach (refs. [106, 107]); – – –, CMSM model potential results (ref. [108]).

In fig. 8 experimental and theoretical R_{01}^0 branching ratios are given for the photoionization of N_2 leading to the $X\,^2\Sigma_g^+$ state of N_2^+. If the dipole matrix elements, $I_{lm\mu}$, were nearly constant as a function of R, then the Franck–Condon approximation would be valid and the vibrational branching ratio $R_{\nu\nu'}^0$ given in eq. (4.17) would be independent of the dipole matrix elements and equal to a ratio of Franck–Condon factors. In the Franck–Condon approximation the branching ratio should be constant as a function of photon energy, and thus the strong energy dependence of the vibrational branching ratios in fig. 8 are characteristic of non-Franck–Condon vibrational effects.

In fig. 8 we compare the accurate FCHF adiabatic-nuclei results obtained using the Schwinger variational expression [100] with the results of the CMSM by Dehmer et al. [109] and with experimental data of West et al. [110]. Above 26 eV, the accurate FCHF results are in very good agreement with the experimental data. The discrepancy in the branching ratios around the photon energy of 23 eV can again be attributed to autoionizing resonances. The poor quantitative agreement between the CMSM branching ratios and the experimental data is an indication of the inaccurate representation of the R dependence of the scattering potential in the CMSM model.

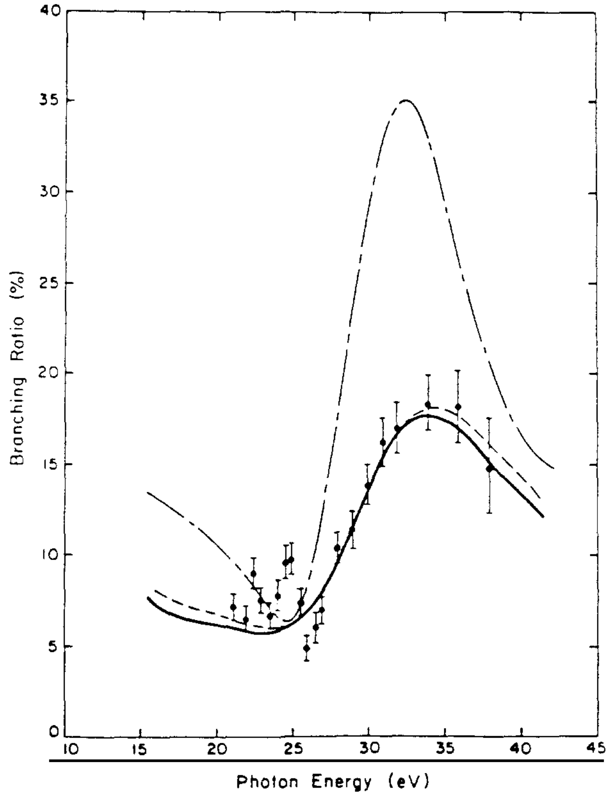

Fig. 8. Branching ratios, R^0_{01}, for the production of the $X^2\Sigma_g^+$ state of N_2^+ by photoionization of N_2: ——, Schwinger variational method results using a FCHF final state and a HF initial state with the dipole length form; ----, Schwinger variational method results with dipole velocity form; — — —, CMSM results from ref. [109]; ●, experimental results from ref. [110].

4.1.3. Photoionization of CO_2

We have also studied the photoionization cross sections of CO_2 using the iterative Schwinger method [92, 111, 112]. The single-center parameters used in these studies were $l_m = l_s^x = l_i^d = 59$, $l_i^x = 38(1\sigma_g)$, $10(2\sigma_g)$, $24(3\sigma_g)$, $16(4\sigma_g)$, $39(1\sigma_u)$, $23(2\sigma_u)$, $15(3\sigma_u)$, $15(1\pi_u)$, $16(1\pi_g)$, $\lambda_m^x = 40$, $\lambda_m^d = \lambda_m^n = 118$, $l_p = 10$, $r_m = 90$ a.u., $h_i = 0.005$–0.16 for a total of 1000 grid points. The variational basis sets contained spherical Gaussian functions with from 32 such basis functions in the $1\pi_u \to k\delta_g$ channel to 14 basis functions in the $1\pi_g \to k\delta_u$ channel [92]. With these basis sets, the photoionization cross sections were converged after one iteration in the iterative Schwinger method. As in the studies of N_2, we also considered the effects of initial state correlation on the photoionization cross sections of CO_2. This was again done by using for the initial state, a "single-plus-doubles" CI wave function constructed from 505 spatial configurations [92].

An interesting feature in the photoionization cross section of CO_2 was a $k\sigma_u$ shape resonance which appears prominently in the $1\pi_g \to k\sigma_u$, $4\sigma_g \to k\sigma_u$, $1\sigma_g \to k\sigma_u$ photoionization channels at a photoelectron kinetic energy of 15–20 eV. Figure 9 shows the cross section for the photoionization of CO_2 leading to the $C^2\Sigma_g^+$ state of CO_2^+. These cross sections contain the resonant $4\sigma_g \to k\sigma_u$ channel. In this figure we compare the FCHF results obtained using the Schwinger method with the CMSM results of Swanson et al. [113] and with the experimental cross sections of Brion and Tan [114]. In fig. 9 we show both the

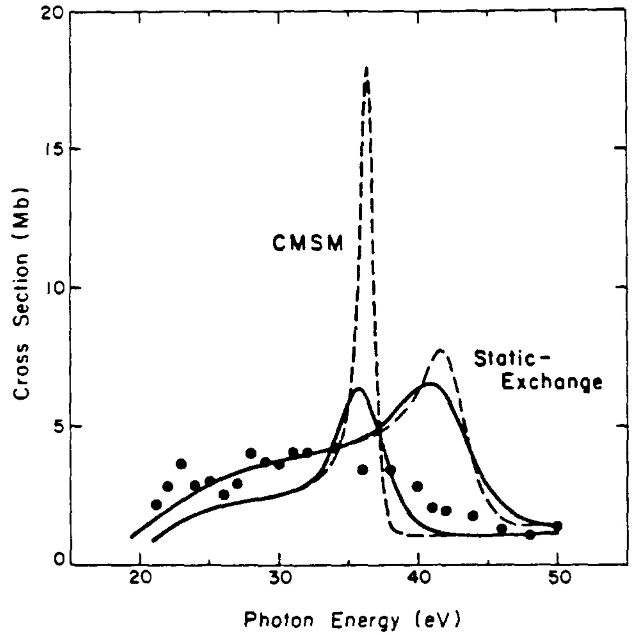

Fig. 9. Photoionization cross sections of CO_2 leading to the $C^2\Sigma_g^+$ state of CO_2^+, including CMSM results of ref. [113] and static-exchange Schwinger variational results of ref. [112]: ——, cross section averaged over the symmetric stretch vibrational mode; ----, equilibrium fixed-nuclei cross section; ●, experimental data from ref. [114].

cross section obtained with the nuclei fixed at their equilibrium value, and the photoionization cross section averaged over the symmetric stretch vibrational mode. The vibrationally averaged cross section is the cross section $\sigma_{\nu,\,\mathrm{ave}}$ summed over all final vibrational states and given by

$$\sigma^{\mathrm{L(V)}}_{\nu,\,\mathrm{ave}} = \frac{4\pi^2}{3c} E \sum_{\substack{lm\mu \\ \nu'}} |\langle \chi^{\nu}_{\mathrm{i}} | I^{\mathrm{L(V)}}_{lm\mu}(R) | \chi^{\nu'}_{f} \rangle|^2 \tag{4.18}$$

which is equivalent to [49]

$$\sigma^{\mathrm{L(V)}}_{\nu,\,\mathrm{ave}} = \langle \chi^{\nu}_{\mathrm{i}} | \sigma^{\mathrm{L(V)}}(R) | \chi^{\nu}_{\mathrm{i}} \rangle \,. \tag{4.19}$$

The results shown in fig. 9 indicate that the CMSM and Schwinger methods predict a narrow resonance in this channel 15–20 eV above threshold. However, the fixed-nuclei results of the CMSM have an unrealistically large-peak resonant cross section and also show a very large effect due to vibrational averaging compared to the exact static-exchange results obtained using the Schwinger method. The deficiency of the CMSM, when vibrational averaging is considered, is analogous to the difficulty in obtaining accurate vibrational branching ratios for the photoionization of N_2 discussed above. It is however interesting that vibrationally averaging the fixed-nuclei CMSM results does seem to compensate for the errors in the original fixed-nuclei result. This combined procedure yields reasonable cross sections, but it will also give unreasonable vibrational branching ratios as seen in N_2.

Although the resonance apparent in the theoretical results of fig. 9 is not present in the experimental total cross sections, there is experimental evidence for this resonance in the photoelectron angular distributions. In fig. 10 we see the presence of a feature in the experimental asymmetry parameters which is in reasonable agreement with the predicted asymmetry parameters in the resonant region. As with the total cross sections, the vibrational averaging of the CMSM results produces a larger quantitative change than does vibrational averaging of the static-exchange results. However, the errors in the CMSM asymmetry parameters seem less serious than in the corresponding cross sections.

A comparison of the results of the STMT method of Padial et al. [115] and the iterative Schwinger method for the resonant $4\sigma_g \to k\sigma_u$ channel is given in fig. 11. Here we give both the underlying Stieltjes density of the variational pseudospectrum [111] as well as the final moment theory cross section [115]. As seen in fig. 11, the low-order moment theory used by Padial et al. [115] has smoothed away the resonant feature which is clearly evident in the variational pseudo-spectrum. Unfortunately, higher-order applications of moment theory, which would show this resonant feature, are not always reliable. To resolve this difficulty one would have to obtain more than the two pseudo-states which are present in the resonance region in the STMT calculation of Padial et al. [115].

Total photoionization cross sections of CO_2 leading to the $A^2\Pi_u + B^2\Sigma_u^+$ and to the $X^2\Pi_g + A^2\Pi_u + B^2\Sigma_u^+ + C^2\Sigma_g^+$ states of CO_2^+ are given in fig. 12. The difference between the FCHF cross sections for photoionization leading to the $A^2\Pi_u + B^2\Sigma_u^+$ states of CO_2^+ and the experimental data is probably due to unresolved autoionization features in the experimental data. As suggested by Lucchese and McKoy [92], the presence of such features might provide an explanation for the difference between the observed fluorescence and photoelectron spectra from photoionization leading to these two states [116].

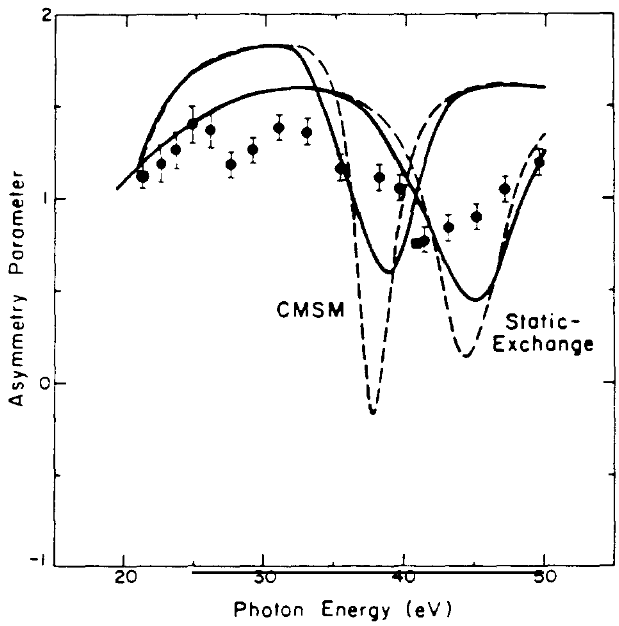

Fig. 10. ITAD photoionization asymmetry parameter, β_k, for the photoionization of CO_2 leading to the $C^2\Sigma_g^+$ state of CO_2^+, including CMSM results [J.R. Swanson, D. Dill and J.L. Dehmer, J. Phys. B. 14 (1981) L207], and static-exchange Schwinger variational results of ref. [112]: ——, cross section averaged over symmetric stretch vibrational mode; ----, equilibrium fixed-nuclei cross section; ●, experimental data [T.A. Carlson, M.O. Krause, F.A. Grimm, J.D. Allen Jr., D.Mehaffy, P.R. Keller and J.W. Taylor, Phys. Rev. A 23 (1981) 3316]; ■, experimental data [S. Katsumata, Y. Achiba and K. Kimura, J. Electron. Spectrosc. Relat. Phenom. 17 (1979) 229].

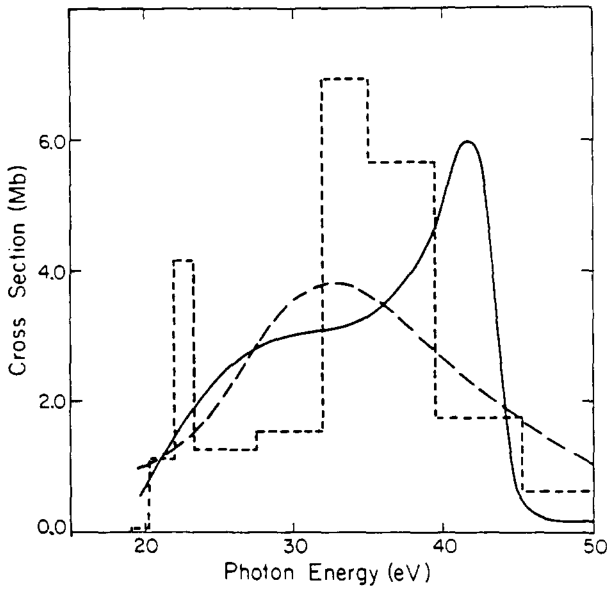

Fig. 11. Comparison of STMT and Schwinger photoionization cross sections of CO_2 in the $4\sigma_g \rightarrow k\sigma_u$ channel: ——, iterative Schwinger results from ref. [59]; ----, STMT results from ref. [115]; ------, Stieltjes density of the variational pseudo-spectrum.

The total cross sections leading to the four valence states of CO_2^+ in fig. 12 show reasonable agreement between theory and experiment. The two main discrepancies are the possible autoionization features below 24 eV, which are not present in the FCHF results obtained using the iterative Schwinger method, and the presence of the resonant feature at 41 eV which is due to the $4\sigma_g \rightarrow k\sigma_u$ resonance. As shown above, in fig. 9, vibrational averaging reduces the discrepancy somewhat, and better agreement between theory and experiment should result when electron correlation effects are included in the final-state wave function.

Figure 13 shows the photoionization cross sections and photoelectron asymmetry parameters for ionization of the oxygen K-shell of CO_2. Again the most prominent feature is a shape resonance which appears in the $1\sigma_g \rightarrow k\sigma_u$ channel. This feature appears clearly in both the cross section and the asymmetry parameter. There is general agreement between the peak in the FCHF cross section and the broad peak in the experimental photoabsorption cross section, but the FCHF resonance peak appears to be too narrow. There are several possible sources of error, including ionic core relaxation, other final-state correlation effects and vibrational averaging. We also compare the iterative Schwinger results with those of the STMT method [115] in fig. 13. As in the $4\sigma_g \rightarrow k\sigma_u$ channel, the STMT results do not show the $k\sigma_u$ resonance. Again, as in the $4\sigma_g \rightarrow k\sigma_u$ channel, this difference in the results of these two methods is probably due to a lack of reliable higher-order moments in the STMT calculations.

4.1.4. Photoionization of C_2H_2

As a final example of our studies of molecular photoionization using the iterative Schwinger method we discuss the cross sections of acetylene [117]. Of particular interest in these studies is the possible nature of the double-peaked structure shown in fig. 14 and which appears in the experimental cross section, leading to the $X^2\Pi_u$ state of $C_2H_2^+$ [118].

In this study of C_2H_2, basis sets containing 22 to 12 spherical Gaussian functions were used [65]. With

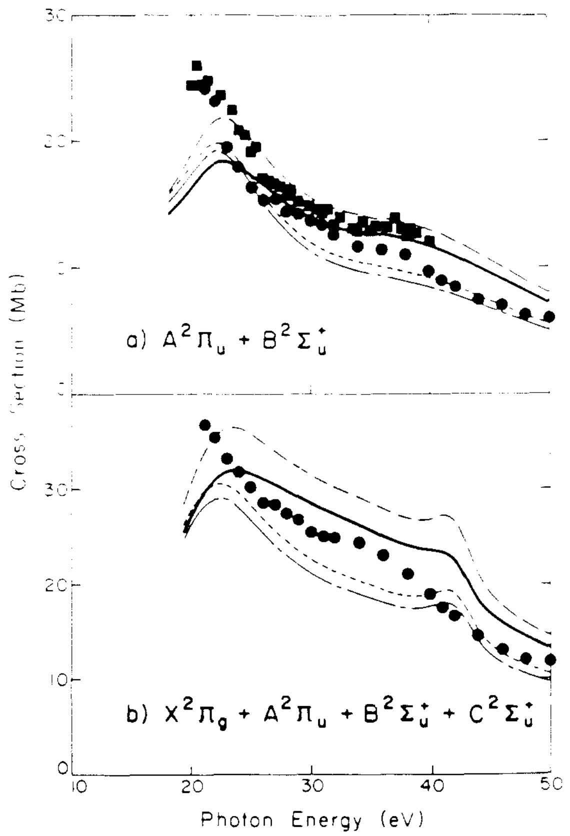

Fig. 12. Total photoionization cross sections of CO_2 leading to the $A^2\Pi_u + B^2\Sigma_u^+$ and $X^2\Pi_g + A^2\Pi_u + B^2\Sigma_u^+ + C^2\Sigma_g^+$ states of CO_2^+; ——, iterative Schwinger results using dipole length form and CI initial-state wave function; ----, iterative Schwinger results using the dipole velocity form and CI initial-state wave function; ———, iterative Schwinger results using the dipole length form and HF initial-state wave function; ------, iterative Schwinger results using the dipole velocity form and HF initial-state wave function; ●, experimental data from ref. [114]; ■, experimental data [T. Gustafsson, E.W. Plummer, D.E. Eastman and W. Gudat, Phys. Rev. A 17 (1978) 175].

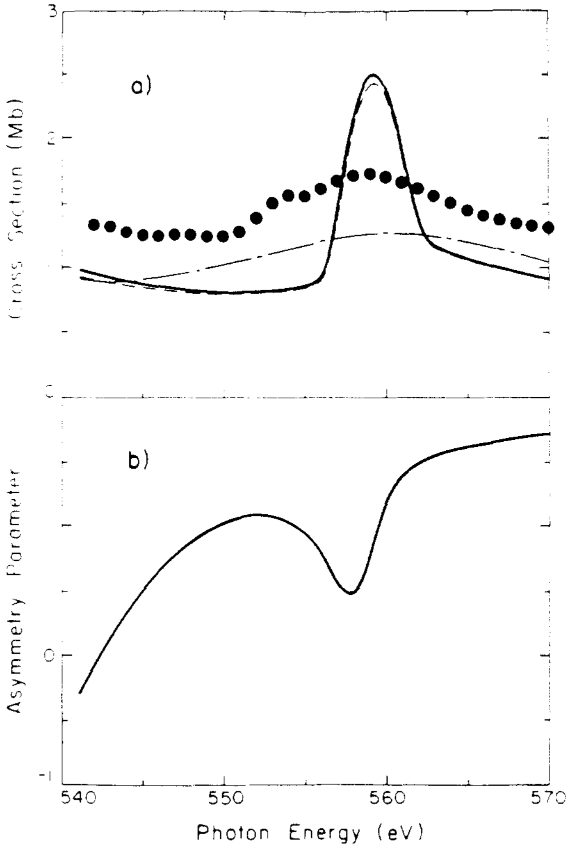

Fig. 13. Photoionization cross sections and ITAD photoelectron asymmetry parameters, β_k, for the oxygen K-shell in CO_2: ——, iterative Schwinger method results using the dipole length form and a HF initial-state wave function; ----, iterative Schwinger method results using the dipole velocity form and a HF initial-state wave function; ----, STMT results from ref. [115]; ●, absolute experimental photoabsorption cross section [D.M. Barrus, R.L. Blake, A.J. Burek, K.C. Chambers and A.L. Pregenzer, Phys. Rev. A 20 (1979) 1045]. Note that length and velocity forms of the asymmetry parameter are nearly identical and we thus only have presented the length form.

basis sets of this size, generally two steps in the iterative procedure were required before the cross sections were satisfactorily converged. The single-center expansion parameters, which gave adequate convergence of the single-center expansions were, $l_m = l_s^x = l_i^d = 30$, $l_i^x = 20(1\sigma_g)$, $12(2\sigma_g)$, $12(3\sigma_g)$, $19(1\sigma_u)$, $11(2\sigma_u)$ and $11(1\pi_u)$, $\lambda_m^x = 30$, $\lambda_m^d = \lambda_m^n = 60$, $l_p = 11$, $r_m = 79$ a.u., $h = 0.01$–0.16 for a total of 1000 grid points. These expansion parameters are very similar to those given above for N_2, with which C_2H_2 is isoelectronic.

One complication, which arises in interpreting the FCHF results of fig. 14 has to do with uncertainty concerning the position of the $1\pi_u \to 1\pi_g$ valence transition. In both N_2 and C_2H_2 the usual FCHF potential, for excitation from the $1\pi_u$ orbital and which has the overall symmetry of ${}^1\Sigma_u^+$, does not have

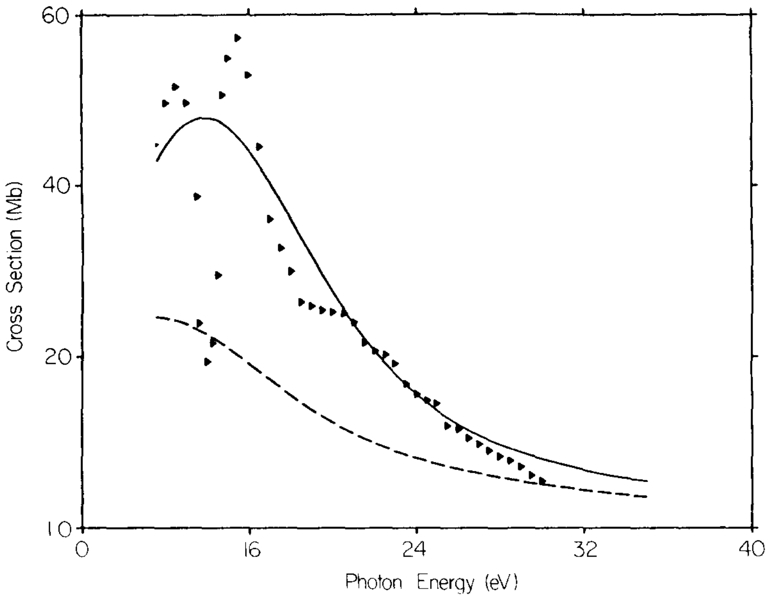

Fig. 14. Photoionization cross sections of acetylene leading to the $X^2\Pi_u$ state of $C_2H_2^+$: ——, iterative Schwinger results using dipole-length form; ----, iterative Schwinger results using dipole-velocity form; ▶, experimental results of ref. [118] normalized to the dipole-length cross section at 24 eV.

a bound excited state corresponding to a $1\pi_u \rightarrow 1\pi_g$ excitation. Thus, in both of these systems a large enhancement of the photoionization cross section, computed using the singlet coupled FCHF potential, appears just above threshold. In N_2, the $1\pi_u \rightarrow 1\pi_g$ transition is known experimentally to lie in the discrete region of the photoabsorption spectrum. To obtain reasonable theoretical cross sections for the $1\pi_u \rightarrow k\pi_g$ photoionization channel of N_2, the FCHF potential must be suitably modified [10, 107]. In the case of acetylene, it is currently unclear where the corresponding $1\pi_u \rightarrow 1\pi_g$ transition energy should be. The results for the photoionization cross section of C_2H_2 shown in fig. 14 were computed assuming that the singlet coupled FCHF potential is correct [117], and the relative experimental cross sections [118] were normalized to the theoretical dipole-length cross sections at 24 eV.

The large difference between the dipole-length and dipole-velocity cross section of fig. 14 reflects the importance of final-state correlation in this system. Final-state correlations can shift the main peak of the FCHF cross section as well as introduce autoionization resonances into this system. At present there seems to be two plausible explanations for the two peaks in this photoionization channel. The first possibility is that these two peaks correspond to two different autoionization resonances, the most likely resonant states being the $3\sigma_g \rightarrow n\sigma_u$ $(^1\Sigma_u^+)$ and $2\sigma_u \rightarrow 1\pi_g$ $(^1\Pi_u)$ states [117, 119]. The second possible explanation is that the first peak is due to intensity from the $1\pi_g \rightarrow k\pi_g$ transition shifted to lower energy by correlation effects and the second peak is due to the autoionizing $2\sigma_u \rightarrow 1\pi_g$ $(^1\Pi_u)$ state [120]. This question cannot be settled using the iterative Schwinger method with the FCHF potential. Extensions of this technique to include final state correlation effects are necessary to resolve questions of the type encountered in the photoionization of acetylene.

4.2. *Padé approximant correction method*

The utility of the Padé approximant correction method discussed in section 2.5.2 has been examined in a study of the photoionization cross sections of CO [40]. The basis sets used in this study contained 28 and 20 spherical Gaussian functions for the σ and π continuum photoionization channels, respectively. The single-center expansion parameters, again similar to the parameters used with N_2, were $l_m = l_c^x = l_i^d = 29$, $l_i^x = 16(1\sigma)$, $16(2\sigma)$, $8(3\sigma)$, $6(4\sigma)$, $5(5\sigma)$ and $5(1\pi)$, $\lambda_m^x = 29$, $\lambda_m^d = \lambda_m{}^n = 58$, $l_p = 7$, $r_m = 100$ a.u., $h = 0.01$–0.16 a.u. for a total of 1000 grid points.

In table 4.2 we compare the results of the Padé approximant method and the iterative Schwinger method at two energies in the $5\sigma \to k\sigma$ photoionization channel of CO. In this table we first note that the $N = 0$ results are identical for both methods since both methods use the same initial basis set approximation to the scattering matrix and dipole matrix elements. Also note that the $N = 1$ results for the eigenphase sums are also equivalent as mentioned in section 2.5.2. Table 4.2 illustrates that the iterative Schwinger method can converge relatively slowly at some energies, compared to the Padé approximant approach while at other energies the two methods have equally rapid convergence.

One interesting point about the Padé approximant approach for computing photoionization cross sections is that the matrix elements $I_{lm\mu}$ given in eqs. (4.6) and (4.7) are defined in terms of $\Psi_{f,klm}^{(-)}$, which is a scattering wave function whose *out-state* is described by the quantum members l and m. However, to avoid having to use complex arithmetic, we generally compute the principle-valued wave function $\Psi_{f,klm}^{(P)}$. The dipole matrix elements can be obtained from the principal-valued wave functions

Table 4.2
Comparison of the Padé approximant and iterative Schwinger (ISM) correction methods. Results are for the $5\sigma \to k\sigma$ photoionization channel of CO. These calculations were based on the use of eq. (4.20) of the text with the M_{33}^{3} (VS', S) variational expression for the K-matrix, and the M_{35}^{3} $(p\phi, S)$ variational expression for dipole matrix elements

	Cross section (Mb)[b]		Eigenphase sum	
N[c]	Padé	ISM	Padé	ISM
		$E = 18$ eV[a]		
0	4.11	4.11	0.88	0.88
1	11.95	2.67	0.84	0.84
2	3.84	3.98	1.00	0.82
3	3.83	3.66	1.00	0.96
4		3.81		0.98
5		3.92		1.01
6		3.81		1.01
7		3.85		1.00
		$E = 25$ eV		
0	4.86	4.86	0.83	0.83
1	4.59	4.63	0.89	0.89
2	4.61	4.61	0.89	0.89
3	4.61	4.61	0.89	0.89

[a] E is the photon energy, with an ionization potential of 14.0 eV.
[b] Dipole length form of the cross section with a HF initial state.
[c] N is either the order of the Padé approximant used to compute M_c as defined in eq. (2.121), or the iteration number as defined in eq. (2.124) for the iterative Schwinger method.

using

$$I_{lm\mu} = (k)^{1/2}\, e^{i\sigma_l} \sum_{l'} \langle \Psi_i | p_\mu | \Psi^{(P)}_{f,\,kl'm} \rangle (1 + iK)^{-1}_{l'l} \tag{4.20}$$

where $p_\mu = r_\mu$ for the length form, $p_\mu = \nabla_\mu / E$ for the velocity form, σ_l is the Coulomb phase shift $\sigma_l = \arg[\Gamma(l + 1 + i\gamma)]$, and $K_{ll'}$ is the scattering matrix.

When the iterative Schwinger method is applied to the calculation of photoionization cross sections, the first modified approach discussed in section 2.5.3 is used. Thus the K-matrix used in evaluating eq. (4.20) is the same at each step of the iterative procedure as the K-matrix which would be obtained from the asymptotic form of the wave function used to compute the dipole matrix element. However, when the Padé approximant correction method is applied to the two types of matrix elements needed to compute $I_{lm\mu}$ via eq. (4.20), corrections to the dipole matrix elements and the K-matrix elements are obtained using different Padé sequences. Thus when the two different matrix elements converge at a different rate, the convergence of the total cross section can be initially quite erratic, as can be seen in the result for 18 eV in table 4.2. This problem is especially noticeable when one of the eigenphases of the K-matrix goes through $\pm\pi/2$ during the iterative procedure. In all cases studied, the Padé correction procedure settled down after a few iterations and then converged rapidly.

Total cross sections for the photoionization of CO leading to the $X^2\Sigma^+$ state of CO^+ are shown in fig. 15. One can see that there is a shape resonance in this system which is analogous to the shape resonance

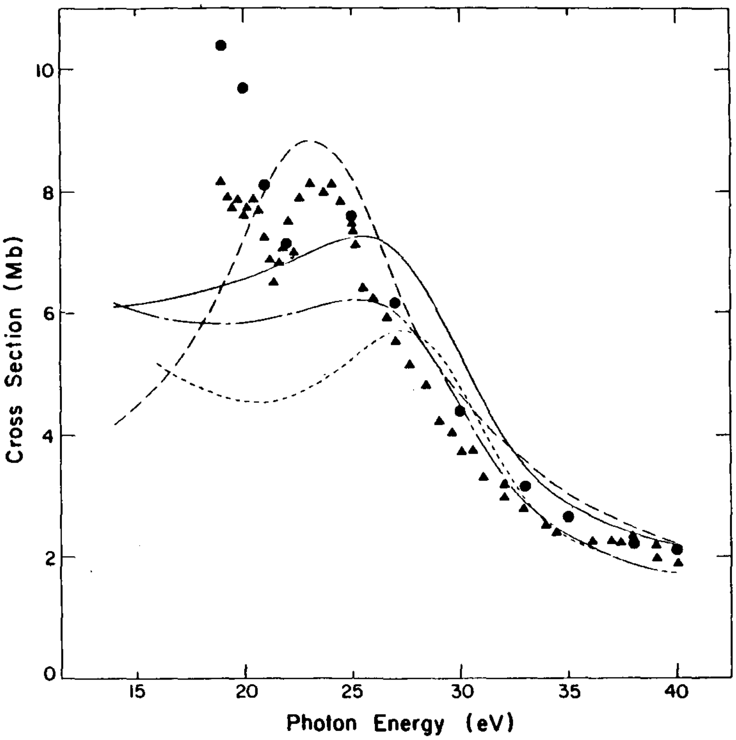

Fig. 15. Photoionization cross section of CO for the production of the $X^2\Sigma^+$ state of CO^+: ——, Padé approximant results using the dipole-length form; ----, Padé approximant results using the dipole-velocity form; ----, CMSM results from ref. [108]; —-—, STMT results from ref. [122]; ●, experimental data [A. Hamnett, W. Stoll and C.E. Brion, J. Electron, Spectrosc. Relat. Phenom. 8 (1978) 2992]; ▲, experimental data [E.W. Plummer, T. Gustafsson, W. Gudat and D.E. Eastman, Phys. Rev. A 15 (1977) 2339].

in the photoionization of N_2 leading to the $X^2\Sigma_g^+$ state of N_2^+. However, near the peak of the resonant cross section of CO at photon energies of 21–24 eV there is additional structure in the experimental data which cannot be accounted for by the accurate FCHF results obtained using the Padé correction procedure. This structure is probably due to autoionization features as suggested by Stephens et al. [121].

In fig. 15 we also compare the Padé correction results to those obtained by the CMSM [108] and by the STMT method [122]. The Padé and the CMSM results are seen to be in reasonable qualitative agreement, with a lower total cross section and a slightly narrower resonance in the CMSM results. The STMT cross sections oscillate about the accurate FCHF results and are qualitatively different in that the shape resonance in the STMT results seems to reproduce all of the structure in the 21–24 eV photon energy region, whereas the Padé results clearly indicate that this structure is due to final state correlation and should not be present in the FCHF approximation used both in the Padé approximant approach and in the STMT method.

In fig. 16 the photoelectron asymmetry parameters for the photoionization of CO are presented. Both the FCHF Padé approximant results and the CMSM results [123] are seen to be in relatively good agreement with experimental data except in the regions where final state correlation effects were important in the total cross sections of fig. 15. These ITAD asymmetry parameters can be compared with those for N_2 in fig. 4, and we can see that the effect of the shape resonance on the asymmetry parameters is much more pronounced in the photoionization of CO than in N_2.

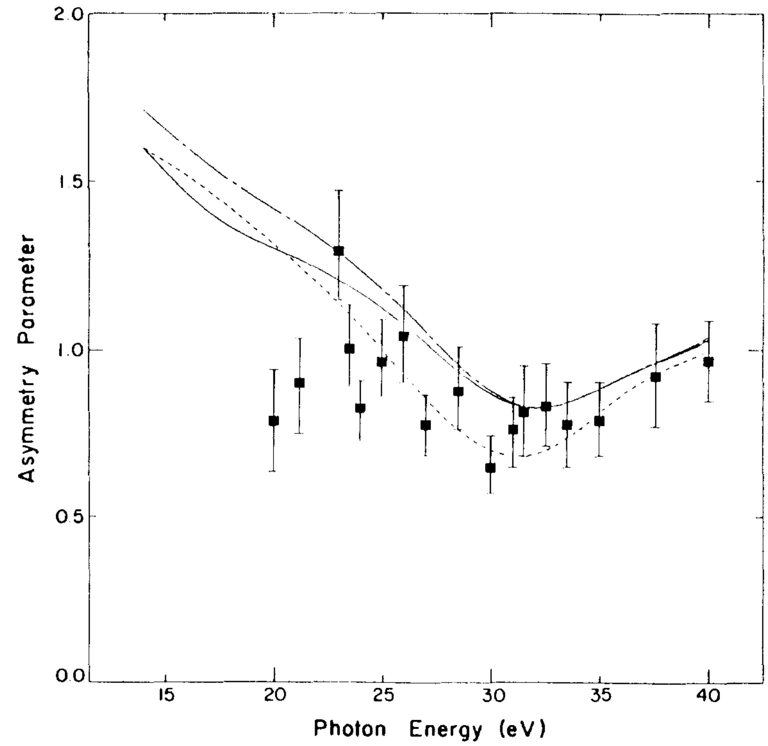

Fig. 16. ITAD photoelectron asymmetry parameters, β_k, for the photoionization of CO leading to the $X^2\Sigma^+$ state of CO^+: ——, Padé approximant results using the dipole-length form; – – – –, Padé approximant results using the dipole-velocity form; - - - -, CMSM results from ref. [123]; ■, experimental data [G.V. Marr, J.M. Morton, R.M. Holmes and D.G. McCoy, J. Phys. B 12 (1979) 43].

4.3. Application of the two-potential technique using the static potential

The two-potential technique discussed in sections 2.5.3 and 3.1.2 has been compared to the iterative Schwinger method in studies of the photoionization of NO [8]. Table 4.3 shows the photoionization cross sections and scattering eigenphase sums for the $2\pi \rightarrow k\sigma$ photoionization of NO, obtained using both the iterated Schwinger method and the iterated two-potential approach. Three different initial basis sets were considered, which were composed of 6 (basis set I), 12 (basis set II) and 18 (basis set III) Cartesian Gaussian functions. The initial basis sets used for the two methods contained the same number of basis functions, but with somewhat different exponents and basis function types. The use of different basis sets was dictated by the differences between the two methods. In the iterative Schwinger method, the basis set must represent the entire scattering potential, whereas in the two-potential method the basis set must only treat the exchange part of the FCHF potential. Thus some attempt has been made to optimize the basis set for the specific method while keeping the same number of basis functions. Further details are discussed in ref. [8]. The single-center expansion parameters in these calculations were $l_m = l_s^x = l_i^d = 20$, l_i^x ranged from $20(1\sigma)$ to $10(1\pi)$, $\lambda_m^x = 20$, $\lambda_m^d = \lambda_m^n = 40$, $l_p = 9$, $r_m = 64$ a.u., $h = 0.01$–0.16 a.u. with a total of 800 points in the grid.

The results of table 4.3 show that even with basis set I, which contains only six functions, the noniterated cross section is within 15% of the correct result for the two-potential method, whereas with the single-potential Schwinger method and an equivalent basis set the error in the photoionization cross section is nearly 50%. The two-potential method provides a rapidly convergent technique for computing photoionization cross sections even when very small basis sets are used. Both the uniterated one- and two-potential methods converge towards the correct results as larger basis sets are employed. When basis set III is used, there is no substantial difference between the results of first iteration of the one- and two-potential methods.

In fig. 17 we show the results of an earlier study of the photoionization of NO using the iterative Schwinger method [124]. In this figure we compare accurate FCHF results with those obtained using the CMSM approach with vibrational averaging [125], and with those obtained using the Tchebycheff

Table 4.3
Comparison of the eigenphase sums and cross sections in the $2\pi \rightarrow k\sigma$ photoionization of NO using one- and two-potential forms of the iterative Schwinger method, at a photoelectron energy of 4.7 eV

	Cross section (Mb)		Eigenphase sum (radians)	
Basis set[a]	$N = 0$	$N = 1$	$N = 0$	$N = 1$
		Two-Potential Method		
I	1.336	1.501	1.616	−0.498
II	1.328	1.509	−0.806	−0.489
III	1.472	1.513	−0.576	−0.488
		One-Potential Method		
I	0.798	0.524	−1.076	−0.853
II	1.239	1.342	−0.869	−0.488
III	1.277	1.524	−0.748	−0.479

[a] Basis set numbers are defined in the text.

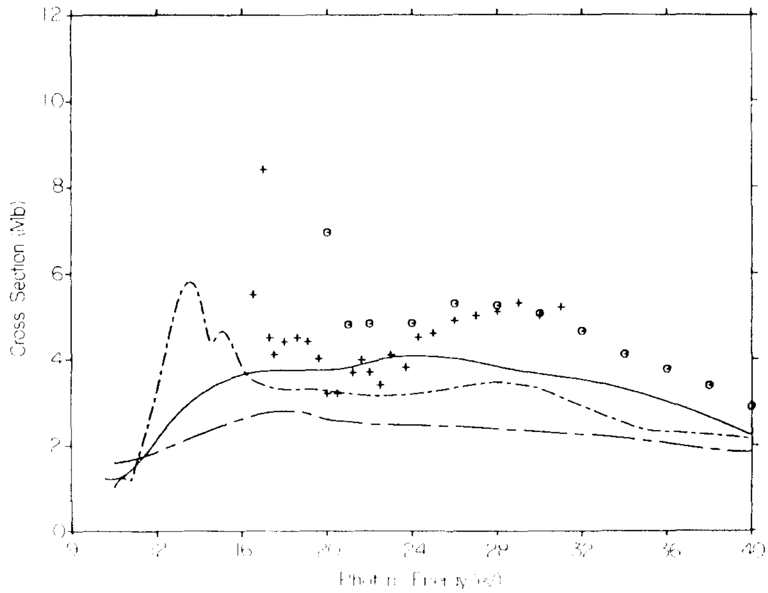

Fig. 17. Photoionization cross section of NO leading to the $X^1\Sigma_g^+$ state of NO^+; ——, fixed-nuclei FCHF results obtained using the iterative Schwinger method from ref. [124]; ----, fixed nuclei FCHF Tchebycheff imaging results from ref. [126]; —·—·—, vibrationally averaged CMSM results from ref. [125]; +, experimental data [S. Southworth, C.M. Truesdale, P.H. Korbin, D.W. Lindle, D.W. Brewer and D.A. Shirley, J. Chem. Phys. 76 (1982) 143]; ⊙, experimental data [C.E. Brion and K.H. Tan, J. Electron. Spectrosc. Relat. Phenom. 23 (1981) 1]. The ionization potential was taken to be 9.3 eV.

imaging method [126] which is related to the STMT method discussed above. The shoulder in the iterative Schwinger method at around 15 eV and the peak in the CMSM cross section at 18 eV are both due to a shape resonance in the $2\pi \rightarrow k\sigma$ channel. The fixed-nuclei iterative Schwinger and the CMSM vibrationally averaged results are in fairly good qualitative agreement as was the case with the results of CO_2 in fig. 9. In contrast to this, the structure in the Tchebycheff cross sections, in large part, comes from the $2\pi \rightarrow k\pi$ and $2\pi \rightarrow k\delta$ channels which are nonresonant in the exact FCHF results obtained using the iterative Schwinger method. These features in the Tchebycheff cross sections are thus spurious [124] and may be due to the inclusion of insufficient or unconverged moments of the oscillator strength distribution [127].

5. Application to electron–molecule collisions

5.1. Iterative Schwinger method applied to H_2 and CO_2

In addition to the applications of the iterative Schwinger method to molecular photoionization, the method has also been applied to electron–molecule collisions for several systems. The first application of the iterative Schwinger method was to e^-–H_2 scattering [5]. In that study, the convergence properties of the iterative Schwinger method were studied extensively. In table 5.1 the K-matrix elements of the iterative Schwinger method are shown for the $^2\Sigma_g$ symmetry in e^-–H_2 scattering. In these calculations no initial basis set was used so that the first trial solutions, ψ_0^S defined by eq. (2.132) were plane waves,

Table 5.1
Convergence of the K-matrix computed using the iterative Schwinger variational method starting from plane waves for e^-–H_2 scattering of $^2\Sigma_g$ symmetry with $k = 0.5$ a.u.

N[a]	K_{00}^N	K_{02}^N	K_{22}^N
0	0.0	0.0	0.0
1	−2.931	0.013	0.016
2	−1.701	0.013	0.016
3	−1.552	0.013	0.016
4	−1.548	0.013	0.016

[a] N is the iteration number as defined by eqs. (2.124)–(2.128).

or in the notation used in section 2.5.2, $|\psi_0^S\rangle = |S\rangle$. As seen from table 5.1, even with no initial basis function, the iterative procedure can converge.

In table 5.2 we show the results of the iterative Schwinger method for four initial basis functions. The four functions were two s-type and two p-type Cartesian Gaussian functions, each with the exponents 0.3 and 1.0. In this table we give results for both the one- and two-potential iterative methods [8]. As in the case of photoionization of NO discussed in section 4.3, the two-potential method clearly gives the superior result at the uniterated level but both methods are essentially converged by the first iteration.

The iterative Schwinger method has also been used to compute the differential and total scattering cross sections for e^-–CO_2 scattering [56]. In these studies spherical Gaussian basis sets containing between 19 and 30 basis functions for the various scattering symmetries were used [56]. The single-center expansion parameters were the same as those used in the study of the photoionization of CO_2 discussed in section 4.1.3. The total elastic cross section for e^-–CO_2 scattering in the static-exchange approximation are given in fig. 18. The most prominent feature in both the theoretical and experimental cross sections is the well-known shape resonance, which lies experimentally at about 3.8 eV [128]. A fit

Table 5.2
Convergence of the K-matrix computed using the one- and two-potential forms of the iterative Schwinger variational method for e^-–H_2 scattering ($^2\Sigma_g$ symmetry with $k = 0.5$ a.u.[a])

N[b]	K_{00}^N	K_{02}^N	K_{22}^N
		Two-Potential	
0	−1.598	0.016	0.015
1	−1.546	0.013	0.016
		One-Potential	
0	−2.079	0.004	0.004
1	−1.545	0.013	0.016

[a] The initial basis sets consisted of four Cartesian Gaussian functions as discussed in the text.
[b] N is the iteration number as defined by eqs. (2.124)–(2.128).

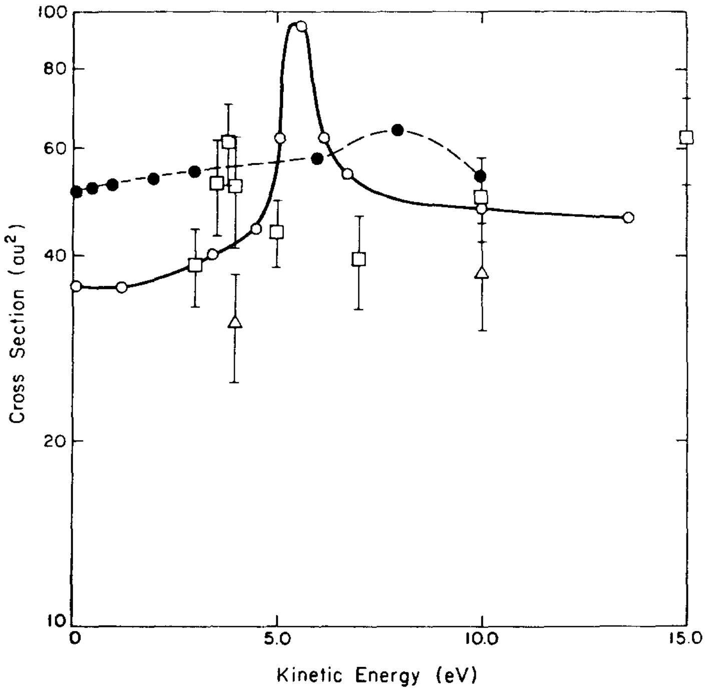

Fig. 18. Total elastic cross section for e^-–CO_2 scattering: —O—, exact static-exchange results using iterative Schwinger method; ---●---, static plus model exchange from ref. [58]; □, experimental data [T.W. Shyn, W.E. Sharp and G.R. Carignan, Phys. Rev. A 17 (1978) 1855]; △, experimental data [D.F. Register, H. Nishimura and S. Trajmar, J. Phys. B 13 (1980) 1651].

of the static-exchange eigenphase sums to a Breit–Wigner form gives a resonance position of 5.39 eV and a width of 0.62 eV. The difference between the static-exchange and experimental results are due to polarization effects in the scattering process. In fig. 18 we also compare the static-exchange results obtained using the iterative Schwinger method to static-local model exchange calculations by Morrison et al. [58]. This comparison shows that the magnitude and shape of the total elastic cross section obtained with the local exchange approximation are quite different from the static-exchange cross sections. An empirical modification of the static-local model exchange potential, such as the addition of a polarization potential, to obtain agreement with experimental data, will hence necessarily have to compensate for both the deficiency of the model local exchange potential as well as electron correlation effects.

Finally, in fig. 19 the elastic differential cross sections for the scattering of 10 eV electrons by CO_2 are presented. The iterative Schwinger static-exchange results are seen to be in reasonable agreement with experimental data.

5.2. Application of the $\tilde{C}$-functional

To illustrate some features of the $\tilde{C}$-functional, particularly its convergence characteristics, we now discuss some results of its applications to electron scattering by the polar molecule LiH. For such targets the noniterative form of the Schwinger principle can show poor convergence with respect to the number of basis functions in the expansion of the trial scattering function [12]. For these strong dipolar

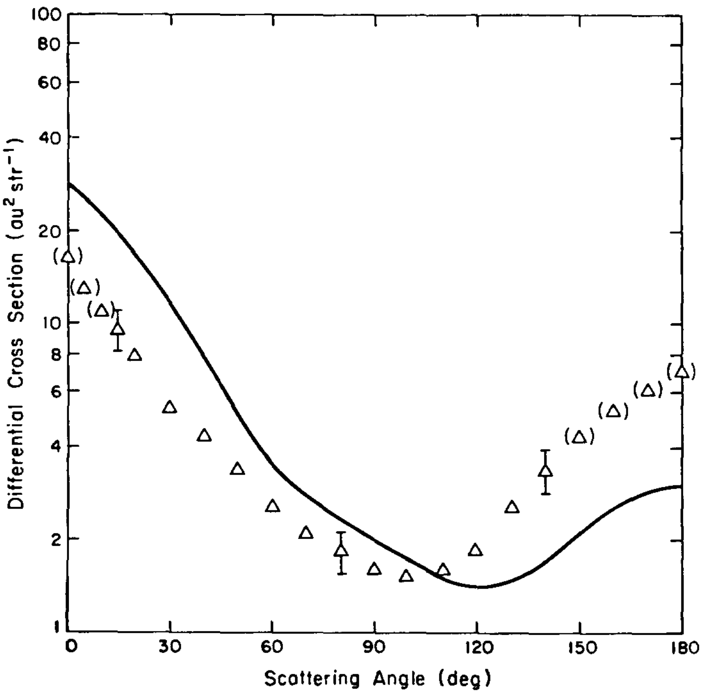

Fig. 19. Elastic differential cross section for scattering 10 eV electrons by CO_2: ——, exact static-exchange obtained using the iterative Schwinger method; △, experimental data [D.F. Register, H. Nishimura and S. Trajmar, J. Phys. B 13 (1980) 1651], (△) indicates an extrapolated value.

potentials long-range basis functions are important. In addition to the regular solutions, regularized irregular solutions such as $(1-\exp(-\alpha r))\, n_l(r)$, where $n_l(r)$ is a Neumann-type function, are often needed in the basis. This regularized trial function can, in turn, interfere distructively with the short-range basis functions [12]. Furthermore, variation of the parameter α has little effect on this adverse feature. In the iterative Schwinger method the iterative procedure takes a basis function η_i and, through the integral equation, introduces $G_0 V\eta_i$ into the scattering function. These $G_0 V\eta_i$ terms play the role of these "regularized" irregular functions in the basis.

In the $\tilde{C}$-functional method the unperturbed solution S, e.g., the regular function, is subtracted out from the total scattering function and treated exactly in the first and second Born terms. We hence expect that basis set expansions of the $\tilde{C}$ trial function should be particularly effective. In fact, we will see that for a given basis the $\tilde{C}$-functional gives a result of comparable accuracy to that obtained after one iteration in the Schwinger iterative method.

In table 5.3 we compare the results of application of the Schwinger and $\tilde{C}$-functionals to e–LiH collisions. We used the SCF orbitals of Watson et al. [55] which gives a dipole moment of 2.369 a.u. at an internuclear distance of 3.015 a.u.. The primary purpose of this example is to illustrate the theoretical prediction that the $\tilde{C}$-functional yields results essentially equivalent to the iterative Schwinger procedure after one iteration [12]. The $^2\Sigma$ and $^2\Pi$ K-matrix elements at $k = 0.5$ and 0.1 a.u., respectively of table 5.3, illustrate this feature quite clearly. We see that the differences between the K-matrices of these two functionals without any iterative improvement can be very substantial and that, moreover, the $\tilde{C}$-functional results can often be quite close to those of the Schwinger principle after one iteration. In general, this behavior is consistent with our prediction [12]. For convenience, in table 5.3

Table 5.3
Illustration of the $\tilde{C}$-functional approach: K-matrix elements and eigenphase sums for e–LiH scattering[a]

	Iterative Schwinger method			$\tilde{C}$-functional	
Iteration	0	1	2	0	1
			$k = 0.5$ $(^2\Sigma)$[b]		
K_{00}	0.268	0.916	0.921	0.904	0.922
K_{01}	−1.475	−0.304	−0.233	−0.346	−0.215
K_{02}	−0.136	0.282	0.316	0.180	0.328
K_{11}	−3.075	−4.227	−4.280	−3.237	−4.120
K_{12}	−1.185	−2.392	−2.614	−2.036	−2.526
K_{22}	−0.258	−0.862	−0.967	−0.773	−0.899
K_{13}	−0.025	−0.724	−0.781	−0.294	−0.736
K_{23}	−0.028	−0.722	−0.817	−0.573	−0.793
K_{33}	−0.008	−0.177	−0.211	−0.094	−0.202
δ_{SUM}[c]	2.558	2.698	2.751	2.733	2.769
			$k = 0.1$ $(^2\Pi)$		
K_{11}	0.009	0.419	0.750	0.493	0.788
K_{12}	−0.001	−0.449	−0.544	−0.504	−0.500
K_{22}	-0.2×10^{-4}	−0.050	−0.053	−0.060	−0.051
K_{13}	-0.1×10^{-4}	0.012	0.071	0.017	0.090
K_{23}	-0.2×10^{-5}	−0.361	−0.417	−0.364	−0.402
K_{33}	−0.0	−0.023	−0.020	−0.022	−0.011
δ_{SUM}	0.009	0.269	0.454	0.304	0.493

[a] With the coordinate system at the center of mass.
[b] Incident momentum.
[c] Eigenphase sum in radians.

we also show the results of the $\tilde{C}$-functional after one iteration. Of course, one could always improve these results by starting with a larger basis but this difference between the two functionals can be very important in applications to systems where numerical iterative procedures are impractical. Finally, a comparison of the eigenphase sums and K-matrices in this table shows that the eigenphase sums can be a very poor criterion for monitoring the accuracy of a calculation.

5.3. Multichannel extension of the Schwinger principle

In this section we discuss some results obtained with our multichannel extension of the Schwinger variational principle, described in section 2.6, for both elastic and electronically inelastic scattering of electrons by H_2. The elastic studies provide a good test of how effectively the formulation can represent the effects of polarization since accurate experimental and theoretical cross sections are available for this system. Moreover, for low-energy collisions the polarization interaction can generally have a significant effect on the overall shape and magnitude of the scattering cross section as well as on the width and position of resonances. Although substantial progress has been made in including these effects for some systems (notably H_2 and N_2 [129]), no single method has yet been demonstrated for general use, i.e., for elastic and electronically inelastic collisions with both linear and nonlinear molecular targets. For electronic excitation, there have also been few studies, beyond the plane-wave and distorted-wave approximations, of these cross sections at low impact energies. Here we will present excitation cross sections for the $X^1\Sigma_g^+ \to b^2\Sigma_u^+$ transition in H_2 for collision energies between 12 and 30 eV. Cross sections for this excitation have been studied using plane-wave theories [130, 131], the distorted-wave approximation [59], and two-state close-coupling methods [63, 64, 132, 133].

We first discuss the results of application of the multichannel Schwinger principle of eq. (2.171) to elastic e–H_2 collisions. Details of these studies along with comparisons of our results with those of other calculations and with experimental data have been given elsewhere [134]. Here we will only present some highlights of these studies. Equation (2.171) provides an analytical approximation to the body-frame fixed-nuclei scattering amplitude $f(\boldsymbol{k}_m, \boldsymbol{k}_n)$. We then expand this amplitude in a partial wave series and make the requisite transformation into the laboratory frame [134]. After accounting for the random orientation of the target, the physical differential cross section is obtained by performing the appropriate average over initial and sum over final spin states. The calculations are carried out in the fixed-nuclei approximation. The scattering wave function is expanded in $(N+1)$-particle determinants and all orbitals appearing in these determinants are in turn expanded in Cartesian Gaussian basis functions. The calculations are hence implemented entirely in an L^2 basis. These elastic cross sections include contributions from determinants of ${}^2\Sigma_g$, ${}^2\Sigma_u$, ${}^2\Pi_u$ and ${}^2\Pi_g$ symmetries.

Figures 20 and 21 show our calculated elastic differential cross sections at 3 and 10 eV both at the static-exchange level and with inclusion of polarization along with the measured values [136–138] and cross sections derived from the results of calculations of Gibson and Morrison [139]. In our studies we used 10 and 8 pseudo-states for the closed channels of Σ and Π symmetry respectively. Differential cross sections can provide greater insight into the scattering process and are a more stringent test of the

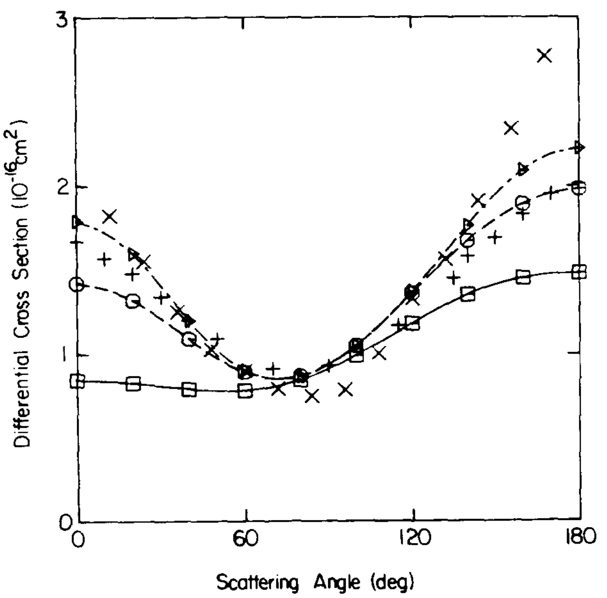

Fig. 20. Differential cross sections for elastic e–H_2 scattering at 3 eV: present static-exchange results, (□); present results including target polarization, (O); results of ref. [139], (Δ); measured values of ref. [136], (+); measured values of ref. [137], (×).

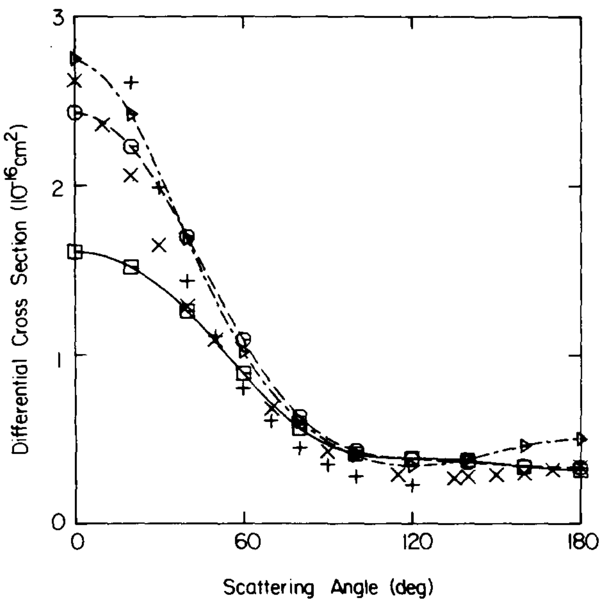

Fig. 21. Differential cross sections for elastic e–H_2 scattering at 10 eV: present static-exchange results, (□); present results including target polarization, (O); results of ref. [139], (Δ); measured values of ref. [140], (+); measured values of ref. [136], (×).

theory than integral cross sections. These results show that the substantial differences in the cross sections due to the inclusion of polarization are reproduced by our method and that the calculated cross sections are in very reasonable qualitative and quantitative agreement with the experimental values. In fig. 22 we compare our integral cross sections obtained with and without target polarization for energies between 1 and 10 eV with the results of other calculations [129, 139] and the measured values [140].

We have also used the Schwinger multichannel formulation of section 2.6 to study the integral and differential cross sections for the $X\,^1\Sigma_g^+ \to b\,^3\Sigma_u^+$ excitation in H_2 for impact energies from 12 to 30 eV [135, 141]. It is convenient to choose this excitation since it is one of the few molecular systems that have been studied by close-coupling techniques [63, 64, 133]. In our studies we include only two open channels and the trial scattering wave function of eq. (2.172) is expanded in $(N+1)$-particle determinants.

Our calculations are performed within the framework of the fixed-nuclei and Franck–Condon approximations. The rotational levels are assumed to be degenerate and the physical cross sections are obtained by averaging the fixed-nuclei results over all molecular orientations. A single electronic transition matrix element is calculated with outgoing electron energy determined by the vertical transition energy from the $\nu = 0$ level of the $X\,^1\Sigma_g^+$ state to the final position on the $b\,^3\Sigma_u^+$ state with the largest Franck–Condon factor. For the ground state we use a self-consistent-field (SCF) wave function obtained with a Cartesian Gaussian basis at an internuclear distance of $1.4003a_0$ [141]. For the $b\,^3\Sigma_u^+(1\sigma_g 1\sigma_u)$ state we make the frozen-core approximation and determine the $1\sigma_u$ orbital by diagonalizing the V_{N-1} potential of the core in the SCF basis. All calculations are carried out in a Cartesian Gaussian basis. Details of the basis are given elsewhere [141]. In the body frame the calculated cross sections include contributions from the Σ_g, Σ_u, Π_g and Π_u symmetries.

Figures 23 and 24 show our calculated differential inelastic cross sections for collision energies of 12

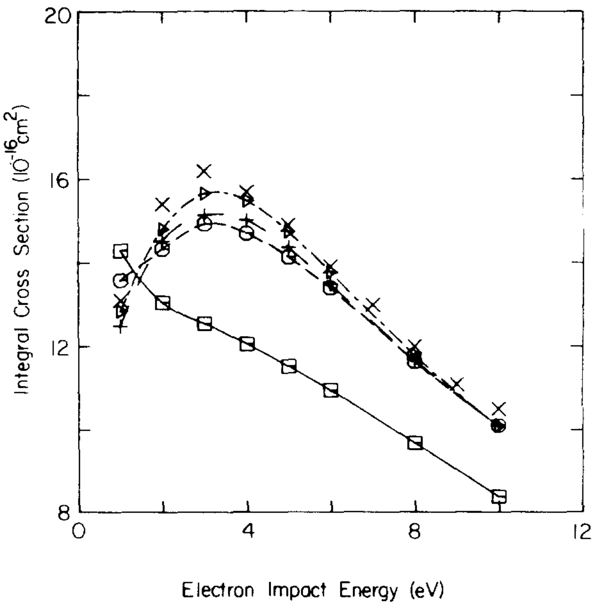

Fig. 22. Elastic cross sections for e–H_2 scattering: present static-exchange results, (□); present results including target polarization, (○); results of ref. [129], (+); results of ref. [139], (△); experimental results of ref. [140], (×).

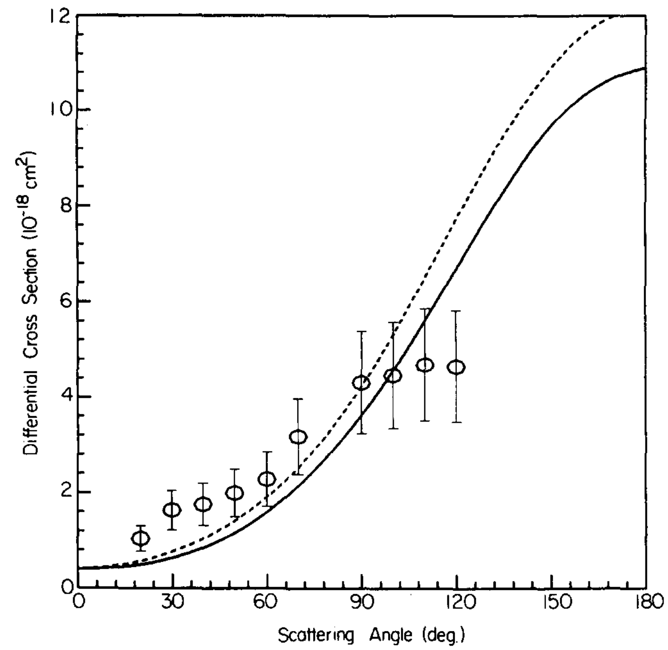

Fig. 23. Inelastic differential cross sections for the $X\,^1\Sigma_g^+ \rightarrow b\,^3\Sigma_u^+$ excitation at 12 eV incident energy: ——, two-state Schwinger multichannel results of ref. [141]; ----, distorted-wave cross sections of ref. [59]; △, experimental results of ref. [142].

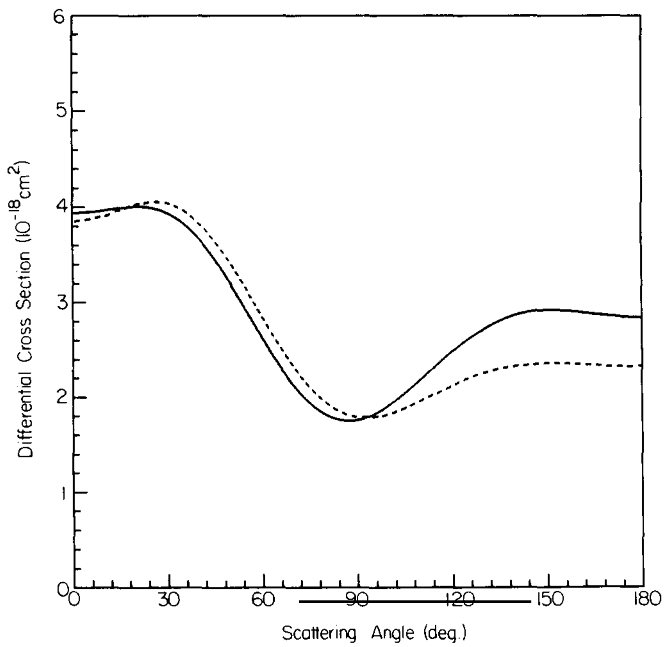

Fig. 24. Inelastic differential cross sections for the $X\,^1\Sigma_g^+ \rightarrow b\,^3\Sigma_u^+$ excitation at 25 eV incident energy: ——, two-state Schwinger multichannel results of ref. [141]; ----, distorted-wave cross sections of ref. [59].

and 25 eV. At both energies we show the differential cross sections of the distorted-wave calculations of Fliflet and McKoy [59]. At 12 eV the measured cross sections of Hall and Andrić [142] are also given. These results along with the differential cross sections at other impact energies and the related integral inelastic cross sections from 12 to 30 eV are discussed in ref. [141] where extensive comparisons to the results of other calculations [63, 64, 130–133] and available data will be presented. For the present purposes we will only comment briefly on a few salient features of the differential cross sections of figs. 23 and 24. At 12 eV the substantial backward peaking of this exchange-type excitation cross section is very evident. The differential cross sections of the present two-state calculations are in surprisingly good agreement with the measured cross sections. Moreover, the distorted-wave cross sections are very similar to those of the two-state calculations both at this energy and at 25 eV. Additional applications of this multichannel theory to excitation of other electronic states of H_2 and to several states of N_2 are under way.

6. Concluding remarks

In this article we have given an overview of our studies of Schwinger-like variational principles and the several important extensions and applications of these variational principles which we have made in recent years. What we believe emerges from these studies is that with current computer capabilities these Schwinger-like variational principles provide a powerful approach to the study of low-energy electron collisions with molecules and molecular photoionization processes. In addition to the examples which we have discussed in this review, these approaches can and are currently being applied to a range of electron collision problems. Among others, these include studies of molecular photoionization of nonlinear molecules and of molecules such as CO and N_2 adsorbed on small clusters of Ni atoms. With these same techniques for determining the electronic continuum orbitals of molecular ions we are studying the cross sections for resonant multiphoton ionization of molecules.

In the area of electron–molecule collisions we have only discussed the results of the first applications of our multichannel Schwinger variational principle to the excitation of H_2. Cross sections for excitation of other electronic states of H_2 and of N_2 are currently being studied both at the two-state level and beyond. Moreover, this formulation of electron collisions has considerable potential for applications to quite general molecular targets. For example, studies of the elastic scattering of low-energy electrons by CH_4 have already been completed at the static-exchange level. An important application of this theory will be to the determination of cross sections for the electronic excitation of oriented molecules adsorbed on small clusters of metal atoms. Such systems are useful models of the behavior of molecules adsorbed on actual metal surfaces.

Finally, we would like to again stress that throughout this discussion we have focused primarily on our own contributions in this area and have not made any attempt to review and assess the many important contributions of others to this field [13].

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